Segmentation (LO-SGT)

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Segmentation (LO-SGT)

Use

Businesses compete today in an increasingly customer-oriented market. Therefore, it becomes necessary to streamline internal processes and logistics around the needs of each market and its unique requirements through appropriate sourcing, planning, and supply of goods to meet market demands.

Some materials are expected to support large volumes of data. Segmentation can be used to order such materials by logically categorizing them based on certain criteria or characteristics.

For example, in the fashion industry, you can distinguish a fashion material such as a t-shirt based on the size and the color. This t-shirt in a particular size and color is called a material variant. You can enter a specific item line for a material variant. Using segmentation, you can also order material variants of different quality levels, for different customer channels, and from different countries of origin.

In material master, you can segment a material using segmentation structure and segmentation strategy.

Implementation Considerations

These functions are a part of the business function **Material Segmentation across Logistics (SD, MM, PP, IM, LE)** (`LOG_SEGMENTATION_03`).

Note

All the segmentation business functions can be activated only on the SAP HANA database.

Integration

Segmentation is integrated in the following applications:

- Material Master
- Sales and Distribution
- Materials Management
- Production Planning and Control

You can assign segments to a material using segmentation structure and segmentation strategy in the material master. You can configure these settings for segmentation in Customizing for **Segmentation** under **SAP Implementation Guide** > **Logistics - General**.

Features

There are two types of segments:

- Requirement segments
- Stock segments

The segment you assign to the materials in the requirements (for example, Sales Order, Planned Independent Requirement, Stock Transport Order) is called requirement segment. This allows you to subdivide your requirements based on certain criteria.

Example

The incoming requirements such as sales orders are divided based on the requested quality - Quality 1, Quality 2, or Quality 3, and by customer channel - Store, Wholesale, or E-Commerce.

Stock segments are used for logical distribution of existing and planned stock. Therefore, stock segments are assigned to both the physical stock and procurement proposals such as planned orders, production orders, purchase requisitions, and purchase orders.

Example

You can divide your stock based on the same criteria as the requirement, that is, by quality - Quality 1, Quality 2, or Quality 3 and by countries of origin - India, China, or Germany.

The segments influence the entire logistics chain. You can assign segments to a material using segmentation structure and segmentation strategy in the material master.

More Information

- Segmentation Structure
- Segmentation Strategy

Segmentation Structure

Definition

A segmentation structure is a combination of two or more segments of any material. For example, a segmentation structure can be a combination of quality, customer channel, and country of origin.

Use

Segmentation structures are defined in the Customizing for **Define Segmentation Structures** under **SAP Implementation Guide** > **Logistics - General** > **Segmentation**

Note

You have defined segmentation relevant characteristics using **Master Data Characteristics Management** (transaction `CT04`). For more information, see Maintain Segmentation Relevance for Functions.

More Information

- Segmentation (LO-SGT)
- Segmentation Strategy

Segmentation Strategy

Use

To efficiently plan your requirement, it is necessary to fulfill newly arising requirement (such as sales orders) with existing or future stock (for example, purchase requisition). If you have defined segments for your material, you can assign the stock to the requirement using these segments. Segmentation strategy determines how requirement segments are assigned to stock segments. This kind of assignment takes place during available-to-promise (ATP) check and material requirements planning (MRP).

Segmentation strategy is also important for the following requirement elements and stock elements:

- For externally procured materials during planning run. For example, during the creation of purchase requisitions.
- For the components of materials, i.e., the dependent requirements and the fulfillment of those requirements.
- For consumption of planned independent requirements using the actual customer requirements during the planning run.

Segmentation strategies are defined in the Customizing for **Define Segmentation Strategies** under **SAP Implementation Guide** > **Logistics - General** > **Segmentation**

Note

You have defined segmentation relevant characteristics using **Master Data Characteristics Management** (transaction CT04). For more information, see Maintain Segmentation Relevance for Functions.

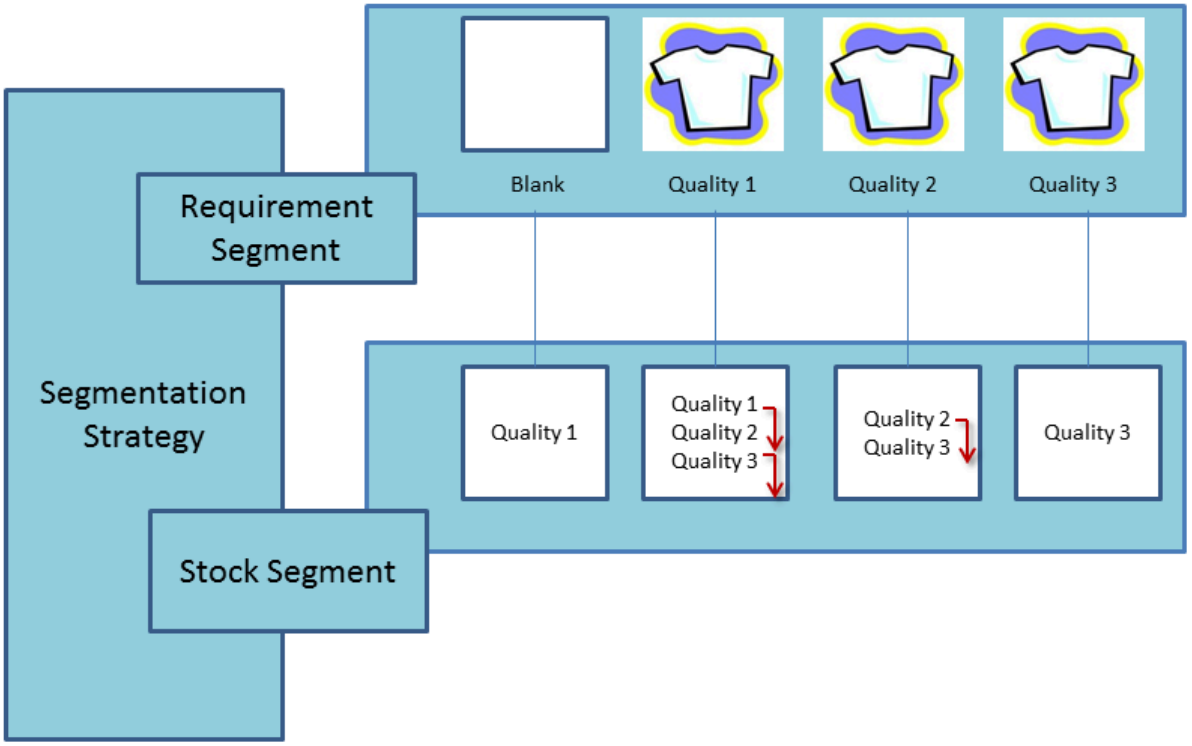
Features

Assignment of Stock or Requirement Segments

If you have assigned segments to a material, you can assign stock to the requirements using segmentation. The assignment of stock to requirements takes place during the planning run. The system determines which requirement segment is covered by which stock segment.

You determine the following sequence for your segmentation using your segmentation strategy:

1. You receive a requirement, such as a sales order. For example, the requirement belongs to customer segment **Wholesale** and is of quality **Quality 1**.
2. Using segmentation strategy, you define that this customer will receive the requested segment first, if you have the segment in stock.
3. If the requested segment is not in stock, you decide that the customer will receive available stock segment in the next level.



It is not mandatory to attach a segmentation structure or a segmentation strategy to a material.

A segmentation strategy can have multiple requirement segments but can be assigned to only one blank stock segment. Such segmentation strategies are identified by Scope in Segmentation Strategy. Scope indicates if the segmentation strategy only contains requirement segments or both requirement and stock segments.

Deletion of Segmentation Values

You can delete segmentation values for segmentation combinations. However, if you wish to delete multiple segmentation values, you can use the transaction `SE38` to execute the report program `SGT_DELETE_CATS`.

More Information

- Segmentation (LO-SGT)
- Segmentation Structure
- Scope in Segmentation Strategy

Scope in Segmentation Strategy

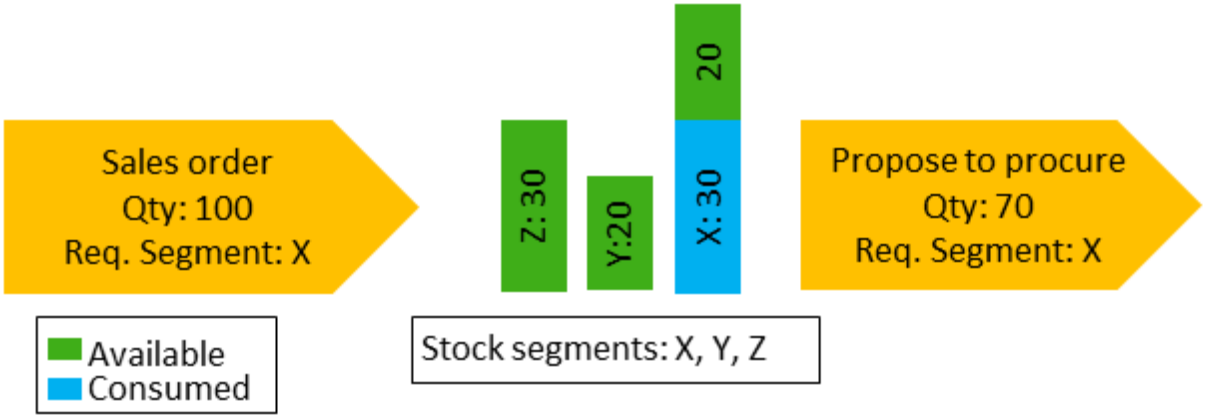
Use

Scope is used to define the mapping rules or segment combinations in segmentation strategy. Scope can either be 1 or 2.

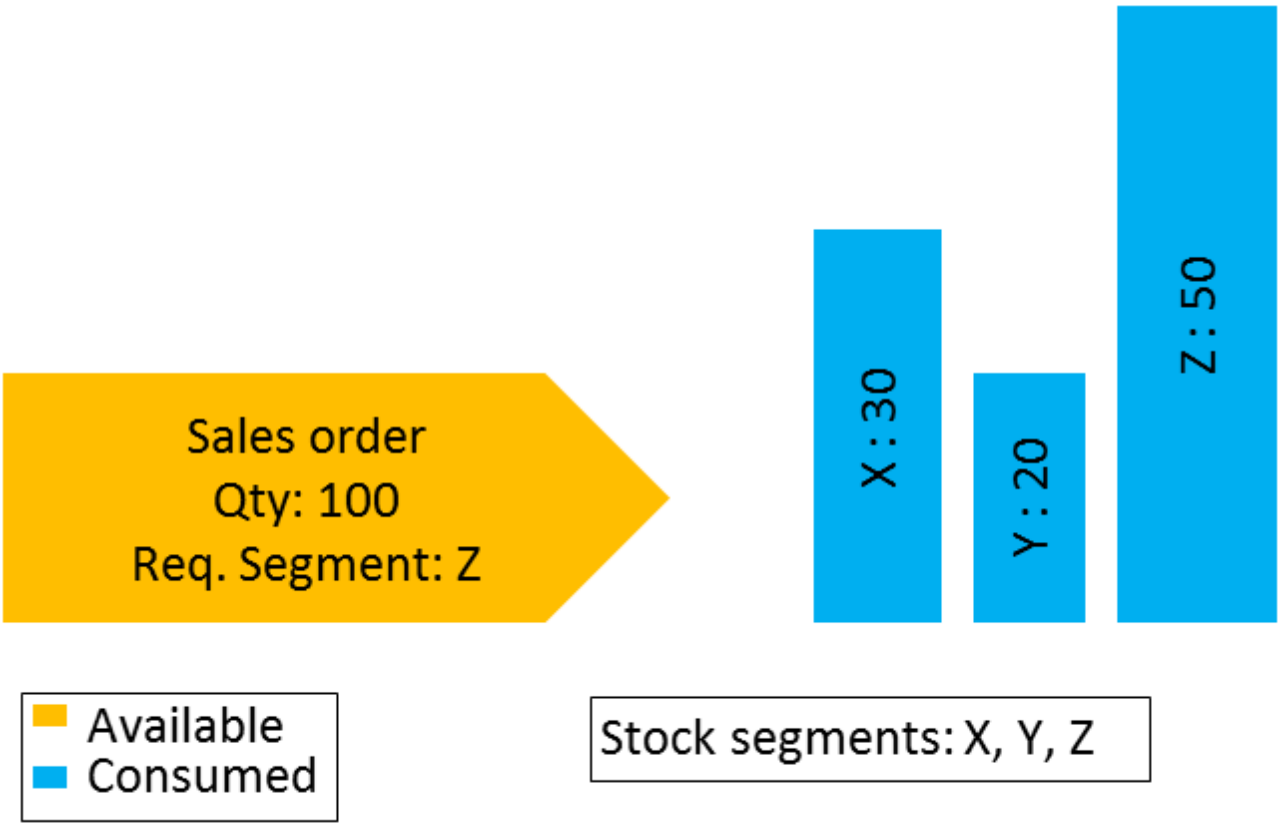
If the scope is 1, then both requirement segment and stock segment are considered during planning run. Mapping of requirement segment and stock segment can be 1:1 or 1:N.

Example

For 1:1 mapping, assume the requirement is for 100 pieces of segment X. As per this mapping, available stock of 30 pieces of segment X is consumed. As the requirement is not fulfilled, the system proposes to procure 70 pieces of segment X.



For 1:N mapping, assume the requirement is for 100 pieces of segment Z. As per this mapping, all available stock of 100 pieces in all three segments X, Y, and Z are procured.



If the scope is 2, then only requirement segment is considered during planning run. Stock segment is not considered. Customer requirements with specific requirement segments are covered by the common pool of stock available.

More Information

- Segmentation (LO-SGT)
- Segmentation Structure
- Segmentation Strategy

Maintain Segmentation Relevance for Functions

Use

This process allows you to maintain segmentation relevance for functions:

Prerequisites

You have defined a characteristics group (in Customizing) using the following steps:

1. In Customizing, navigate to Cross-Application Components > Classification System > Characteristics > Define Characteristics Groups.
2. Click New Entries.
3. Enter SGT_SAP-C under the Group field.

4. Enter **Characteristics for Segmentation** under the **Characteristics Group Description** field.
5. Save the entries. If the system prompts for the changes to be saved under a transport request, save the changes under a customizing transport request.

Process

You make you of the following steps to maintain segmentation relevance for functions:

1. Execute transaction `CT04`.
2. Enter the name of the **Characteristics**.
3. On the **Basic Data** tab page, enter the **Description**.
   a. Enter **Characteristics for Segmentation** as the **Chars Group**.
   b. Enter **Character Format** as the **Data Type**

      .

   c. Enter **Number of Chars**.
4. On the **Values** tab page, enter **Char. Value** and a corresponding **Description**.
5. On the **Addnl data** tab page, choose the **Relevance** pushbutton.

   Define Relevance for Functions overview screen in displayed.
6. Select the applications relevant for segmentation.
7. Choose **Save**.

ℹ Note

You can define segmentation structure and segmentation strategy for an application only if the application is made relevant for segmentation.

More Information

- Segmentation Structure
- Segmentation Strategy

Enabling Segmentation Relevancy

Use

You use this report to enable segmentation relevancy for the existing characteristics of a segmentation structure. The relevancy can be enabled to maintain segmentation relevant data for the following fields:

- Bill of Material (BOM)
- Routing (ROU)
- Season (SEA)
- Pricing (PRI)

- MRP General Data (MRM)
- MRP Quantity based Data
- Warehouse Management Level (WM1)
- Warehouse Management Storage Level (WM2)
- EAN

Prerequisites

- You have defined segmentation relevant characteristics using Master Data Characteristics Management (transaction `CT04`).
- You have defined segmentation structures for characteristics in the Customizing for **Define Segmentation Structures** under **Logistics - General** > **Segmentation** > **Segmentation Structures**

Features

On the report entry screen, you specify the range of segmentation structures that should be made segmentation relevant for `BOM`, `ROU`, `WM1`, and `WM2`. Based on your selection, the report displays the characteristics available in the segmentation structures along with an option to enable segmentation relevancy.

This is a one-time report. You can set the relevancy of the existing characteristics that were created for SAP Fashion Management 1.0 and SAP Fashion Management 1.0 SP 03 releases only.

Activities

1. Execute the program `SGT_SETRELEVANCE`.
2. Enter the range of segmentation structures to be enabled for relevancy
3. Select the relevancy checkbox for the selected structures.
4. Execute the report.

   The characteristics of the selected structures are displayed.
5. Save the changes. You cannot revert the changes after you have set the relevancy.

Segmentation: Enhancements to Material Master

Use

Segmentation logically categorizes materials based on certain criteria or certain characteristics of the material. From SAP ERP 6 EHP 7 SP 04 onwards, SAP has introduced a new business function `LOG_SEGMENTATION` which allows the concept of segmentation in core ERP processes. These enhancements to material master can be broadly divided into changes made to the client data and changes made to the plant data.

Features

Client Data

In client data, you must mark a material relevant for segmentation before assigning segmentation characteristics and their features to the material.

In the material master, you can specify if a material is relevant for segmentation or not by selecting the **Segmentation Relevant** checkbox. If you make a material relevant for segmentation, then you have to maintain the segmentation structure and strategy for that material. The **Segmentation Structure** and **Segmentation Strategy** fields appear only for segmentation relevant materials.

You can define if the **Segmentation Structure** and **Segmentation Strategy** fields should be mandatory or optional fields based on the Customizing activity **Define Attributes of System Messages** under **Logistics - General** > **Material Master** > **Basic Settings**.

After you set the segmentation structure and segmentation strategy, the **Segmentation Strategy Scope** maintained for the segmentation strategy is displayed in material master.

If you create a new material as a copy of an existing material, then the segmentation relevance, segmentation structure, and the segmentation strategy of the newly created material will be the same as that of the existing, reference material.

Segmentation structure and segmentation strategy can be changed in the change mode as long as segmentation is not fixed at the plant level. If segmentation is fixed for even one plant, segmentation structure and segmentation strategy cannot be changed.

Plant Data

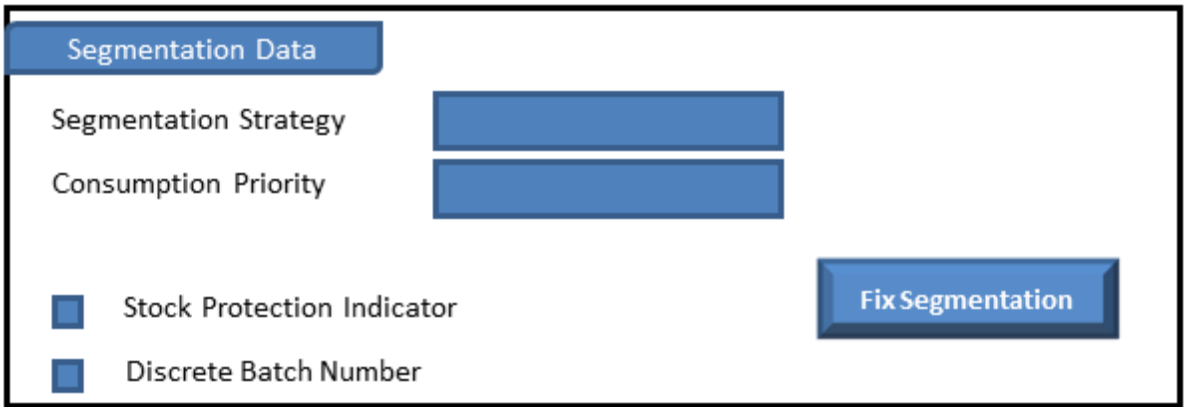
Segmentation Structure, Segmentation Strategy, Segmentation Strategy Scope, and Fix Segmentation

You can switch off segmentation in a plant, but you cannot set a different segmentation strategy for the same material in different plants.

You can change the segmentation strategy for plant data based on the following criteria:

- If the material is marked as segment relevant in client data
- **Segmentation Structure** and **Segmentation Strategy** fields are maintained in client data and the **Segmentation Strategy** is not fixed in plant data

You can then use the **Fix Segmentation** pushbutton to fix segmentation values for plants.



Material Master: Fix Segmentation

In some scenarios, you can change the segmentation assignment and fixing of segmentation strategy to a material using the **Conversion of Material** report. To access this report, on the **SAP Easy Access** menu, choose **Logistics** > **Materials Management** > **Material Master** > **Material** > **Conversion of Material**. You can also use the transaction `SGT_CONVERT_MATERIAL` to execute the report.

ℹ Note

You can use this report to change segmentation at the plant level only if no follow-on documents, for example sales orders, purchase orders, exist for the material.

If you create a new material as a copy of an existing material, then the segmentation strategy of the newly created material will not be copied from the reference material. This gives you the flexibility to decide the relevance of segmentation strategy at the plant level.

Consumption Priority

Consumption priority is used to whether planned independent requirements (PIRs) are consumed first based on the segments and then based on the consumption mode, or the other way round. With segmentation, you have one more option for consumption priority, which is, only if requirement segments match. You can maintain the **Consumption Priority** only if you have maintained segmentation strategy for a material. The **Seg. Strategy Scope** can be 1 or 2.

Stock Protection Indicator

Stock protection is used in demand-supply matching processes that protect the stock for demands from high-priority requirement segments even when the actual demand is not yet available in the system. Stock protection allows businesses using a common stock pool for satisfying demands with different priorities to reserve stock for higher-priority demands against lower-priority demands that come in earlier. You can maintain the **Stock Protection Indicator** only if you have maintained segmentation strategy for a material. The **Seg. Strategy Scope** can be 1 or 2.

Discrete Batch Number

Batch Management is also mandatory when a segmentation strategy is maintained for a material and the **Seg. Strategy Scope** is 1.

Batch management also results in generation of batch numbers in sales and purchase documents. If you want to generate unique batch numbers during batch management, you can do so by selecting the **Discrete Batch Number** checkbox. This checkbox cannot be used on its own. It can only used in combination with the **Batch Management** checkbox and the **Scope** if the segmentation strategy is 1.

More Information

Segmentation (LO-SGT)

Conversion of Material

Use

You can use this report to change the relevancy of a material for segmentation. You can also change the relevancy for order allocation run (ARun) and modify the season usage of a material.

You can make this change in the material master but it is a time-consuming process. Using this report, you can convert a material in a lesser span of time.

Features

You can convert a material using the following options:

- **Segmentation**: You can use this option to make a material relevant for segmentation (if the material was not relevant for segmentation when it was created) and vice versa. For example, if you make a material not relevant for segmentation, and

later on, want to make the same material relevant for segmentation, it can be using this report at both the client and plant levels.

- **Order Allocation Run**: You can use this option to make a material ARun-relevant (if the material was not ARun-relevant when it was created), and vice versa.
- **Season**: You can use this option to change season usage of a material. If you have already assigned a season usage for a material when it was created in material master and later on, you want to change the season usage, it can be done using this report.

  For example, you had created a material with the season usage as **Use Season Year/Season** and now want to change the season usage as **Season Year / Season / Collection / Theme**, you can do this using the **Season** checkbox on the report.

i Note

This report only allows you to modify the indicators. If you want to assign a segmentation strategy or a segmentation structure to a material, it can only be done in the material master assignment. Similarly, you can change the season and ARun information of a material in the material master only.

You can use this report only if no follow-on documents, for example sales orders, purchase orders, exist for the material. Also, the presence of a purchasing info record does not allow a change in the segmentation assignment of an article. To enable segmentation for articles with follow-on documents, see 2018157.

Activities

Follow these steps to change the relevancy of a material for segmentation:

1. In the SAP Easy Access menu, choose **Logistics** > **Materials Management** > **Purchasing** > **Master Data** > **Subsequent Settlement** > **Vendor Rebate Arrangements** > **Environment** > **Condition/Arrangement** > **Environment** > **Material** > **Segmentation** > **Segmentation, ARun and Season Relevancy Conversion in Material**.
2. Enter a material number.
3. Select the required indicator to change the relevancy of a material.

Mass Maintenance of Material Master Data

Use

Maintaining master data can be time-consuming if you have to make each change individually. It is quicker to make one change and have the system automatically update all the affected records automatically. The mass maintenance tool enables you to change a large group of master data records at once.

You can select the mass maintenance tool within a number of master data transactions. All these mass maintenance tools are based on the same basic framework and user interface.

Features

You can mass maintain data related to the following segmentation related enhancements made to material master:

- segmentation structure
- segmentation strategy

- fix segmentation
- consumption priority
- stock protection indicator
- discrete batch management

More Information

Segmentation: Enhancements to Material Master

Interfaces for Material Segmentation

Use

IDoc (for intermediate document) is a standard data structure for electronic data interchange (EDI) between application programs written for the SAP business system or between an SAP application and an external program.

The following sections contain information on the interfaces relevant for material segmentation.

More Information

For more information about the **IDoc Interface/Electronic Data Interchange**, see **SAP Library** for **SAP ERP Central Component** on **SAP Help Portal**.

Material Master

In SAP ERP Logistics, you can associate the material master with segmentation characteristics by performing the following in the material master:

- Mark the material as relevant for segmentation by selecting the **Segmentation Relevan**t checkbox.
- Assign segmentation structure and segmentation strategy for the materials at both the client data level and plant data level.

For a material that is relevant for segmentation, you must assign and fix the segmentation strategy. After fixing the segmentation strategy for a material, you cannot change the segmentation strategy assignment for the material. For a material that is relevant for segmentation, you can maintain segmentation-specific master data such as weights and volumes, sales block status, split valuation types and so on.

Technical Information: Material Master (MATMAS05)

The following section describes the enhanced standard IDoc segments of IDoc `MATMAS05`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment E1MARA1: Additional Fields for E1MARAM

Field	Short text	Explanation
SGT_CSGR	Segmentation Structure	Required only if the article is segmentation relevant. Contains all stock segment and requirement segment values valid for certain materials.
SGT_COVSA	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SGT_REL	Segmentation Relevant	Required only if the article is segmentation relevant. Indicates if the article is relevant for segmentation.

Segment E1MARCM: Master material C segment (MARC)

Field	Short text	Explanation
SGT_COVS	Segmentation Strategy	Required only if the article is segmentation relevant and the plant supports segmentation Strategy, else the value can be blank. Non Blank Segmentation Strategy should be same as that in client data.
SGT_STATC	Segmentation Status	Fix Status.
SGT_PRCM	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
SGT_CHINT	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
SGT_STK_PRT	Stock Protection Indicator	Indicates whether stock protection is enabled.
SGT_DEFSC	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Technical Information: Create and Change Material Master Data (MATMAS_BAPI02)

The following section describes the enhanced standard IDoc segments of IDoc `MATMAS_BAPI02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment E1BP_MARC1: Material Data at Plant Level

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Segment E1BP_MARCX: Checkbox Structure for BAPI_MARC

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.

Field	Short text	Explanation
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Technical Information: Create and Change Material Master Data (MATMAS_BAPI03)

The following section describes the enhanced standard IDoc segments of IDoc MATMAS_BAPI03. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment E1BP_MARA1: Material Data at Client Level

Field	Short text	Explanation
SEGMENTATION_STRUCTURE	Segmentation Structure	Required only if the article is segmentation relevant. Contains all stock segment and requirement segment values valid for certain materials.
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_RELEVANCE	Segmentation Relevant	Required only if the article is segmentation relevant. Indicates if the article is relevant for segmentation.

Segment E1BP_MARAX: Checkbox Structure for BAPI_MARA

Field	Short text	Explanation
SEGMENTATION_STRUCTURE	Segmentation Structure	Required only if the article is segmentation relevant. Contains all stock segment and requirement segment values valid for certain materials.
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.

Field	Short text	Explanation
SEGMENTATION_RELEVANCE	Segmentation Relevant	Required only if the article is segmentation relevant. Indicates if the article is relevant for segmentation.

Segment E1BP_MARC1: Material Data at Plant Level

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Segment E1BP_MARCX: Checkbox Structure for BAPI_MARC

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based

Field	Short text	Explanation
		on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Technical Information: BAPI for Mass Maintenance of Material Data

The following section describes the enhanced standard IDoc segments of the following IDocs:

- MATMAS_MASS_BAPI01
- MATMAS_MASS_BAPI02

The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment E1BPE1MARC1: Material Data at Plant Level

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for

Field	Short text	Explanation
		all segment value combinations

Segment E1BPE1MARCX: Checkbox Structure for BAPIE1MARC

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Technical Information: BAPI for Mass Maintenance of Material Data (MATMAS_MASS_BAPI03)

The following section describes the enhanced standard IDoc segments of IDoc `MATMAS_MASS_BAPI03`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment E1BPE1MARA1: Material Data at Client Level

Field	Short text	Explanation
SEGMENTATION_STRUCTURE	Segmentation Structure	Required only if the article is segmentation relevant. Contains all stock segment and requirement segment values valid for certain materials.
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock

Field	Short text	Explanation
		materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_RELEVANCE	Segmentation Relevant	Required only if the article is segmentation relevant. Indicates if the article is relevant for segmentation.

Segment E1BPE1MARAX: Checkbox Structure for BAPIE1MARA

Field	Short text	Explanation
SEGMENTATION_STRUCTURE	Segmentation Structure	Required only if the article is segmentation relevant. Contains all stock segment and requirement segment values valid for certain materials.
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_RELEVANCE	Segmentation Relevant	Required only if the article is segmentation relevant. Indicates if the article is relevant for segmentation.

Segment E1BPE1MARC1: Material Data at Plant Level

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.

Field	Short text	Explanation
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Segment E1BPE1MARCX: Checkbox Structure for BAPIE1MARC

Field	Short text	Explanation
SEGMENTATION_STRATEGY	Segmentation Strategy	Required only if the article is segmentation relevant. A strategy that is a collection of business rules that define which stock materials should satisfy which demand objects. This is done based on the mapping between stock segment associated with a material in stock and requirement segment associated with a demand object.
SEGMENTATION_STATUS	Segmentation Status	Fix Status.
CONSUMPTION_PRIORITY	Consumption Priority	Indicates whether planned independent requirements (PIRs) are consumed based on the segments or the consumption mode, and in which sequence.
DISCRETE_BATCH_FLAG	Discrete Batch Number	Decides whether batch numbering of materials with segmentation is active or not.
STOCK_PROTECTION_IND	Stock Protection Indicator	Indicates whether stock protection is enabled.
DEFAULT_STOCK_SEGMENT	Default Stock Segment value	Segment value, which is (from the material valuation point of view) representative for all segment value combinations

Materials Management

The following sections contain information on the area of Materials Management.

Purchasing Info Record

Use

Purchasing info records can be created and updated using IDoc interfaces. SAP ERP Logistics IDocs are enhanced primarily in the areas of processing material segmentation.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1EINAM`: Purchasing Info Record: General Data
  - `E1EALTH`: Text Identification for Purchasing Info Record: General Data
  - `E1EINEM`: Purchasing Info Record: Purchasing Organization Data
    - `E1EELTH`: Text ID for Purchasing Info Record: Purchasing Org. Data
    - `E1ESEGM`: Purchasing Info Record: Segmentation Specific Data

Technical Information: Purchasing Info Record (INFREC02)

The following section describes the material segmentation-specific segment of IDoc `INFREC02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Document Item: Purchasing Info Record: Segmentation Specific Data (E1ESEGM)

To enter material segmentation information for the materials when you create a purchasing info record, you can use the standard IDoc segment `E1ESEGM`.

Field	Short text	Explanation
STK_SEGMENT	Stock Segment	Segment assigned to a stockable item.

Technical Information: Purchasing Info Records (Mass IDOC) (INFRECMASS02)

The following section describes the material segmentation-specific segment of IDoc `INFRECMASS02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Document Item: Purchasing Info Record: Segmentation Specific Data (E1ESEGM)

To enter material segmentation information for the materials when you create a purchasing info record, you must use the standard IDoc segment `E1EINAM` to pass the purchasing info record details and use the IDoc segment `E1ESEGM` to pass segmentation-specific details.

Field	Short text	Explanation
STK_SEGMENT	Stock Segment	Segment assigned to a stockable item.

Purchase Requisition

Purchasing requisitions can be created and updated using IDoc interfaces. SAP ERP Logistics IDocs are enhanced primarily in the areas of processing material segmentation.

Technical Information: Create Enjoy Purchase Requisition (PREQCR102)

Use

The following section describes the material segmentation-specific segment of IDoc PREQCR102. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PREQCR1`: Header Segment
  - `E1BPMEREQHEADER`: Transfer Structure for Enjoy Purchase Req. - Header
  - `E1BPMEREQHEADERX`: Change Parameter for Enjoy Purchase Requisition - Header
  - `E1BPMEREQITEMX`: Change Parameter for Enjoy Purchase Requisition - Item Data
    - `E1BPMEREQITEM`: Transfer Structure for Enjoy Purchase Req. - Item Data
      - `E1BPMEREQITEM1`: Transfer Structure for Enjoy Purchase Req. - Item Data

Document Item: Transfer Structure for Enjoy Purchase Req. - Item Data (E1BPMEREQITEM1)

To enter material segmentation information for the materials when you create a purchase requisition, you can use the standard IDoc segment `E1BPMEREQITEM1`.

Field	Short text	Explanation
STK_SEGMENT	Stock Segment	Segment assigned to a stockable item.

Technical Information: Change Enjoy Purchase Requisition (PREQCHANGE02)

Use

The following section describes the material segmentation-specific segment of IDoc `PREQCHANGE02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PREQCR1`: Header Segment
  - `E1BPMEREQHEADER`: Transfer Structure for Enjoy Purchase Req. - Header
  - `E1BPMEREQHEADERX`: Change Parameter for Enjoy Purchase Requisition - Header
  - `E1BPMEREQITEMIMP`: Change Toolbar for Enjoy Purchase Req. - Item
  - `E1BPMEREQITEMX`: Change Parameter for Enjoy Purchase Requisition - Item Data
    - `E1BPMEREQITEM`: Transfer Structure for Enjoy Purchase Req. - Item Data
      - `E1BPMEREQITEM1`: Transfer Structure for Enjoy Purchase Req. - Item Data

Document Item: Transfer Structure for Enjoy Purchase Req. - Item Data (E1BPMEREQITEM1)

To enter material segmentation information for the materials when you change a purchase requisition, you can use the standard IDoc segment `E1BPMEREQITEM1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.
`REQ_SEGMENT`	Requirement Segment	Indicates additional information about the type of requirement.

Technical Information: Get Details for One MEREQ Object (PREQGETDETAIL02)

Use

The following section describes the material segmentation-specific segment of IDoc `PREQGETDETAIL02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PREQGETDETAIL`: Header Segment
  - `E1BPMEREQITEM`: Transfer Structure for Enjoy Purchase Req. - Item Data
    - `E1BPMEREQITEM1`: Transfer Structure for Enjoy Purchase Req. - Item Data
  - `E1BPMEREQACCOUNT`: Transfer Structure for Enjoy Purchase Req. - Acct Assignment

Document Item: Transfer Structure for Enjoy Purchase Req. - Item Data (E1BPMEREQITEM1)

To enter material segmentation information for the materials when you change a purchase requisition, you can use the standard IDoc segment `E1BPMEREQITEM1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.
`REQ_SEGMENT`	Requirement Segment	Indicates additional information about the type of requirement.

Purchase Order

Purchasing documents can be created and updated using IDoc interfaces. SAP ERP Logistics IDocs are enhanced primarily in the areas of processing material segmentation.

Technical Information: Purchase Order Item (PORDCR103)

Use

The following section describes the segmentation-specific segment of IDoc PORDCR103. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PORDCR1`: Header Segment
  - `E1BPMEPOHEADER`: Purchase Order Header Data
  - `E1BPMEPOHEADERX`: Purchase Order Header Data (Change Parameter)
  - `E1BPMEPOITEM`: Purchase Order Item

Document Item: Purchase Order Item (E1BPMEPOITEM1)

To enter segmentation information for the materials when you create a purchasing document, you can use the standard IDoc segment `E1BPMEPOITEM1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.

Technical Information: Change Purchase Order (PORDCH03)

Use

The following section describes the segmentation-specific segment of IDoc `PORDCH03`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PORDCR1`: Header Segment
  - `E1BPMEPOHEADER`: Purchase Order Header Data
  - `E1BPMEPOHEADERX`: Purchase Order Header Data (Change Parameter)
  - `E1BPMEPOITEM`: Purchase Order Item
    - `E1BPMEPOITEM1`: Purchase Order Item
  - `E1BPMEPOITEMX`: Purchase Order Item Data (Change Parameter)

Document Item: Purchase Order Item (E1BPMEPOITEM1)

To enter segmentation information for the materials when you create a purchasing document, you can use the standard IDoc segment `E1BPMEPOITEM1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.
`REQ_SEGMENT`	Requirement Segment	Indicates additional information about the type of requirement.

Technical Information: Detail Information for a Purchase Order (PORDGD102)

Use

The following section describes the segmentation-specific segment of IDoc `PORDGD102`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1PORDGD1`: Header Segment
  - `E1BPMEPOITEM`: Purchase Order Item
    - `E1BPMEPOITEM1`: Purchase Order Item

Document Item: Purchase Order Item (E1BPMEPOITEM1)

To enter segmentation information for the materials when you create a purchasing document, you can use the standard IDoc segment `E1BPMEPOITEM1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.
`REQ_SEGMENT`	Requirement Segment	Indicates additional information about the type of requirement.

Goods Movement

Use

The section describes the IDoc required for various movement types.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1MBGMCR`: Header Segment
  - `E1BP2017_GM_HEAD_01`: BAPI Communication Structure: Material Document Header Data
  - `/SPE/E1BP2017_GM_REF_EWM`: BAPI Communication Structure: Material Document EWM Ref.
  - `E1BP2017_GM_ITEM_CREATE`: BAPI Communication Structure: Create Material Document Item
    - `E1BP2017_GM_ITEM_CREATE1`: BAPI Communication Structure: Create Material Document Item

Technical Information: Post Goods Movements with MB_CREATE_GOODS_MOVEMENT (MBGMCR03)

The following section describes the enhanced standard IDoc segments of IDoc `MBGMCR03`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Document Item: Post Goods Movements with MB_CREATE_GOODS_MOVEMENT (E1BP2017_FSH_GM_ITEM_CREATE1)

To enter segmentation information for the materials in the goods movement document, you can use the standard IDoc segment `E1BP2017_GM_ITEM_CREATE1`.

Field	Short text	Explanation
`STK_SEGMENT`	Stock Segment	Segment assigned to a stockable item.

Purchasing and Sales

Use

Purchasing documents and sales documents can be created and updated using IDoc interfaces. SAP ERP Logistics IDocs are enhanced primarily in the areas of processing material segmentation.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1EDK01` IDoc: Document header general data
- `E1EDK14` IDoc: Document Header Organizational Data
- `E1EDK03` IDoc: Document header date segment
- `E1EDKA1` IDoc: Document Header Partner Information
- `E1EDP01` IDoc: Document Item General Data
    - `E1EDP20` IDoc schedule lines
    - `E1EDP19` IDoc: Document Item Object Identification

Technical Information: Purchasing/Sales (ORDERS05)

The following section describes the enhanced standard IDoc segments of IDoc `ORDERS05`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment: IDoc: Document Item General Data (E1EDP01)

To enter segmentation values for the materials in the sales document or the purchasing document, you can use the standard IDoc segment `E1EDP01`.

Field	Short text	Explanation
`SGT_RCAT`	Requirement Segment	Indicates additional information about the type of requirement.
`SGT_SCAT`	Stock Segment	Segment assigned to a stockable item.

Sales and Distribution

The following sections contain information on the area of Sales and Distribution.

Delivery Interface

Use

Outbound IDocs contain all relevant delivery information. Standard IDocs for outbound are enhanced primarily in the area of processing material segmentation data.

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- E1EDL20 Delivery header
- E1EDL21 Delivery Header Additional Data
- E1EDT13 IDoc: Deadline (delivery)
- E1EDL33 Export Data Delivery Header
- E1EDL24 Delivery Item
    - E1EDL26 Delivery Item Additional Data

Technical Information: Delivery Interface (DELVRY07)

The following section describes the enhanced standard IDoc segments of IDoc `DELVRY07`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment: Delivery Item Additional Data (E1EDL26)

To enter segmentation information for the materials in the delivery document, you can use the standard IDoc segment `E1EDL26`.

Field	Short text	Explanation
`SGT_RCAT`	Requirement Segment	Indicates additional information about the type of requirement.
`SGT_SCAT`	Stock Segment	Segment assigned to a stockable item.

Invoice or Billing Document

Use

Invoice or Billing Document

Prerequisites

You created the following standard IDoc segments before supplying data for the segments specific to material segmentation:

- `E1EDK01` IDoc: Document header general data
- `E1EDKA1` IDoc: Document Header Partner Information
- `E1EDK02` IDoc: Document header reference data
- `E1EDK03` IDoc: Document header date segment
- `E1EDK14` IDoc: Document Header Organizational Data
- `E1EDP01` IDoc: Document Item General Data

- `E1EDP03` IDoc: Document Item Date Segment
- `E1EDP19` IDoc: Document Item Object Identification
- `E1EDP26` IDoc: Document Item Amount Segment

Technical Information: Invoices (INVOIC02)

The following section describes the enhanced standard IDoc segments of IDoc `INVOIC02`. The fields contained in the segments are all listed with the field name, short text, and functional description.

Segment: IDoc: Document Item General Data (E1EDP01)

To enter segmentation information for the materials in the invoice document, you can use the standard IDoc segment `E1EDP01`.

Field	Short text	Explanation
`SGT_RCAT`	Requirement Segment	Indicates additional information about the type of requirement.
`SGT_SCAT`	Stock Segment	Segment assigned to a stockable item.

Planning and Production

The following sections contain information on Planning and Production (PP).

Master Data: Material BOM

For an integrated materials management and shop floor control, BOMs and work plans contain critical basic or master data. In the engineering/design department the functional and production conform design takes place for a new product to be produced. The results of this product phase are drawings and a list of all necessary parts, the BOM.

BOMs are found in varying forms everywhere where end products are composed of several basic parts or materials.

One of the steps you should carry out when you start up PP in the SAP system is the transfer of BOMs from a legacy system. This step is required because it is not usually possible to create initial data manually in the R/3 system due to the multitude of BOMs used.

Typical application examples are:

- Transfer of BOMs from a non-SAP system
- Transfer of data from data pools

Segmentation Special Features

You can use the new IDoc type `BOMMAT07` to create material BOMs with segmentation information.

- **Mapping of Segmentation**: Using ALE, segmentation information are mapped in the same way as in the dialog transaction.

Technical Information: Bill of Materials

Technical Description of the IDoc Type

Document Type	Direction	Logical Message	IDoc Type	Process Code
Bill of material	Inbound	BOMMAT	BOMMAT07	BOMM

Segmentation-Specific Fields for Bills of Materials

Segment	Field	Explanation	Origin
E1SGTBOM	MSGFN	Function	Same processing like in standard • 003-Delete: Message contains objects to be deleted • 004-Change: Message contains changes • 005-Replace: This message replaces previous messages • 009-Original: First message for process • 023-Wait/Adjust: Do not import data • 018-Resend
	STOCK_SEGMENT	Stock Segment	Segmentation value for requirement segment or stock segment which contains one value of a segmentation field or a combination of values for all segmentation fields.
	SEGMENTATION_INCLUDE	Segmentation : Flag Including	Includes or excludes components or routings for selected segment values of the header material.
	SEGMENTATION_EXCLUDE	Segmentation: Flag Excluding	Includes or excludes components or routings for the selected segment values of the header material.

Master Data: Sales Order BOM

For an integrated materials management and shop floor control, BOMs and work plans contain critical basic or master data. In the engineering/design department the functional and production conform design takes place for a new product to be produced. The

results of this product phase are drawings and a list of all necessary parts, the BOM. BOMs are found in varying forms everywhere where end products are composed of several basic parts or materials.

One of the steps you should carry out when you start up PP in the SAP system is the transfer of BOMs from a legacy system. This step is required because it is not usually possible to create initial data manually in the R/3 system due to the multitude of BOMs used.

Typical application examples are:

- Transfer of BOMs from a non-SAP system
- Transfer of data from data pools

Segmentation Special Features

You can use the new IDoc type `BOMORD03` to create order BOMs with segmentation information.

- **Mapping of Segmentation**: Using ALE, segmentation information are mapped in the same way as in the dialog transaction.

Technical Information: Bill of Materials

Technical Description of the IDoc Type

Document Type	Direction	Logical Message	IDoc Type	Process Code
Bill of material	Inbound	`BOMORD`	`BOMORD03`	BOMM

Segmentation-Specific Fields for Bills of Materials

Segment	Field	Explanation	Origin
`E1SGTBOM`	`MSGFN`	Function	Same processing like in standard • 003-Delete: Message contains objects to be deleted • 004-Change: Message contains changes • 005-Replace: This message replaces previous messages • 009-Original: First message for process • 023-Wait/Adjust: Do not import data • 018-Resend
	`STOCK_SEGMENT`	Stock Segment	Segmentation value for requirement segment or stock segment which contains one

Segment	Field	Explanation	Origin
			value of a segmentation field or a combination of values for all segmentation fields.
	SEGMENTATION_INCLUDE	Segmentation : Flag Including	Includes or excludes components or routings for selected segment values of the header material.
	SEGMENTATION_EXCLUDE	Segmentation: Flag Excluding	Includes or excludes components or routings for the selected segment values of the header material..

Materials Management

Materials Management is a comprehensive solution for mapping logistical processes. The tools in **Materials Management** such as material requirements planning (**MRP**), **Purchasing**, **Goods Receipt**, **Inventory Management**, and **Invoice Verification** supports you during the entire procurement process and in inventory management. If you activate the business function **Material Segmentation Across Logistics**, you can procure and manage materials at the level of segmentation during the entire procurement process. Reports and screens have been adjusted and master data has been enhanced so that you can manage materials segmentation (such as various qualities and country of origin).

Segmentation: Purchasing Documents

Use

Segmentation allows you to logically segment materials based on certain attributes, for example, **Quality** or **Customer segment**. When you use materials with segmentation, you can also process quantities and conditions in purchasing documents for individual stock segments.

For more information about processing of purchasing documents in the standard ERP system, see Purchasing (MM-PUR).

Prerequisites

- You have activated the business function **Material Segmentation Across Logistics** (`LOG_SEGMENTATION`)
- You have created the characteristics for segmentation using transaction `CT04`
- You have defined segmentation structures in Customizing for **Logistics – General** > **Segmentation** > **Define Segmentation Structures**
- You have defined segmentation strategies in Customizing for **Logistics – General** > **Segmentation** > **Define Segmentation Strategies**
- You have created materials and assigned a segmentation strategy in the material master (transaction `MM01`)

Process

To procure materials with segmentation, the system allows you to maintain data in purchasing documents for the corresponding stock segment. You can manually enter the stock segment or the system copies the value from various sources such as other

purchasing documents, or determine during requirement planning (MRP).

The system does not allow you to enter or maintain stock segment for a material if one of the following condition is true:

- Material is not relevant for segmentation
- Plant to which the material belongs is not relevant for segmentation
- Material is relevant for segmentation but you did not assign or fix the segmentation strategy and the segmentation structure
- Scope is **Only Requirement Segments Maintained** in the segmentation strategy

For more information about how segmentation is handled in individual purchasing documents, see the following:

Purchasing Info Record

Purchase Requisition

Contract

Purchase Order

Goods Issue

Goods Receipt

Inbound Delivery

Invoice Verification

Physical Inventory

Reservations

More Information

Segmentation Structure

Segmentation Strategy

Purchase Requisition

Definition

A purchase requisition is a request or instruction to **Purchasing** to procure a certain quantity of a material or a service so that it is available at a certain point in time.

Use

The system allows you to maintain instructions to procure a certain quantity of different stock segments of a material.

When you manually create a purchase requisition, the system allows you to enter a valid stock segment for a material. During MRP if the required quantity of stock is not available, the system proposes a purchase requisition and determines stock segment for the

material. The system determines the stock segment based on the segmentation strategy you assigned for the material of the plant.

When you create purchase orders with reference to a purchase requisition, the system allows you to change the stock segment and ordered quantities for a material that are copied from the reference document.

The system determines the requirement segment for a stock transport requisition based on the stock segment and the segmentation strategy you assigned for the material in the material master.

Structure

On the **Item Overview** screen, you can specify the following information for different stock segments of a material:

- Purchase requisition quantities
- Delivery date

The system also displays the schedule line number and ordered quantities.

More Information

Segmentation: Purchasing Documents

Purchasing Info Record

Definition

Stores master data such as vendor and material, for a purchasing organization or a plant

Use

The system allows you to maintain data for individual stock segment for a combination of vendor and material.

When you create a purchase order with reference to documents, for example purchasing info record, the system copies the segment values from the reference document.

You can maintain purchase info record (under delivery and over delivery tolerance, minimum and maximum purchase order quantities) for individual segments of a material in the material master. Assume that you did not maintain these values for a material in the material master and try creating a purchase order for the same material. In such case, the system copies the purchase info record details for the individual segment of a material from the purchasing info record document that you already created for the material.

Structure

You can maintain the following information for different stock segments of a material while creating the purchasing info record:

- Under delivery tolerance
- Over deliver tolerance
- Minimum purchase order
- Maximum purchase order quantities

- Rounding profile for each segment

Contracts

Definition

A contract is a type of outline purchase agreement against which release orders (releases) can be issued for agreed materials or services as and when required during a certain overall time-frame. You can create quality based contracts (MK) or value based contracts (WK).

Use

The system allows you to create and process contracts for different stock segments of a material.

Structure

The contract consists of items defining the individual materials, material groups with prices and target quantities for different stock segments of a material. For example, you can create contract with vendor-specific quantities or order value of the same material for different stock segment.

If you maintain values for different stock segments in the material master under **Additional Data**, the system displays the valid weights and volumes in the detail view for the different stock segments of the material.

More Information

Weights and Volumes in Purchasing Documents

Purchase Orders

Definition

Purchase order is a formal request or instruction from a purchasing organization to a vendor or a plant to supply or provide a certain quantity of goods or services at or by a certain point in time.

Use

Segmentation at item level allows you to order materials with different attributes, for example, quality and/or customer channel. You can maintain information such as delivery tolerances, purchase quantities, and so on, for different segmentation value of a material. The system copies such information from a reference document, such as purchasing info record, purchase requisition, and contracts to a purchase order. The system determines the valid stock segment for a material based on the segmentation structure and the segmentation strategy you assigned for the material.

For a material in purchase order and stock transport order, the system allows you to maintain only stock segment and determines the requirement segment based on the segmentation strategy.

Structure

The system allows you to maintain stock segments for a material in the **Item Detail** screen only if you maintain the plant is relevant for segmentation. You can also maintain purchase order quantities, delivery date, and so on, for the different stock segments of a material

ℹ Note

For transfer of stock between plants, the system determines the requirement segment of the issuing plant using the segmentation strategy and the stock segment of the receiving plant. For more information about stock transfer between plants, see Stock Tansport Order for Materials with Segmentation

Purchase Order Confirmations

With SAP ERP 6 Enhancement Package 7 Support Package Stack 10 release onwards, you can process and view order confirmations for the individual segmentation values in the **Confirmation** tab of purchase order for materials that are relevant for segmentation. When you create an order confirmation manually, you enter the quantities and the corresponding segmentation values and save the document. In addition, when you create an inbound delivery with reference to a purchase order, you can view the inbound delivery item details such as stock segment and quantity in the **Confirmation** tab.

Purchase Order History

In the standard ERP, on the purchase order history tab, the system displays the history of goods receipts and invoices relating to the purchase order. In addition, with SAP ERP 6 Enhancement Package 7 Support Package Stack 10 release onwards, you also can view the segmentation details for segmentation relevant materials in the history of goods receipts and invoices in this **History** tab.

More Information

Weights and Volumes in Purchasing Documents

Changing Stock Segment Using Fast Change

Prerequisites

You have created materials and assigned the segmentation strategy in the material master.

Context

Fast change enables you to change several items of a document simultaneously. In addition, you can change the stock segment for several items of a purchase order using the fast change option. For example, you notice that you must procure only materials of segment X and not of segment Y. You can use fast change to change segment X to segment Y. You can change the stock segment for all the selected items or for all the items in purchase order.

For more information about fast change in the standard SAP ERP system, see Purchase Order – Single-Screen Transaction (ME21N, ME22N, and ME.

Procedure

1. Create or open an existing purchase order.
2. Enter a material and all the necessary details including the relevant stock segment.
3. Choose one or more line items for which you want to change stock segment.
4. In the **Change Layout** dialog box, choose **Stock Segment** from the list of **Hidden fields**. If you have already selected the **Stock Segment** column, this dialog box does not appear.

5. In the **Fast Change** dialog box, choose a different stock segment.
6. Choose the icon for **Fast Change**.

Results

Stock segment is changed for all the selected line items or for all the items in the document based on the selection in step 3. If any of the material is not relevant for segmentation, stock segment is not assigned to the material as a result of fast change execution.

Example

You have the following items in purchase order:

- `Shirt_T` with stock segment X
- `Shirt_T1` with stock segment Y
- `Shirt_T2` with no stock segment

For all the three items you want to change stock segment to Z. The result of fast change is as follows:

- `Shirt_T` with stock segment Z
- `Shirt_T1` with stock segment Z
- `Shirt_T2` with no stock segment

In addition, the system validates the newly assigned stock segment. The system displays a message if the stock segment assigned for `Shirt_T1` as a result of fast change is not valid. You must choose only that line item and choose a valid stock segment for that material.

Return Purchase Order

Use

In a business scenario, the company will have to send the goods back to the vendor if the stock is of poor quality or if there is excess stock. The return to stock is done with the help of a document called **Return Purchase Order**.

Process

The goods to be returned are marked as `Return Items` in the purchase order.

If a plant that is relevant for segmentation returns the stock to vendor, the system determines the stock segment for the supplying plant based on the settings you maintained in Customizing for **Purchase Orders** under **Logistics - General** > **Segmentation** > **Determine Segment Value for Automatic Determination**. The requirement segment is not applicable.

If a plant that is not relevant for segmentation returns the stock to vendor, segmentation is not applicable and the system does not determine the stock segment or the requirement segment.

Goods Issue

Definition

The system allows you to post a material withdrawal or a material issue. A goods issue leads to a reduction in warehouse stock.

Structure

When you create goods issue for a material, you can specify quantities that you like to withdraw for each stock segment. Since negative stock is not permitted, you cannot enter a goods issue for stock segments that exceed the available stock. If the material is managed in batches, the system assigns a batch number to the material when you post a goods receipt. The batch number the system determines is based on the batch indicator you set in the material master. For more information about batch management, see Batch Management for Material Segmentation.

Goods Receipt

Definition

With the goods receipt (GR) you post the physical inward movement of goods from an external vendor or from production and then complete a goods movement, which leads to an increase in the warehouse stock.

Structure

The system allows you to process goods receipt for different segments of the material. You can maintain the quantity delivered for each stock segment on the **Item Detail** screen. In addition, you can maintain the stock segment for materials both in the **Item Overview** screen and in the **Batch** tab of the **Item Detail** screen. Before posting the goods, you can change stock segment for a material only by using the transaction `MIGO`.

If the receiving plant is not relevant for segmentation, then stock segment for a material is invisible in the following scenarios:

- For goods movement with reference to documents, for example `Purchase Order`
- For transfer posting
- For goods movement with reference to **others**

You must use batch management for materials with stock segment. The system assigns a batch number to the material when you post a goods receipt. The batch number the system determines is based on the batch indicators you set in the material master. For more information about batch management, see Batch Management for Material Segmentation.

Inbound Delivery

Definition

Inbound delivery is enhanced to handle materials with segmentation. The system copies the stock segment from the purchase order.

You must use batch management for materials with stock segment. Batch determination happens in inbound delivery and the batch number is proposed. After you post the goods receipt, the system creates a new batch or post the quantities to an existing batch. For more information about batch management, see Batch Management for Material Segmentation.

More Information

Weights and Volumes in Purchasing Documents

Batch Determination for Inbound Deliveries

Use

You can enable the system to check if the batch entered in the inbound delivery document already exists in the system. To do this, you can use the **No Batch Check** checkbox in the Customizing activity **Define Item Categories for Deliveries** under **Logistics Execution** > **Shipping** > **Deliveries**. A batch split describes the situation where more than one batch is used to cover a particular delivery item. The batch split function in the delivery includes a batch selection screen and an availability check.

Features

During the batch determination process, for a material that is relevant for segmentation the system proposes the new batch number according to the segmentation value entered in the document.

- If you select this checkbox, the system does not update the new batch number in the document and does not perform a batch split, although the batch is relevant for segmentation. However, if the system determines an existing batch number, the system updates the batch number in the document and performs a batch split.
- If you do not select this checkbox, the system determines a new batch number or an existing batch number and updates the determined batch number in the document and performs a batch split.

Invoice Verification

You can use the logistics invoice verification to check incoming invoices for their functional, price, and mathematical correctness. You can run this verification for materials with segmentation at the segment level. The segment values are copied from purchase order and you cannot change them.

Physical Inventory

The system allows you to carry out a physical inventory of your company's warehouse stocks for balance sheet purposes. For materials with segmentation, you can maintain the inventory for different stock segments. Stock segment is attached to a batch and thus batch management is mandatory in **Physical Inventory**.

When you enter the materials for which you want to carry out physical inventory, the system determines the stock segment based on the batch number.

You can filter the batches based on the stock segments.

Reservations

Definition

A request to the warehouse or plant to keep a material ready for issue at a future date for a certain purpose

Use

Reservations are enhanced to handle materials with segmentation. If the material is relevant for batch management, the system determines the stock segment based on the material, batch, and plant. You cannot change the stock segment. For more information, see Batch Management for Material Segmentation

Quotation

Definition

A quotation is an offer by a vendor to a purchasing organization regarding the supply of materials or performance of services subject to specified conditions.

> **Note**
> The system only validates the stock segment you enter for a material in Quotation when you enter a plant and a material that is relevant for segmentation. If you do not enter a plant but enter an invalid stock segment for a material that is relevant for segmentation, the system allows you to save the document.

Structure

A quotation is legally binding on the vendor for a certain period. The quotation is the vendor's response to a request for quotation issued by a purchasing organization.

A quotation consists of items in which the total quantity and delivery date of an offered material or service are specified. For materials relevant for segmentation, the system allows you to maintain quantity and delivery data for the different stock segments of the material.

Request for Quotation

Definition

A request for quotation (RFQ) is an invitation extended to a vendor by a purchasing organization to submit a quotation (bid) for the supply of materials or performance of services. The system allows you to create request for quotation to supply different stock segments of a material.

> **Note**
> The system only validates the stock segment you enter for a material in RFQ when you enter a plant and a material that is relevant for segmentation. If you do not enter a plant but enter an invalid stock segment for a material that is relevant for segmentation, the system allows you to save the document.

Structure

An RFQ consists of the RFQ header and the items.

- RFQ header

  Contains general information on the RFQ, such as the vendor's address.

- Items

Contain the total quantities and delivery dates for the materials or services specified in the RFQ. You can maintain quantities and delivery dates for individual stock segments of a material.

An item of a quotation may contain a delivery schedule made up of a number of schedule lines in which the total quantity is broken down into smaller quantities to be delivered on the specified dates over a certain period.

Scheduling agreement

Definition

A form of outline purchase agreement under which materials are procured on predetermined dates within a certain time period.

Structure

A scheduling agreement consists of a number of items, for each of which a procurement type is defined. The following procurement types exist:

- Standard
- Subcontracting
- Consignment
- Stock transfer

Delivery of the total quantity of material specified in a scheduling agreement item is spread over a certain period in a delivery schedule, consisting of lines indicating the individual quantities with their corresponding planned delivery dates. For materials relevant for segmentation, you can maintain the quantities and the planned delivery dates for individual stock segments of the material.

When you maintain delivery schedule for the scheduling agreement of materials relevant for segmentation, the segmentation details are adopted from the scheduling agreement and you cannot edit the stock segment.

Stock Transport Order for Materials with Segmentation

Use

You can carry out stock transfers between storage locations within a plant, between different plants, and between company codes. You cannot transfer materials with segmentation between plants with different segmentation strategies. You can only transfer materials with segmentation between plants with the same segmentation strategies. You can also transfer materials with segmentation between plants if one of the plants is not relevant for segmentation. If both the receiving plant and the supplying plant is assigned with the same segmentation strategy, you must define the mapping between the stock segment and the requirement segment of the plants involved.

Features

- If both the receiving plant and the supplying plant are relevant for segmentation, the stock segment in the stock transport order is mapped to the corresponding requirement segment of the supplying plant. This mapping is based on the Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**.

- If the supplying plant is relevant for segmentation and the receiving plant is not relevant for segmentation, the stock transport order does not have a stock segment. The system determines the requirement segment for the supplying plant in one of the following ways:
    - Priority 1: Based on the segment value assignment you define in the Customizing for **Define Segment Value for Automatic Determination** under **Logistics – General > Segmentation**.
    - Priority 2: If you do not define the segment value assignment in the Customizing activity **Define Segment Value for Automatic Determination**, the system uses the default stock segment maintained in the Customizing for **Define Segmentation Structures** under **Logistics – General > Segmentation**.
    - Priority 3: If the above options fail, the system determines the requirement based on the logic you define in the BAdI `BADI_SEG_SPEC_PROD_MM` (under **Logistics - General > Segmentation > Enhancements for Segmentation > BAdI: Segment Values in Purchase Order**.
- If the supplying plant is not relevant for segmentation and the receiving plant is relevant for segmentation, the stock transport order has the stock segment value and the material is posted to the supplying plant without any segmentation value.

Example: Stock Transport Order

You have specified a material in the following plants:

Plant	Segmentation Enabled	Company Code
DC1	Yes	001
DC2	No	001
DC3	Yes	002

Scope is maintained as Both Requirement and Stock Segment Maintained in the segmentation strategy

Scenario 1: DC3 supplies to DC1. Both the supplying plant and the receiving plant are relevant for segmentation. You defined the stock segment in the stock transport order. The system determines the requirement segment in one of the following ways:

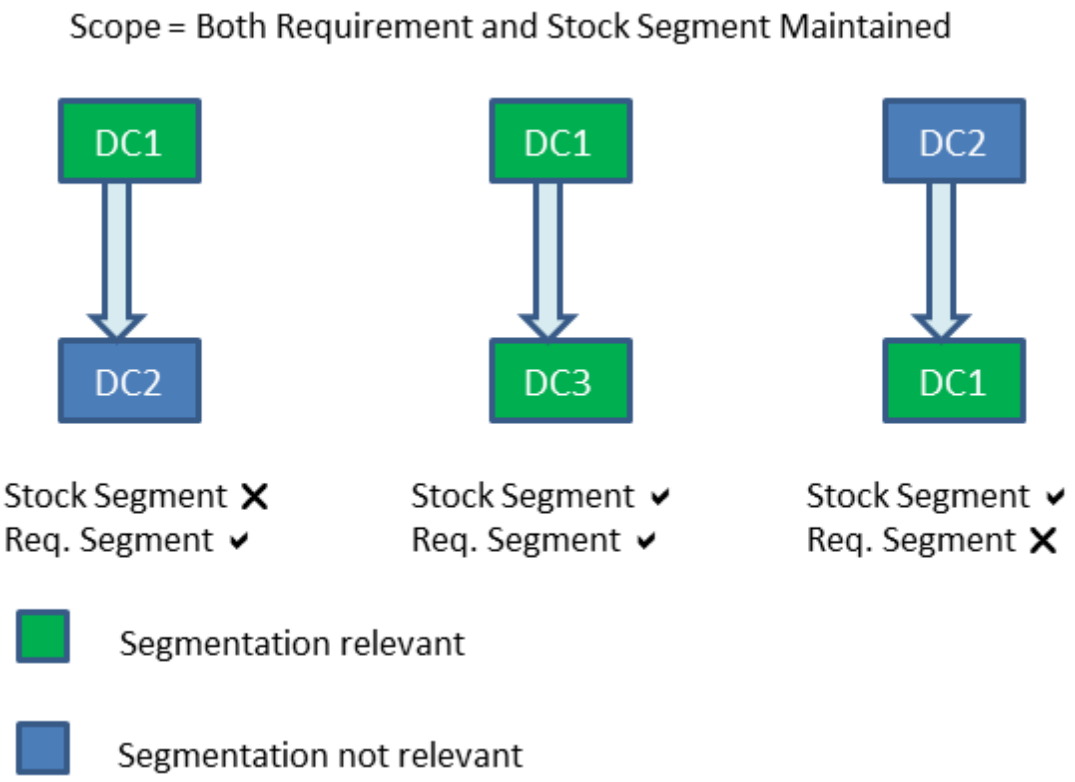
- Priority 1: Based on the segment value assignment you define in Customizing for **Stock Transfer** under **Logistics – General > Segmentation > Define Segmentation Strategies**.
- Priority 2: If you do not define the segment value assignment in Customizing activity **Define Segmentation Strategies**, the system determines the requirement segment based on the logic you define in the BAdI `BADI_SEG_SPEC_PROD_MM` (under **Logistics - General > Segmentation > Enhancements for Segmentation**).
- Priority 3: If you do not specify the logic in the BAdI, the system determines the requirement segment based on the default stock segment you define in Customizing for **Define Segmentation Structures** under **Logistics – General > Segmentation**.
- Priority 4: If BLANK is not a valid segmentation value and no segmentation value is found, the system displays an error message.

Scenario 2: DC3 supplies to DC2. The receiving plant is not relevant for segmentation. You did not define the stock segment in the stock transport order. The system determines the requirement segment in one of the following ways:

- Priority 1: Based on the segment value assignment maintained in Customizing for Define Segment Value for Automatic Determination under Logistics – General > Segmentation >.
- Priority 2: If you do not specify the segment value assignment in Customizing activity Define Segment Value for Automatic Determination, the system determines the requirement segment based on the default stock segment you define in Customizing for Define Segmentation Structures under Logistics – General > Segmentation >.
- Priority 3: If BLANK is not a valid segmentation value and no segmentation value is found, the system displays an error message.

Scenario 3: DC2 supplies to DC1. The supplying plant is not relevant for segmentation. You define the stock segment in the stock transport order. As the supplying plant is not relevant for segmentation, the system does not determine the requirement segment and the material is posted.

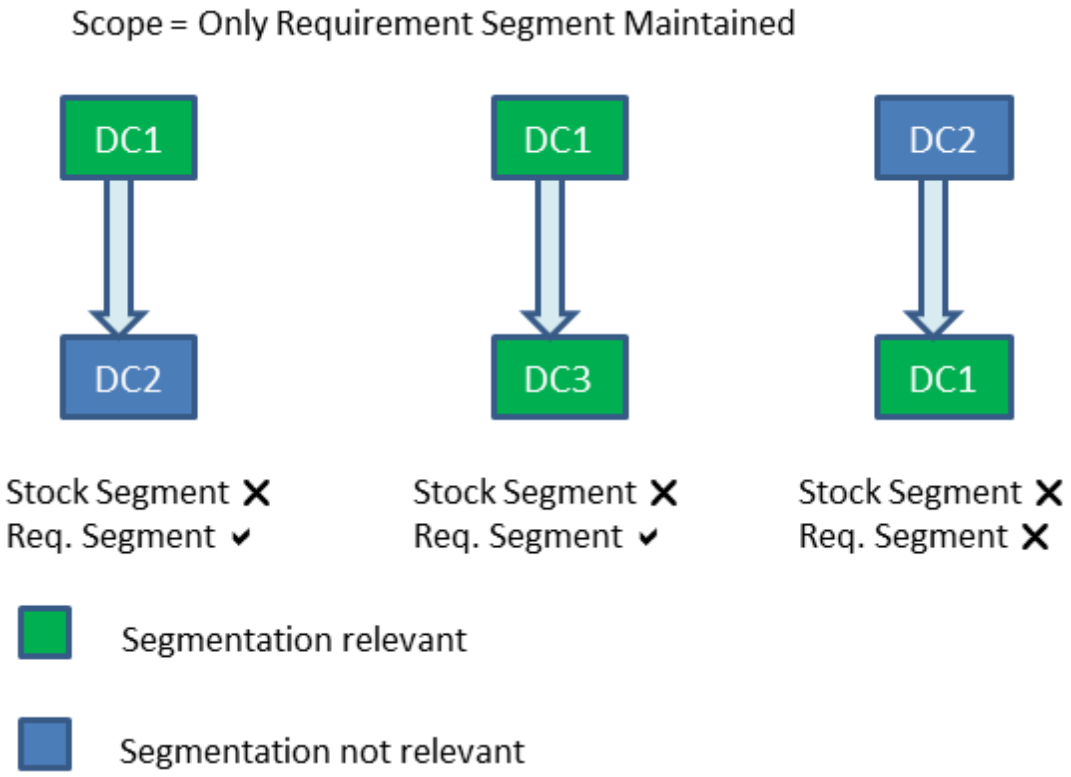
The following image illustrates the transfer of stock between plants with different segmentation strategies:



Scope is maintained as Only Requirement Segment is Maintained in the segmentation strategy

Scenario 4: DC3 supplies to DC1. Both the supplying plant and the receiving plant are relevant for segmentation. However, as the scope is Only Requirement Segment is Maintained in the segmentation strategy, you do not define the stock segment. The system determines the requirement segment based on the segment value assignment you define in Customizing for Define Segment Value for Automatic Determination under Logistics – General > Segmentation >.

The following image illustrates the transfer of stock between plants with different segmentation strategies:



Mass Maintenance of Materials with Segmentation

Use

You can use mass maintenance for changing large volumes of data in a number of different purchasing documents, in a single step. You can change header data, and item data such as the plant or the terms of payment.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION`
- You have created materials and assigned a segmentation strategy to the materials in the material master

Features

If you have activated the business function `LOG_SEGMENTATION` and have assigned a segmentation strategy to your materials, you can use mass maintenance function to change the stock segment in the following documents in a single step:

- Purchase order
- Stock transport order
- Contracts
- Purchase requisitions
- Stock transport requisitions
- Purchasing info record

The system validates the segmentation value when you save the document, based on the segmentation strategy assigned to the material.

Weights and Volumes in Purchasing Documents

Use

Weights and volumes is an additional data you can maintain for different stock segments of a material.

Features

You can view the valid weights and volumes for different stock segments in the detail view of the material. The system displays the weights and volumes based on the values maintained in the material master under additional data.

When you create a delivery, the weights and volumes for the material in the purchase document are transferred accordingly to the batch split item of the delivery document. If you do not maintain the weights and volumes in the material master, the system determines the weights and volumes for the material.

The system displays the weights and volumes for materials in the following purchasing documents:

- Purchase Order
- Contracts
- Inbound Delivery

Batch Management for Material Segmentation

Use

You can define batch management depending on plants and materials in material master. Batch management is supported for the plant and the batch number is determined when you perform a goods issue, outbound delivery, and transfer order. Batch management is mandatory, if the material is relevant for segmentation and the **Scope** is **Only Requirement Segments Maintained** in segmentation strategy.

Features

You have the following options for the internal batch management:

- Standard Batch Numbering
- Discrete Batch Numbering

The batch number is determined when posting the goods receipt, dependent on which batch indicator is set in material master.

For materials with segmentation, the batch management is at plant level only. Thus the batch number is

- unique in connection with a plant and a material
- only known in the plant in which it was created
- not known in other plants in which the material also exists

During a stock transfer to another plant, the batch is not known in the receiving plant and the system does not transfer any data. The system determines the batch in the receiving plant according to the batch management selected there.

Standard Batch Numbering

Use

The standard internal batch numbering procedure for materials with segmentation, assigns one batch number for each combination of material, plant, and stock segment during goods movement. The system will use the same batch number for all goods movements of the same combination of material, plant, and stock segment.

Example

Assume a material with different stock segments X, Y, and Z. At the goods receipt of the corresponding stock segments, the batch numbers are assigned as follows: 001 for X, 002 for Y and 003 for Z. If the stock segment Z already exists in the system with the batch number 023, it is posted to the already existing batch 023, and not assigned to the new batch 003.

Discrete Batch Numbering

Use

The discrete batch numbering enables you to trace individual batches in the system, since a separate batch number is issued for each received quantity of a material with stock segmentation. Discrete batch numbering is available only for the materials with segmentation and if the materials are managed in batches. The system assigns an internal batch number to each item of a goods receipt that is posted. Each received quantity of a material with segmentation will then have an individual batch number, without reusing batch numbers for already existing stock segment.

Prerequisites

If you want to activate discrete batch numbering, choose the **Disc. batch num** option in material master.

Example

Assume a material with different stock segments X, Y, and Z. At the goods receipt of the corresponding stock segments, the batch numbers are assigned as follows: 001 for X, 002 for Y and 003 for Z.

Split Valuation of Materials in Purchasing Documents

Use

You can use materials that you enable for split valuation in as many different partial stocks as you want. The system uses the split valuation type to apply a split valuation to a partial stock. The partial stocks refer to the batches or stock segments of materials with segmentation. The system updates the total stock, in addition to the partial stock. The value calculation of the total stock results from the total of inventory value and inventory quantity of the individual pieces of partial stock (moving average price).

Integration

The system determines the valuation type for different stock segments of the material in the following purchasing and inventory documents:

- Purchase order

- Goods issue
- Goods receipt
- Inbound delivery
- Invoice verification
- Batch management

Prerequisites

- You have assigned and fixed a segmentation strategy for a material in the material master
- You have assigned a valid split valuation type by choosing valuation type X for **Automat. (Batch)** in **Logistics DC: 1 > Accounting** view of the material in the material master
- You have created characteristics that are relevant for segmentation (Material Valuation (VAL) using the transaction `CT04`) and used these characteristics as segment values

Features

You create batches for different stock segments of a material using the transaction `MSC1N`. When you create batches in this way, the system uses the stock segment to determine the valuation type. After creating a batch, you cannot change the valuation type in the transaction `MSC2N`. If you create a batch (when you perform the first goods receipt), the system updates the valuation type to the batch master record of the relevant stock segment and defines the valuation of the batch.

In these purchasing and inventory documents, the system uses the stock segment you enter to determines the valuation type. To reapply a valuation type for the batches, you must run the report `SGT_MSC2`.

If the material is relevant for segmentation and the accounting view is not generated, the system does not determine the valuation type.

Viewing Stock Overview of Materials with Segmentation

Use

You can display the stock overview of a material with segmentation in a tree structure. The system displays the nodes in the tree structure in the following sequence:

Plant > Storage Location > Stock Segment > Batch

The system displays the stock quantity for individual batch number.

Procedure

1. Goto transaction `RWBE`.
2. In the **Stock Overview** screen, enter the material.
3. Choose the **List Display** tab.
4. In the **List as** section, choose **Tree Structure**.

   In the **Display Levels** section, the system selects **Base Plant Group**, **Plant**, and **Storage Location** by default.

5. To view the stock of a material at the stock segment level, you must select the **Batch** option.

   You can only view the stock overview for the storage location of a material, If you do not select the **Batch** option. If you do not select **Storage Location** and you only select the **Batch** option, the **Storage Location** option and the plant details are automatically selected.

6. Enter the stock segment to filter the stock overview for the article at the stock segment level.

For materials with segmentation, you must use batch management. When you select the **Batch** option, the system automatically selects the **Storage Location** option. The system also displays the stock segment details for the material.

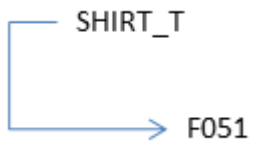
Example

The following table describes the system settings for the material `SHIRT_T`, of type **Single Article**:

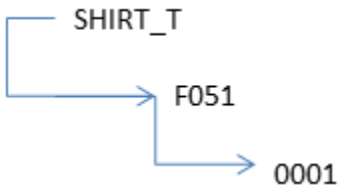
Plant	Storage location	Stock segment	Batch number
F051	0001	LOWRET	00000176

The system can display stock of `SHIRT_T` in the tree view as follows:

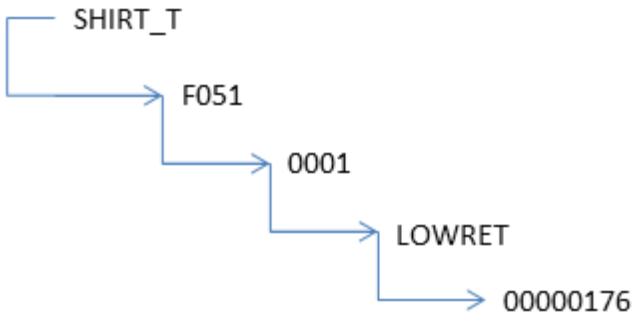
- By default:



- By storage:



- By batch, if the material is managed in batches:



Reporting in Purchasing

Use

Reports are used to access information from the database, and to list the results based on the selection criteria.

Features

If you have activated the business function `LOG_SEGMENTATION`, the following reports display the segmentation values in the result screen:

Purchase orders

Transaction	Description
ME2L	Purchasing Documents per Vendor
ME2M	Purchasing Documents for Material
ME2K	Purchasing Documents per Account Assignment
ME2J	Purchasing Documents per Projects
ME2C	Purchasing Documents for Material Group
ME2B	Purchasing Documents per Requirement Tracking Number
ME2N	Purchasing Documents per Document Number
ME2W	Purchasing Documents per Supplying Plant

Purchase requisitions

Transactions	Description
ME5A	Purchase Requisitions: List Display
ME5K	Requisitions by Account Assignment
ME5J	Purchase Requisitions for Projects
ME5R	Archived Purchase Requisitions

Outline Agreement

Transaction	Description
ME3L	Outline Agreements per Vendor
ME3M	Outline Agreements by Material
ME3K	Outl. Agreements by Acct. Assignment
ME3J	Outline Agreements per Project
ME3C	Outline Agreements by Material Group
ME38	Maintain Sched. Agreement Schedule
ME3N	Outline Agreements by Agreement No.

RFQ/Quotation

Transaction	Description
ME4L	RFQs by Vendor
ME4M	RFQs by Material
ME4S	RFQs by Collective Number
ME4C	RFQs by Material Group
ME4B	RFQs by Requirement Tracking Number
ME4N	RFQs by RFQ Number

More Information

Reporting in Purchasing

Segmentation based EAN in Purchasing Documents

Use

For materials that are relevant for segmentation, the system uses the stock segment to determine the EAN value in the purchasing documents (such as goods receipt, stock transport order, purchase order).

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_03`.
- You have created characteristics that are relevant for EAN using transaction `CT04`.
- You have created the segmentation strategy and the segmentation structure using the characteristics that are relevant for EAN.
- You have created a material and maintained EANs for different segmentation values of Unit of Measure (UoM). You can maintain EAN in the Additional EAN tab of the Additional Data screen in material master.
- You have maintained the Vendor Main EAN for the material.

Features

Purchase Orders

When creating or changing purchase orders, the system determines the Vendor Main EAN according to the stock segment and the Unit of Measure (UoM) of the purchase order item. If you change one of the following values for a purchase order item in the document, the system redetermines the EAN number accordingly:

- Unit of Measure
- Stock segment
- Vendor
- Plant

ℹ Note

You can change the EAN value manually for a purchase order item. The system does not validate the new EAN value along with the segmentation value and allows you to save the document.

Stock Transport Orders

In case of a stock transport order, the system uses the requirement segment to determine the EAN value for the item. The system always determines the main EAN value if you have maintained multiple EAN numbers. When you create an outbound delivery with reference to a stock transport order, the system does not copy the EAN details of the material from the reference document. Instead, the system determines the EAN value according to the stock segment value. If you enter a stock segment for a batch item or change the sales UoM for a batch split item in the batch split screen, the system redetermines the EAN value according to the stock segment. If you change the sales UoM for the generic material in the item overview screen, the system redetermines the EAN value for the material.

ℹ Note

You can change the EAN value manually for a stock transport order item. The system does not validate the new EAN value along with the segmentation value and allows you to save the document.

Inbound Delivery

When you create an inbound delivery with reference to a purchase order, the system copies the EAN details of the material from the reference document. If you enter or change a stock segment for a batch item or change the sales UoM for a batch split item in the batch split screen, the system redetermines the EAN value according to the stock segment. If you change the sales UoM for the material in the item overview screen, the system redetermines the EAN value for the material. You can change the EAN value manually for an item in the inbound delivery. The system does not validate EAN and allows you to save the document.

Goods Receipt

When you create a goods receipt with reference using transaction `MIGO`, the system copies the EAN number from the reference document. You can change the EAN value manually and the system does not validate the new EAN value along with the segmentation value and allows you to save the material document. If you change the stock segmentation value, the system redetermines the EAN number.

When you create a goods receipt using the BAPI, the system performs the following:

- Determines the material and the corresponding segmentation value using the EAN number sent through the BAPI. You must have maintained the segmentation value for the EAN in material master.
- Compare the determined segmentation value with the segmentation value of the batch number provided. If the segmentation values are equal, the system saves the material document and if the values are not equal, the system displays an error message.

More Information

For more information about the master data settings, see **Segmentation based EAN** in SAP Help Library.

Determination of Pricing in Purchasing Documents based on Segmentation

Use

The standard condition technique has been enhanced with segmentation-specific pricing. For example, you can create the additional condition records according to the segmentation value needed for the pricing. To do this, you should use segmentation-specific condition types.

Prerequisites

You have activated the business function `LOG_SEGMENTATION_03`.

Features

In addition to the existing standard condition type, condition table, access sequence, and pricing procedure available for the purchasing documents, you can use the following segmentation-specific condition type, condition table, access sequence, and pricing procedure in the purchasing documents. These are standard condition tables delivered by SAP. You can use them appropriately in the existing configurations to enable the segmentation-specific pricing.

Condition Type	`PBSG`
Condition Table	Vendor/Material/Purch. Organization/Plant/Info record category/Stock segment: `A4AG` Purch. Organization /material/plant/Info record category /Stock segment: `A4AH`
Access Sequence	`SG02`
Pricing Procedure	`SM0000`

You can only maintain condition records for a segmentation value that is relevant for pricing. When you create characteristics for segmentation using transaction `CT04`, you need to mark whether the segmentation value is relevant for pricing. When you create purchasing documents, the system determines the pricing according to the segmentation value you have entered in the document. If the segmentation value is not relevant for pricing or you have not maintained the condition records for the corresponding segmentation value entered in the document, the system does not determine the pricing record in the purchasing documents.

ℹ Note

Segmentation-specific pricing is not applicable in purchasing contracts.

More Information

Conditions and Price Determination (MM-PUR-VM)

Maintain Segmentation Relevance for Functions

Origin of Costing Data

Segmentation: Subcontracting

Use

This section describes how to procure materials relevant for segmentation using a Subcontracting Purchase Order. It describes how subcontract orders are processed in **Purchasing and Inventory Management**. By procuring a material that is relevant for

segmentation using subcontracting, you provide your vendor with materials needed to produce the finished product you have ordered.

Prerequisites

You have activated the business function `LOG_SEGMENTATION_03`.

Features

You can procure a material using subcontracting by performing the following steps:

- You can order the finished product with a subcontracting purchase order. You can enter the components required by the vendor for the production of the finished product in the purchase order or derive them automatically from the relevant Bill of Materials (BOM).
- You can provide the required components to the vendor by directly posting a goods issue or by using the outbound delivery. This physical stock is located at your vendor, but remains the property of your company (tracked using **Special Stocks with Vendor**) until the goods receipt of the finished product.
- The vendor provides the service and delivers the ordered material. With the goods receipt of the finished product, the backflush consumption of the components is automatically completed.

More Information

Segmentation (LO-SGT)

Master Data

Subcontracting in Purchasing

Subcontracting in Inventory Management

Master Data

Use

For more information about creating bill of material, see Bill of Materials.

More Information

Segmentation: Subcontracting

Subcontracting in Purchasing

Use

You can use the subcontracting items (SC items) in purchase requisitions and purchase orders to order material to be produced in subcontracting. If you want to procure materials that are relevant for segmentation using subcontracting, you can manually enter the type and quantity of the components to be provided for the individual segmentation value of the components in the document or if you have already created a BOM for the ordered material, the system automatically copies the relevant data from the BOM.

Features

The standard SAP ERP system supports the procurement of materials relevant for segmentation using subcontracting at the segment level and is available for subcontracting (SC) purchase requisitions and SC orders.

More Information

Segmentation: Subcontracting

Creating or Changing Subcontracting Purchase Requisitions or Purchase Orders

Creating or Changing Subcontracting Purchase Requisitions or Purchase Orders

Use

You can start the subcontracting process in purchasing with one of the following:

- A purchase requisition that has been created manually or as a result of MRP
- Purchase order that has been created manually

ℹ Note

In both cases, the purchasing document item should have a specific item category for subcontracting (L).

To create a subcontracting purchase requisition or purchase order manually, perform the following steps:

Procedure

1. Enter the material you want to order and the item category for subcontracting (L) in the order item.
    - If the material is relevant for segmentation, you need to specify the stock segment for the purchasing document item.
2. You can enter the list of components to be provided manually or through BOM explosion.
    - If you have not maintained a BOM for the material, you can enter the components in the **Components Overview** screen manually.
    - If you have maintained a BOM for the material, the system automatically explodes the BOM in the **Components Overview** screen. You can view the components and/or add/change components and the quantities.
        - If the header material and the components are relevant for segmentation, the system determines the requirement segment for the individual components depending on the mapping defined in the **Dependent Requirements/Stock Transfer** table in the Customizing activity **Define Segmentation Strategies** under **Logistics - General** > **Segmentation**. The stock segment of the header material is used to determine the requirement segment for the components.
        - If the header material is relevant for segmentation and the components are not relevant for segmentation, the system does not determine the requirement segment for the components.
        - If the header material is not relevant for segmentation and the components are relevant for segmentation, the system determines the requirement segment from the relevant BOM.

Note

The system allows you to change the requirement that is already determined through the BOM or the segmentation strategy. However, you only can enter the valid segmentation value based on the selected segmentation strategy.

More Information

Subcontracting in Purchasing

Subcontracting in Inventory Management

Use

Inventory Management supports the following functions for the special procurement type subcontracting:

- Provision of components
- Goods receipt of the finished product (with consumption of components)

More Information

Segmentation: Subcontracting

Provisioning of Components

Goods Receipt of the Finished Product

Provisioning of Components

Use

You can initiate the process of provisioning of the components using the **Subcontracting Cockpit**.

You can provide the components to the subcontractor by directly creating an outbound delivery or by posting the goods issue. When you provide the components, you transfer post the materials from the unrestricted-use stock to the 'special stock at vendor (subcontracting stock)'.

When you create an outbound delivery or post the goods issue, the system allows you enter a batch number according to the segmentation value defined in the segmentation strategy assigned for the material. The system displays an error message if the batch number does not match the requirement segment. Alternatively, the system can use the batch determination process to determine the batch according to the stock segment.

More Information

Subcontracting in Inventory Management

Goods Receipt of the Finished Product

Use

After manufacturing the product, the subcontractor delivers the finished product. You can receive the finished product by creating an inbound delivery or directly posting the goods receipt for the subcontracting purchase order.

When you receive the goods, the unrestricted-use stock of the finished product increases in the receiving site and stock of the component materials decreases at the special stock at vendor.

More Information

Subcontracting in Inventory Management

Sales and Distribution

Sales allows the realization of different business transactions based on sales documents stored in the system. When creating a sales order, you can carry out an availability check to determine whether the desired materials are in stock. This also checks whether the stock is sufficient to fill the order by the desired delivery date. A confirmed delivery date is determined based on the results of the availability check. The material requirements planning (MRP) assigns the requirement quantities to the available stock, or creates procurement proposals. For stock that is not available the internal or external procurement (purchase requests or planned orders) are created.

Segmentation in Sales Documents

Use

Segmentation is available in sales and distribution (SD). Segmentation at the item level allows you to you to sell materials with different attributes, for example, quality of the material and its country of origin. To sell materials with segmentation, you can provide data in sales documents using requirement segments.

For more information about processing sales documents in the standard ERP system, see Sales (SD-SLS).

Prerequisites

You have made the following settings:

- Segmentation structure and segmentation strategy are maintained in the Customizing for **Segmentation** under **SAP Implementation Guide** > **Logistics - General**
- The material is made segmentation relevant in material master
- Fix segmentation has been chosen for the corresponding distribution center (DC)

Features

To sell materials with segmentation, the system allows you to maintain data in sales documents for the corresponding requirement segment. You can maintain requirement segment in sales document in one of the following ways:

- manually enter requirement segment
- copy requirement segment from preceding document

- default determination of requirement segment based on the default value maintained for the characteristic maintained for the application area, in this case, sales order is the application area

For detailed information about how segmentation is handled in individual sales documents, see the following:

- Segmentation: Sales Orders, Inquiries, Quotations, and Contracts
- Segmentation: Delivery and Billing

More Information

Creation of a Sales Document with Reference

Incompletion Log

Segmentation: Sales Orders, Inquiries, Quotations, and Contracts

Use

Segmentation is available in sales and distribution (SD). Segmentation at the item level allows you to you to sell materials with different attributes, for example, quality of the material and its country of origin. To sell materials with segmentation, you can provide data in sales documents using requirement segments.

Prerequisites

You have maintained the following settings in Customizing under **SAP Implementation Guide** > **Logistics - General** > **Segmentation**:

- segmentation structure
- segmentation strategy

Features

For sales documents such as sales order, inquiry, quotation and contract, you can carry out the following functions:

- specify requirement segment for a material
- change requirement segment for a material

i Note

You can maintain blank as the segmentation value to be used in combination with a segmentation strategy. In such cases, materials will have blank as segmentation value in sales documents.

More Information

- Segmentation (LO-SGT)
- Segmentation Structure
- Segmentation Strategy

Segmentation: Delivery and Billing

Use

Segmentation is available in sales and distribution (SD). Segmentation at the item level allows you to you to sell materials with different attributes, for example, quality of the material and its country of origin. To sell materials with segmentation, you can provide data in sales documents using requirement segments and stock segments.

Prerequisites

You have maintained the following settings in Customizing under **SAP Implementation Guide** > **Logistics - General** > **Segmentation** >:

- segmentation structure
- segmentation strategy
- requirement segment for the sales document

Features

For delivery, you can carry out the following functions based on the type of delivery:

- Outbound delivery with reference

  Requirement segment will be adopted from the preceding sales document and stock segment will be adopted from the batch.

- Outbound delivery without reference

  You specify the requirement segment and stock segment will be adopted from the batch. There is incompletion log for this type of deliveries and this will be triggered like sales order when you save segmentation relevant article without specifying requirement segment values.

For billing, requirement segments and stock segments are adopted from preceding sales documents.

i Note

You can maintain blank as the segmentation value to be used in combination with a segmentation strategy. In such cases, materials will have blank as segmentation value in sales documents.

More Information

- Segmentation (LO-SGT)
- Segmentation Structure
- Segmentation Strategy

Batch Determination for Outbound Deliveries

Use

You can activate automatic batch determination for delivery item categories using the Customizing activity **Activate Automatic Batch Determination in SD** under **Logistics - General** > **Batch Management** > **Batch Determination and Batch Check**. For such delivery items, batch determination is triggered automatically in outbound deliveries. A batch split describes the situation when more than one batch is used to cover a particular delivery item. The batch split function in the delivery includes a batch selection screen and an availability check.

Prerequisites

You have defined the necessary settings in the Customizing activities under **Logistics - General** > **Batch Management**.

Features

Segmentation is now integrated in the batch determination process. If you use materials that are not relevant for segmentation, the batch is determined in the same way as in the batch determination process in standard SAP ERP.

Scenario 1

If you create an outbound delivery with reference to a sales order, a stock transfer order, a return purchase, or without any reference, the batch is determined as follows:

Automatic Batch Determination is OFF

If you have not activated automatic batch determination, you can enter the batch number in the delivery documents manually. You can only enter the batches that belong to the valid stock segment value that is maintained in the mapping rule in the **Requirement/Stock Allocation** table in the Customizing activity **Define Segmentation Strategy**.

Automatic Batch Determination is ON

For a material relevant for segmentation, The system filters the batch numbers according to the segmentation value and proposes a suitable batch number(s) with batch split.

i Note

For the requirement segment provided in the reference document, the corresponding stock segment is determined using the mapping rule defined in the **Requirement/Stock Allocation** table in the Customizing activity **Define Segmentation Strategy** under **Logistics - General** > **Segmentation**. Then, the batch is determined based on the stock segment.

Scenario 2

If you create an outbound delivery with reference to a return sales order or return stock transfer order, the batch is determined is the same way as the determination process described in scenario 1 with the following differences:

- If the system cannot determine a batch in the standard batch determination process, you can manually enter a new batch number that is not yet available in the system.
- For a material relevant for segmentation, the system proposes the first entry of the segmentation value in the mapping rule defined in the **Requirement/Stock Allocation** table in the segmentation strategy. A new batch is created for the proposed stock segmentation. If the mapping rule is not maintained in customizing, the system cannot create the batch.

Creation of a Sales Document with Reference

Use

You can use the requirement segment of an existing sales document as the basis for a new sales document by making use of **Create with Reference**.

The requirement segment of the source document is copied to the target sales document.

Prerequisites

The source sales document was created with requirement segment.

Features

- You can copy requirement segment from one sales document type to a new sales document by making use of **Create with Reference**.
- You can then modify the requirement segment of the new sales document (target). For example, if you have copied a requirement segment from an inquiry to a quotation, you can later on change the requirement segment of the quotation but not that of the inquiry.
- If the source sales document type is the same as the target sales document type, then the requirement segment can be modified in both the source or target sales document. For example, if you have copied a requirement segment from one quotation to another quotation, you can later on change the requirement segment of both the quotations.

Incompletion Log

Use

If you try to save a sales document without having maintained the requirement segment for segmentation relevant material, then the system displays the incompletion log for the sales document. The incomplete sales documents can be saved in the system.

> **i Note**
> Sales documents can be created for materials without segmentation strategy. No incompletion log is created for such sales documents.
>
> If the requirement segment is blank and valid, no incompletion log will be generated.

Activities

- All sales documents with incomplete data, except delivery documents created with reference and billing documents, are stored in the incompletion log.
- From the incompletion log, you maintain requirement segment for the corresponding material.

Special Business Processes in Sales

In addition to carrying out routine sales order processing in the **SAP Sales & Distribution** module, you can also implement more specialized business processes, such as the processing of third-party business transactions, purchase-to-order transactions, and consignment processing.

For more information about special business processes in the standard sales processes, see Special Business Processes in Sales.

Segmentation in Special Business Processes

Use

You can use materials with segmentation while executing specialized business processes if the business function `LOG_SEGMENTATION` is activated.

Process

Consignment Processing

When you process the consignment stock, if you use materials that are relevant for segmentation, segment values are considered during order creation and the orders are saved with the segmentation values.

Third-Party Order (TPO) Processing

In third-party order processing, a purchase requisition is created for each order item. For the requirement segment in the sales order the matching stock segment in purchase requisition is determined based on segmentation strategy defined in Customizing for Logistics – General > Segmentation > Define Segmentation Strategies > Requirement to stock conversion.

Reporting in Sales

Use

Reports are used to access information from the database, and to list the results based on the selection criteria.

Features

If you have activated the business function `LOG_SEGMENTATION`, the following reports also display the segmentation values in the result screen:

Sales Documents

Transaction	Description
VA05	List of Sales Orders
VA15	List of Inquiries
VA25	List of Quotations
VA45	List of Contracts
VF05	List of Billing Documents

More Information

For more information about reporting in sales, see Reports and Analyses (SD-IS-RE) in SAP Library for SAP ERP Central Component on SAP Help Portal.

Segmentation based EAN in Sales Documents

Use

For materials that are relevant for segmentation, the system uses the requirement segment to determine the EAN number in the sales documents such as sales order and contracts.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_03`.
- You have created characteristics that are relevant for EAN using transaction `CT04`.
- You have created the segmentation strategy and the segmentation structure using the characteristics that are relevant for EAN.
- You have created the material and maintained EANs for different segmentation values of Unit of Measure (UoM). You can maintain EAN in the Additional EAN tab of the **Additional Data** screen in the material master.

Features

Sales orders

When creating or changing sales orders, the system determines the EAN number for the combination of material, Unit of Measure (UoM), and requirement segment. If there are multiple EAN numbers, the system selects the main EAN entry you maintained for the material. If you change one of the following values for a sales order item in the document, the system redetermines the EAN number accordingly.

- Unit of Measure
- Requirement segment
- Plant

i Note

You cannot change the EAN number manually in a sales document.

Material determination enables the automatic substitution of materials in sales documents during sales order processing. In addition, you can use material determination if you want the system to substitute the EAN numbers with your own material numbers automatically.

You can enable material determination by making the following settings:

- Maintain the material determination master record using transaction VBxx
- Assign the material determination procedure to the document type in the customizing activity **Assign Procedures To Sales Document Types** under **Sales and Distribution** > **Basic Functions** > **Material Determination (transaction OV14)**.

If you maintained a substitute material in transaction `VBxx`, the system determines a substitute material for the material you enter during sales order creation. The system also displays the substitution reason in the document.

When you enter the EAN number instead of the material in the document, the system determines the material for the EAN number maintained in the transaction `VBxx`. In addition, the system determines the segmentation value of the material for the EAN number if maintained in the material master.

When you enter both EAN number and segmentation value in the document, the system performs the following:

- Determines the material for the EAN value maintained in transaction `VBxx`.
- Accepts the segmentation value you entered and saves the document if the segmentation characteristics are not relevant for EAN.
- Determines the segmentation value for the EAN value and compares with the segmentation value you entered if the segmentation characteristics are relevant for EAN. If the segmentation values are the same, the system saves the document. If the segmentation values are different, the system does not allow you to save the document and displays an error message.

i Note

The system follows the similar process when you use BAPI or IDoc.

Outbound Delivery

When you create an outbound delivery for segmentation relevant materials with reference to a sales order, the system does not copy the EAN details of the material from the reference document. Instead, the system determines the EAN value according to the stock segment. If you enter the stock segment for a batch item or change the UoM for a batch split item in the batch split screen, the system redetermines the EAN value according to the updated stock segment. When you create an outbound delivery for the segmentation relevant materials without a reference, the system determines the EAN value according to the stock segment.

More Information

For more information about the master data settings, see **Segmentation based EAN** in SAP Help Library.

Determination of Pricing in Sales Documents based on Segmentation

Use

The standard condition technique has been enhanced with the segmentation-specific pricing. For example, you can create the additional condition records according to the segmentation value needed for the pricing. To do this, you should use the segmentation-specific condition types.

Prerequisites

You have activated the business function `LOG_SEGMENTATION_03`.

Features

In addition to the existing standard condition type, condition table, access sequence, and pricing procedure available for the sales documents, you can use the following segmentation-specific condition type, condition table, access sequence, and pricing procedure in the sales documents. These are standard condition tables delivered by SAP. You can use them appropriately in the existing configurations to enable the segmentation-specific pricing.

Condition Type	PRSG
Condition Table	Sales org./Distribution channel/Customer/Material/Requirement segment: `A4AE`

	Sales org./Distribution channel/Customer/Material/Stock segment: A4AF
Access Sequence	PRSG
Pricing Procedure	SVAA01

You can only maintain condition records for a segmentation value that is relevant for pricing. When you create characteristics for segmentation using transaction CT04, you need to mark whether the segmentation value is relevant for pricing. When you create sales documents, the system determines the pricing according to the segmentation value you have entered in the document. If the segmentation value is not relevant for pricing or you have not maintained the condition records for the corresponding segmentation value entered in the document, the system does not determine the pricing record in the sales documents.

More Information

Pricing and Conditions

Maintain Segmentation Relevance for Functions

Origin of Costing Data

Production Planning and Control

Production Planning is a part of Logistics functional area. It helps manage broad level business activities, such as Sales and Operation Planning, and Material Requirement Planning. The requirements planning processes uses the data available in the forecast or the Sales & Operations Planning to fulfill the requirements, such as sales orders, with the existing or planned stock.

Material segmentation is now supported in the planning process.

Segmentation: Planned Independent Requirements (PIRs)

Use

Planned independent requirements are planned and organized in **Demand Management**. A planned independent requirement contains one planned quantity and one date or a number of planned independent requirements schedule lines. That is, one planned quantity split over time according to dates.

Using segmentation, you can maintain separate planned independent requirements for a combination of material, plant, version, requirements plan number, and external requirements plan ID. For more information about PIRs in the standard ERP system, see Planned Independent Requirements (PIRs).

More Information

Creating PIRs for Materials with Segmentation

Creating PIRs with Reference

Consumption of PIRs with Segmentation

Example: Consumption of PIRs with Segmentation

Creating Planned Independent Requirements for Materials with Segmentation

Prerequisites

- You have created characteristics for segmentation using transaction `CT04`
- You have defined segmentation structures in Customizing for **Define Segmentation Structures** under **Logistics – General** > **Segmentation**
- You have defined segmentation strategies in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**
- You have created materials and assigned a segmentation strategy in the material master (transaction `MM01`)
- You have selected **Plnng with assembly (101)** in the **Planned Independent Requirements: Internal Number Assignm** view to maintain the number range for the requirement plan ID

  You can access this view using transaction `OMPP` or in Customizing for **Define Internal Number Assignment per Reqmts Class** under **Production** > **Production Planning** > **Demand Management** > **Planned Independent Requirements** > **Default Values and Presettings**.

Procedure

To create planned independent requirements for materials with segmentation, proceed as follows:

1. Go to transaction `MD61`.
2. Enter the material with segmentation, **Product group**, or **Ext Req Plan ID** in the screen for creating planned independent requirements.
3. Enter the plant and choose Enter.
4. In the **Items** tab, enter the material and requirement segment.
5. In the **Table** tab, enter the requirement on a monthly basis for the material at the requirement segment level.
6. In the **Sched. Lines** tab, the system displays the external requirement plan ID and the requirement segment.

Result

A new requirements plan number (`PBDNR`) is generated internally for the unique combination of material, plant, requirement type, version number, external requirement plan ID, and requirement segment.

You can use the external requirements plan ID to select the planned independent requirements for materials with segmentation.

Creating Planned Independent Requirements with Reference

Prerequisites

- You have activated the business function **Material Segmentation across Logistics** (`LOG_SEGMENTATION`)
- You have performed the planning run and have the results to use as reference

Context

You can create PIRs using a reference as already available in the standard SAP ERP system. In addition, you can create PIRs using a reference to PIRs with material segmentation.

Procedure

1. In the initial screen for creating planned independent requirements, enter the material number, the name of the product group, or the requirements plan. If the business function `LOG_SEGMENTATION` is activated, you can enter the external requirements plan ID.
2. Enter the plant and / or the material.
3. Enter the version.
4. Choose the **Reference** Icon. The **Copy Reference** dialog box appears.
5. Choose **Planned independent requirements of materials with segmentation** as the reference type.

Results

If you have activated the business function `LOG_SEGMENTATION`, materials with segmentation are also considered when you create PIRs using **Reference**.

In addition to the existing behavior in the standard SAP ERP system, the following scenarios are applicable if you have activated the business function `LOG_SEGMENTATION`:

Scenario 1: Copying data from a material without segmentation to a material with segmentation.

The system copies the item details from the source material to the target material. The system determines the requirement segment in one of the following ways:

- If you have maintained a default value for the requirement segment in the segmentation structure of the target material, the system copies this default value to the requirement segment.
- If you have not maintained a default value for the requirement segment, the system assigns the value BLANK – if it is a valid requirement segment.
- If the above conditions are not met, an error message is displayed.

Scenario 2: Copying data from a material with segmentation to a material with segmentation.

The system copies the item details from the source material to the target material. The system copies the requirement segment in one of the following ways:

- If you have maintained the same segmentation structure to both materials, then the system copies the requirement segment.
- If you have maintained a different segmentation structure to each material, the following is valid:
    - If you have maintained a default value for the requirement segment in the segmentation structure of the target material, the system copies this default value to the requirement segment.
    - If you have not maintained a default value for the requirement segment, the system assigns the value BLANK – if it is a valid requirement segment.
- If the above conditions are not met, an error message is displayed.

Scenario 3: Copying data from a material with segmentation to a material without segmentation.

- The item details are copied from the source material to the target material and the requirement segment is not copied.

Consumption of Planned Independent Requirements with Material Segmentation

Use

For materials with segmentation, the consumption of the planned independent requirements depends on the consumption mode and the consumption priority.

Prerequisites

You have defined one of the following planning strategy groups for materials with segmentation in the material master:

- **40 – Planning with Final Assembly**
- **20 – Make-to-Order**

Process

In the material master, you define the sequence in which the system consumes the PIRs based on the segments, consumption mode, and the consumption priority.

Using the consumption priority, you can define how the planned independent requirements are to be consumed based on segments or the consumption mode. You can choose the following option:

- 1 – Based on Consumption Mode and then Segmentation
- 2 – Only if Requirement Segments Match
- 3 – Based on Segmentation and then Consumption Mode

The system consumes the requirements based on the consumption mode and whereby the requirement segments of the customer requirement and the PIR must match.

Result

Withdrawal Quantity

The system updates the **withdrawal quantity** column when you create a delivery for a customer requirement, which has consumed planned independent requirements with material segmentation.

Reorganize PIRs with Segmentation

You can reorganize the PIRs with segmentation using the combination of the key date and the external requirement plan ID.

More Information

Example: Consumption of PIRs with Segmentation

Example: Consumption of PIRs with Segmentation

Assume the following settings:

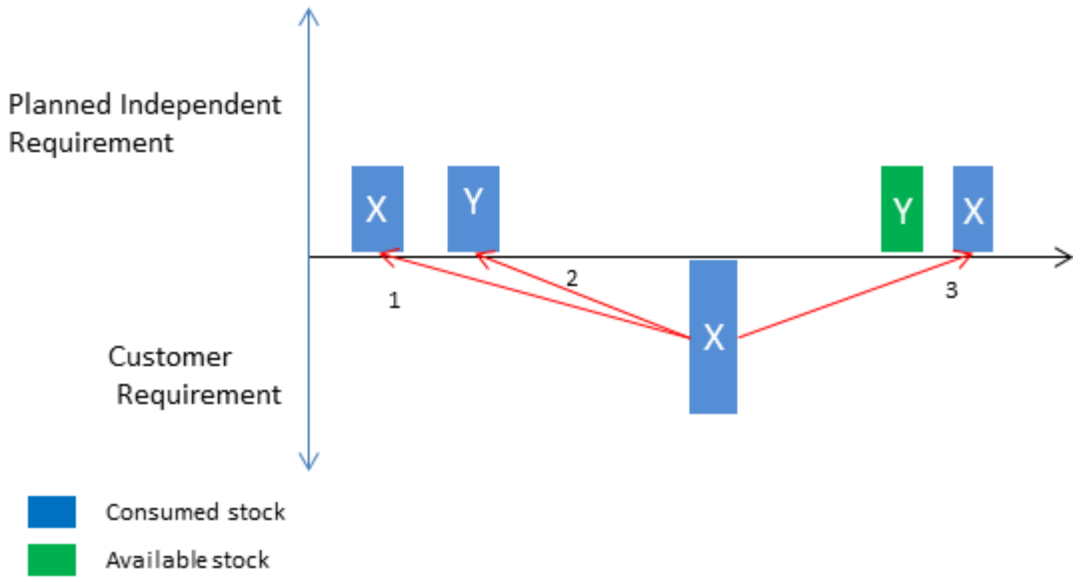
Consumption Mode = 2 (Backward/Forward Consumption)

Segments: X, Y

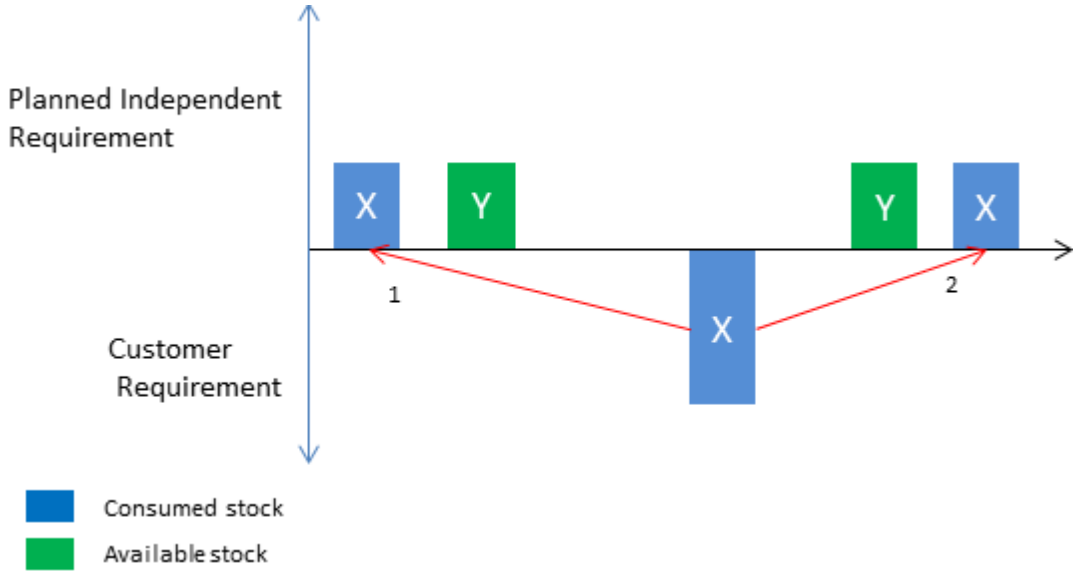
Segmentation Strategy:

Requirement segment in sales order	PRC planning level	Consecutive number for segmentation of equal status	Planned independent requirement segment
X	1	1	X
X	2	2	Y

1. If the consumption priority = 1, the system consumes the requirements based on the consumption mode first and then based on segments. In this example, the customer requirement first consumes the planned independent requirements available before the requirement date per segmentation strategy. If the customer requirements cannot be completely fulfilled during backward consumption, the system switches to forward consumption and then consumes the planned independent requirements available after the requirement date per segmentation strategy. See the following graphic:

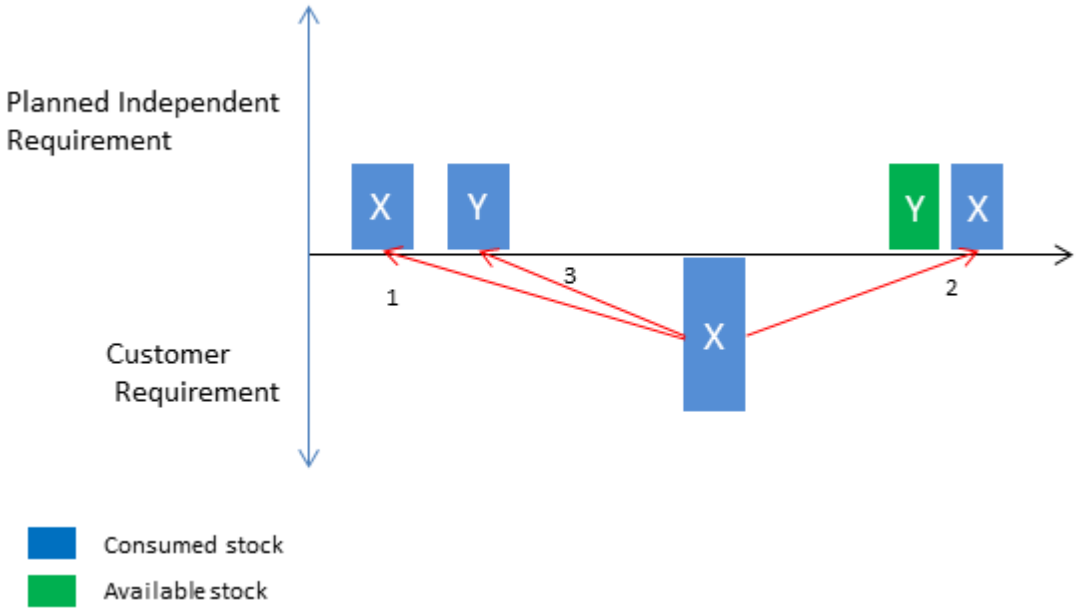


2. If the consumption priority = 2, the system consumes the requirements based on the consumption mode whereby the requirement segments of the customer requirement and the PIR must match. In the example below, the system first consumes the planned independent requirements backwards on the time axis with the same requirement segment and if a rest quantity still remains, the system switches to forward consumption and consumes the planned independent requirements forwards on the time axis with the same requirement segment. Any planned independent requirements with different requirement segments are not considered in this case. Customizing for the segmentation strategy is not considered for consumption. See the following graphic:



PIR consumption when consumption priority = 2

3. If the consumption priority = 3, the system consumes the requirements based on the requirements segment first and then based on the consumption mode. In the example below, the customer requirement first consumes the planned independent requirements with the matching requirement segment backwards on the time axis. If not all customer requirements are fulfilled, the system switches to forward consumption and consumes the planned independent requirements with the same segment forwards on time axis. If a rest quantity still exists, the system then looks for the planned independent requirements with different requirement segments as defined in the segmentation strategy. See the following graphic:



Segmentation: Material Requirements Planning (MRP)

Use

The main function of material requirements planning is to guarantee material availability. That is, it is used to procure the requirement quantities on time both for internal purposes and for sales and distribution. This process involves the monitoring of stocks and, in particular, the automatic creation of procurement proposals for purchasing.

The material requirements planning process has now been enhanced to guarantee material availability at segment level.

The enhanced process uniquely assigns requirement quantities to available stock using segmentation values and / or creates procurement proposals, such as purchase requisitions, at segment level for nonexisting stock. The system determines the

requirement segment in the proposals based on the rules defined in the segmentation strategy.

Prerequisites

- You have maintained the transaction `MD01N` in the table `PPH_HANA_ACTIVE` so that you can use HANA DB
- You have maintained the user parameter `HANA_ACCESS = 02` in the user profile using transaction `SU3`

Process

For materials with segmentation, the system executes the planning processes based on the following mapping rules defined in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**:

- Requirement/Stock Allocation
- Stock Transfer
- Requirement to Stock Conversion
- PIR Consumption

More Information

Defining Segmentation Strategy for MRP

Planning Process

Defining the Segmentation Strategy for MRP

Use

Using the segmentation strategy, you can map the requirement segments to the stock segments. The system considers this mapping during material requirements planning for materials with segmentation.

Prerequisites

- You have created characteristics for segmentation using the transaction `CT04`
- You have defined the segmentation structure in Customizing for **Define Segmentation Structures** under **Logistics – General** > **Segmentation**

Procedure

You define your segmentation strategies in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**.

1. Choose the segmentation structure.
2. Define the list of valid segment combinations. The system validates the segment values that are defined in transaction `CT04`.
3. Specify how the requirement segments consume the stock segments during material requirements planning in the **Requirements/Stock Allocation** table.

4. Define the assignment of stock segments to requirement segments between two plants in the **Stock Transfer** table.
5. Define the stock segment to be proposed for a specific requirement when the system creates new proposals during the lot-size calculation in the **Requirement to Stock Conversion** table.
6. Define which requirement segments of planned independent requirements are consumed by which requirement segments of the customer requirement in the **Planned Independent Requirement Consumption** table.

If you do not want to use specific segment values in the segmentation strategy for the above tables, you can use:

- * to indicate a 1:1 mapping

  Requirement segment X is mapped to stock segment X

- # to indicate a 1:N mapping

  Requirement segment X is mapped to multiple stock segments as defined in the segmentation strategy

Planning Process

Use

During planning, the system carries out the following processes for materials with segmentation:

- Net requirements calculation
- Lot sizing
- Determines procurement proposals

Prerequisites

You have activated the business function **Material Segmentation across Logistics** (`LOG_SEGMENTATION`).

Process

Master Data

The master data component (`PP-MRP-MD`) forms the basis for MRP. Master data for MRP includes:

- Segmentation structure
- Segmentation strategy
- Planning strategies
- MRP type: **PD** and **ND**

Net Requirements Calculation

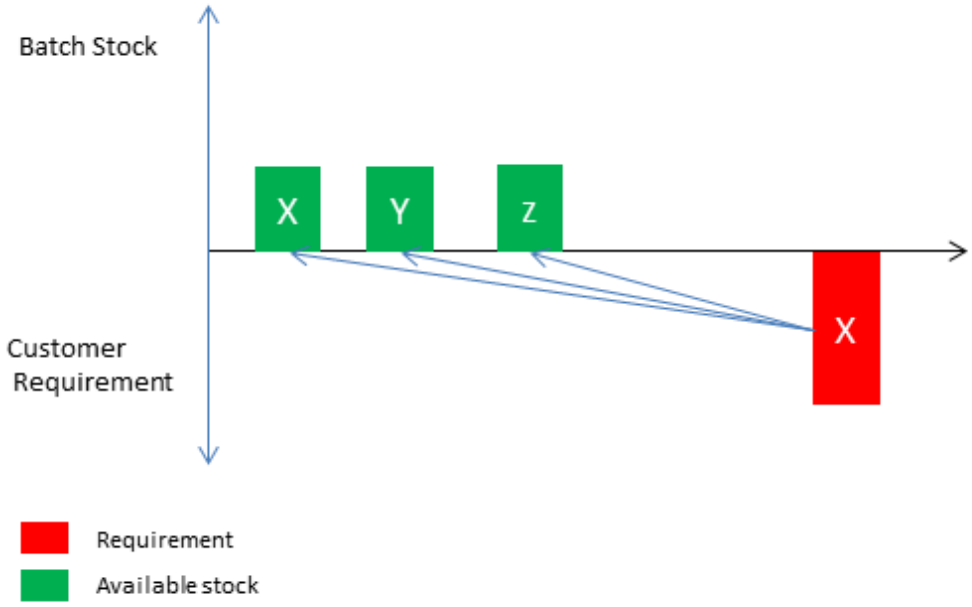
The net requirements calculation can be carried out for both standard materials and materials with segmentation.

In MRP, requirement quantities are maintained in the system as planned independent requirements, customer requirements, as well as forecast requirements. For more information about net requirements calculation in the standard ERP system, see Net Requirements Calculation.

For materials with segmentation, requirement to stock allocation depends on the settings you have defined in the **Requirement/Stock Allocation** table in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**.

The system considers available stock and future receipts. If there are multiple stock segments that can be consumed by a certain requirement segment, you can define the sequence for consuming the different stock segments using the **Consec.No for Eq Seg** option in the **Requirement/Stock Allocation** table.

Segmentation Strategy (Requirement/Stock Allocation)	
Scope: 1 (Maintain both requirement and stock segments)	
Mapping rule: 1:N	
Req. Segment	Stock Segment
X	X
X	Y
X	Z



Requirement/Stock Allocation

If stock is available, the system determines the requirement segment that should be assigned to the stock segment. If the required quantity of a particular stock segment is not available, the system may use another segment if it is defined in the segmentation strategy.

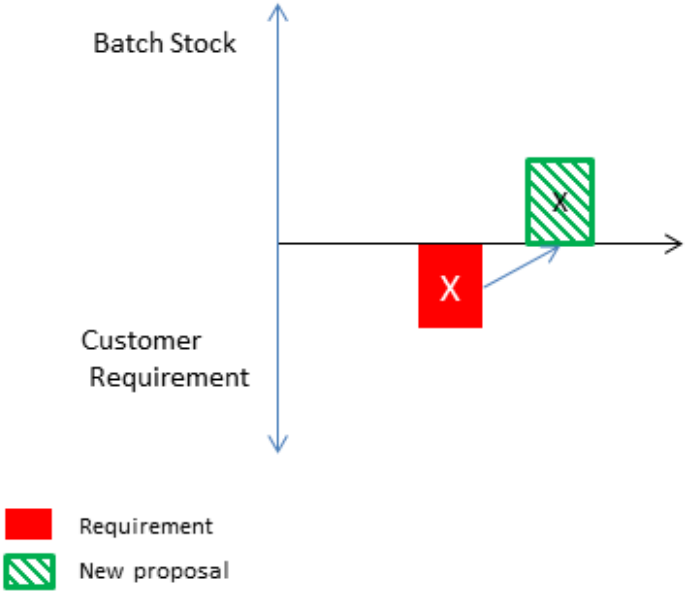
If no stock is available, the system determines the stock segment to be procured.

Lot-Sizing Procedures

MRP calculates the procurement quantities for purchase requisitions using the lot-sizing procedures. For more information about lot-sizing procedures in the standard ERP system, see Lot-Sizing Procedures.

For materials with segmentation, requirement to stock conversion depends on the settings you have defined in the **Requirement to Stock Conversion** table in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**. The system determines the stock segment to be procured.

Segmentation Strategy (Requirement to Stock Conversion)	
Scope: 1 (Maintain both requirement and stock segments)	
Mapping rule: 1:1	
Req. Segment	**Stock Segment**
X	X
Y	Y
Z	Z



Requirement to Stock Conversion

Determine Procurement Proposal

Procurement proposals, for example, purchase requisitions are used to procure the shortage quantities. Procurement proposals are internal planning elements that can be changed, rescheduled, or deleted at any time. For more information about the determination of procurement proposal in the standard ERP system, see Procurement Proposal (PP-MRP-PP).

During a stock transfer between plants, if the supplying plant is not relevant for segmentation and the receiving plant is relevant for segmentation, the system determines the requirement segment according to the settings you have defined in the Customizing activity **Purchase Requisitions** under **Logistics – General** > **Segmentation** > **Define Requirement Segment for Supplying Plants**.

i Note

For materials with segmentation, the system only supports external procurement and does not support in-house production.

More Information

Stock Transfer

Stock/Requirements List

Safety Stock Split

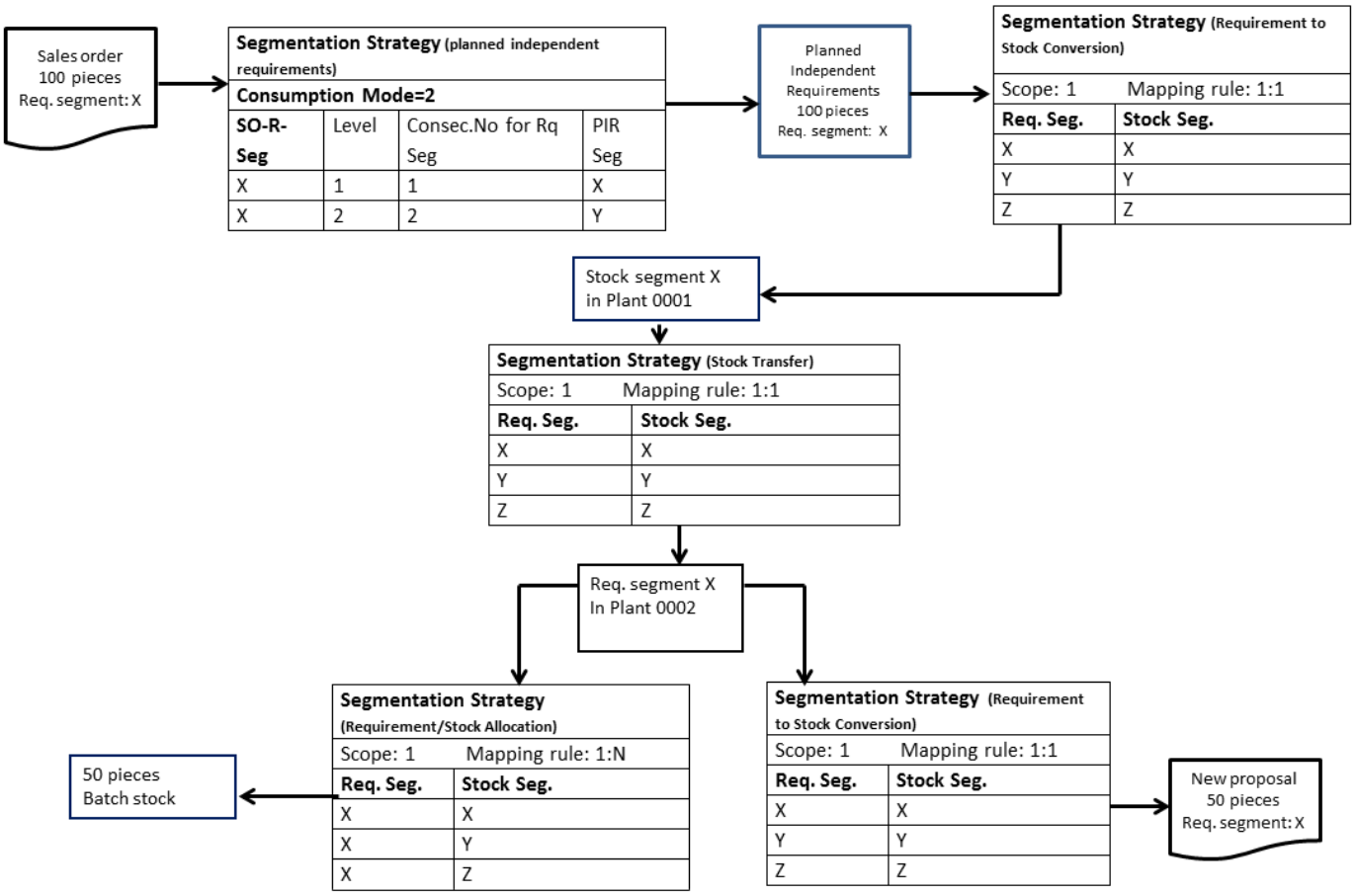
Planning Strategies

Stock Transfer During MRP

Use

You define the assignment of stock segments to requirement segments between two plants in the **Stock Transfer** table. You can access this table in Customizing under **Logistics — General** > **Segmentation** > **Define Segmentation Strategies**. The system uses the assignment information in this table when creating the dependent requirement.

Example



Stock Transfer

1. Plant 0001 receives a customer requirement (for example, a sales order) for 100 pieces in requirement segment X.
2. According to the settings defined in Customizing for **Planned Independent Requirements Consumption** in the segmentation strategy, the customer requirement consumes the planned independent requirements of 100 pieces in requirement segment X.
3. According to the settings defined in Customizing for **Requirement to Stock Conversion** in the segmentation strategy, requirement segment X is mapped to stock segment X.
4. According to the settings defined in Customizing for **Stock Transfer** in the segmentation strategy, the stock in segment X of plant 0001 is mapped to the requirements in segment X of plant 0002.
5. According to the settings defined in Customizing for **Requirement/Stock Allocation** in the segmentation strategy, after the planning run, the batch stock of 50 pieces in stock segment X is consumed and the system creates a new proposal for 50 pieces in requirement segment X.

Stock/Requirements List

Use

In the stock/requirements list, the system displays the most up-to-date stock and requirements situation. For more information about the stock/requirements list in the standard ERP system, see MRP List and Stock/Requirements List.

Features

For materials with segmentation, you can analyze the planning result using the stock/requirements list.

The list maps the stock elements to the requirements elements based on the mapping rules defined in Customizing for **Logistics — General** > **Segmentation** > **Define Segmentation Strategies** > **Requirement/Stock Allocation**. You can filter the list based on the stock segment.

You must specify the stock segment to display the list. The list supports the following mapping rules:

- 1:1 (a requirement segment fulfilled by a single stock segment)
- N:1 (multiple requirement segments fulfilled by a common pool of stock)

For materials with segmentation for which the scope **Only Requirements Segments Maintained** is defined in the segmentation strategy, the system only displays the requirement segment in the stock/requirements list. In this case, the system does not display the stock segment filter.

For every combination of plant, material, and stock segment, the system displays the aggregation of the batch stock as a single line entry similar to the plant stock.

Safety Stock Split

Use

Safety stock is the quantity of stock to satisfy high requirements in the stocking-up period. The purpose of the safety stock is to prevent material shortage. During MRP, if the materials with segmentation are planned, the system splits the safety stock equally across different requirement segments.

Prerequisites

- You have assigned and fixed a segmentation strategy (with scope = **Requirement and Stock Segments Maintained**) for your material
- You have activated the business function **Material Segmentation across Logistics** (`LOG_SEGMENTATION`)

Features

During MRP, the system splits the safety stock based on the valid segment combinations you maintain in the segmentation strategy. If you do not maintain segment combinations, the system proposes the value **BLANK** as the valid segment value for the entire safety stock.

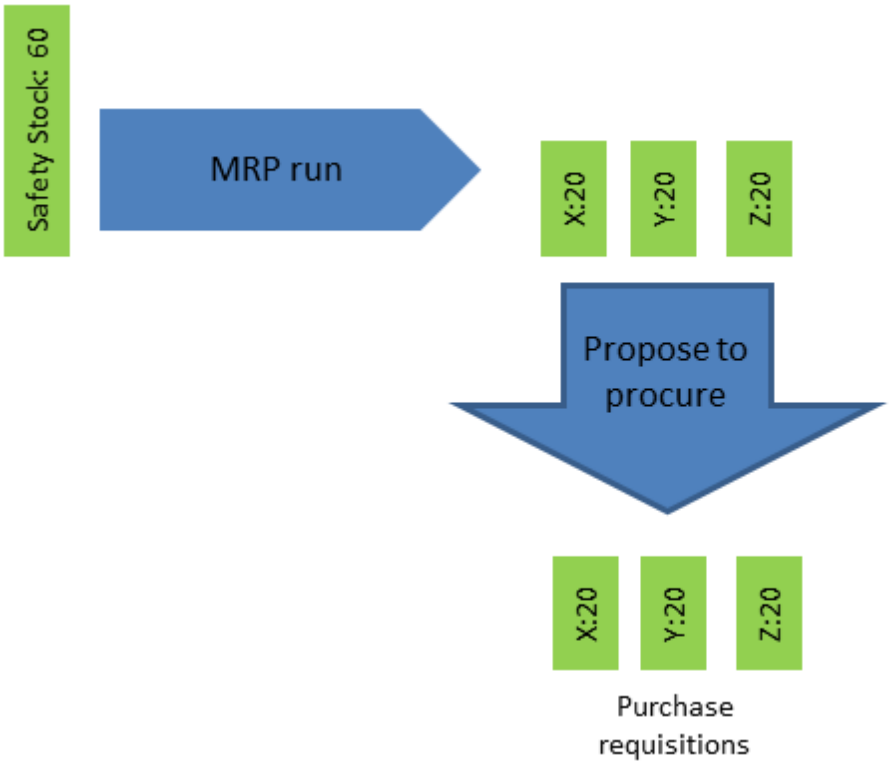
Example

You have the material Shirt_Mens with a safety stock of 60 pieces. The valid segment combinations X, Y, and Z are maintained in the segmentation strategy that is assigned to the material.

For the mapping rule 1:1, the segmentation strategy with scope = Requirements and Stock Segments Maintained is as follows:

Requirement Segment	Stock Segment
X	X
Y	Y
Z	Z

During MRP, the system splits the safety stock between segments X, Y, and Z. As a result, the system proposes new purchase requisitions for all segments X, Y, and Z.

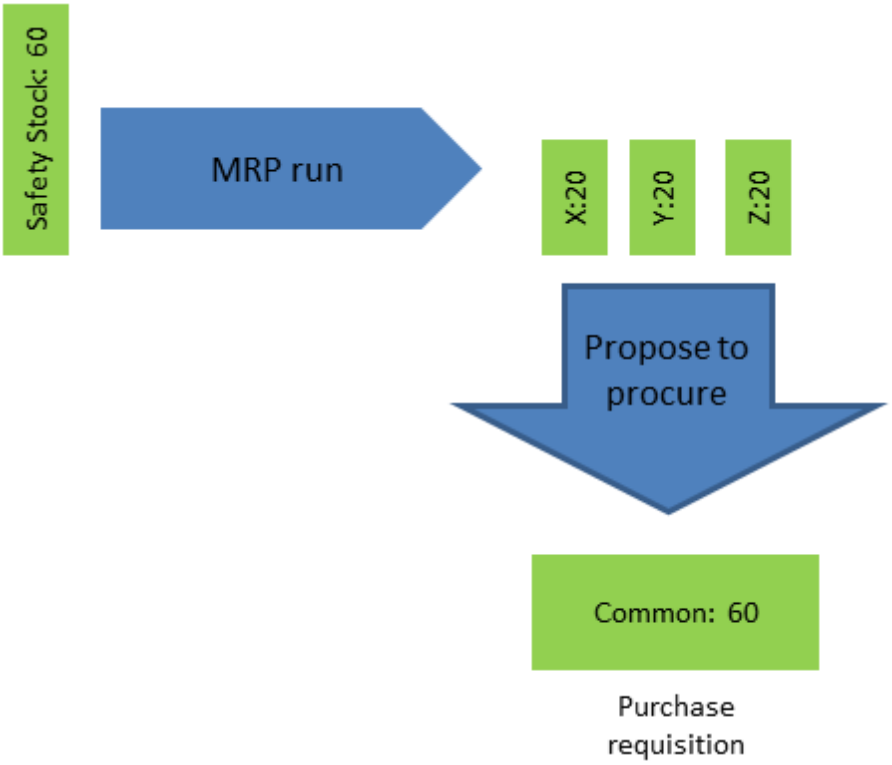


Safety stock split and individual purchase requisitions

For the mapping rule N:1, the segmentation strategy with scope = Requirements and Stock Segments Maintained is as follows:

Requirement Segment	Stock Segment
X	Common
Y	Common
Z	Common

During MRP, the system splits the safety stock between segments X, Y, and Z. However, based on the mapping rule, any requirement segment is covered by the common stock. As a result, the system proposes one new purchase requisition for the common stock.



Safety stock split and single purchase requisition

More Information

[Segmentation: Material Requirements Planning (MRP)]

Default Segment Value Determination During MRP

Use

You can carry out stock transfers between storage locations within a plant, between different plants, and between company codes. You cannot transfer materials with segmentation between plants with different segmentation strategies. You can only transfer materials with segmentation between plants with the same segmentation strategies. You can also transfer materials with segmentation between plants if one of the plants is not relevant for segmentation. If both the receiving plant and the supplying plant are assigned with the same segmentation strategy, you must define the mapping between the stock segment and the requirement segment of the plants involved.

Features

- If both the receiving plant and the supplying plant are relevant for segmentation, the stock segment in the stock transport order is mapped to the corresponding requirement segment of the supplying plant. This mapping is based on the Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**.
- If the supplying plant is relevant for segmentation and the receiving plant is not relevant for segmentation, the stock transport order does not have a stock segment. The system determines the requirement segment for the supplying plant in one of the following ways:
    - Priority 1: Based on the segment value assignment you define in the Customizing activity **Define Segment Value for Automatic Determination** under **Logistics – General** > **Segmentation**.
    - Priority 2: If you do not define the segment value assignment in the Customizing activity **Define Segment Value for Automatic Determination**, the system uses the default segment maintained in the Customizing activity **Define Segmentation Structures** under **Logistics – General** > **Segmentation**.

- Priority 3: If BLANK is a valid segment value and no segment value is defined in the Customizing activities, Define Segment Value for Automatic Determination and Define Segmentation Structures, the system proposes BLANK as the segment value.

- If the supplying plant is not relevant for segmentation and the receiving plant is relevant for segmentation, the stock transport order has the stock segment value and the material is posted to the supplying plant without any segmentation value.

Example: Default Segment Value Determination

You have specified a material in the following plants:

Plant	Segmentation Enabled	Company Code
DC1	Yes	001
DC2	No	001
DC3	Yes	002

Scope is maintained as Both Requirement and Stock Segment Maintained in the segmentation strategy

Scenario 1: DC3 supplies to DC1

Both the supplying plant and the receiving plant are relevant for segmentation. You defined the stock segment in the stock transport order. The system determines the requirement segment based on the segment value assignment you define in Customizing for Stock Transfer under Logistics – General > Segmentation > Define Segmentation Strategies.

- If BLANK is a valid segment value and no segmentation strategy is defined, the system proposes BLANK as the default segment value.

Scenario 2: DC3 supplies to DC2

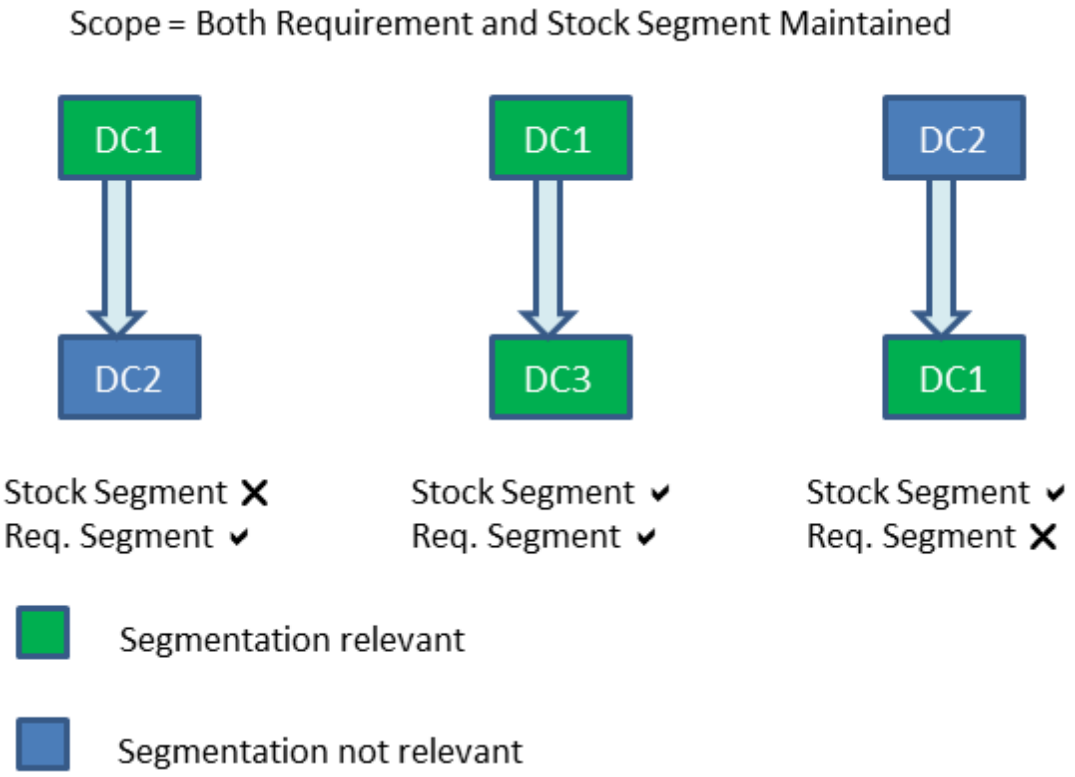
The receiving plant is not relevant for segmentation. You did not define the stock segment in the stock transport order. The system determines the requirement segment in one of the following ways:

- **Priority 1**: Based on the segment value assignment maintained in the Customizing activity Define Segment Value for Automatic Determination under Logistics – General > Segmentation.
- **Priority 2**: If you do not specify the segment value assignment in the Customizing activity Define Segment Value for Automatic Determination, the system determines the requirement segment based on the default stock segment you define in the Customizing activity Define Segmentation Structures under Logistics – General > Segmentation.
- **Priority 3**: If BLANK is a valid segment value and no segment value is define in the Customizing activities Define Segment Value for Automatic Determination and Define Segmentation Structures, the system proposes BLANK as the default segment value.

Scenario 3: DC2 supplies to DC1

The supplying plant is not relevant for segmentation. You define the stock segment in the stock transport order. As the supplying plant is not relevant for segmentation, the system does not determine the requirement segment and the material is posted.

The following image illustrates the transfer of stock between plants with different segmentation strategies:

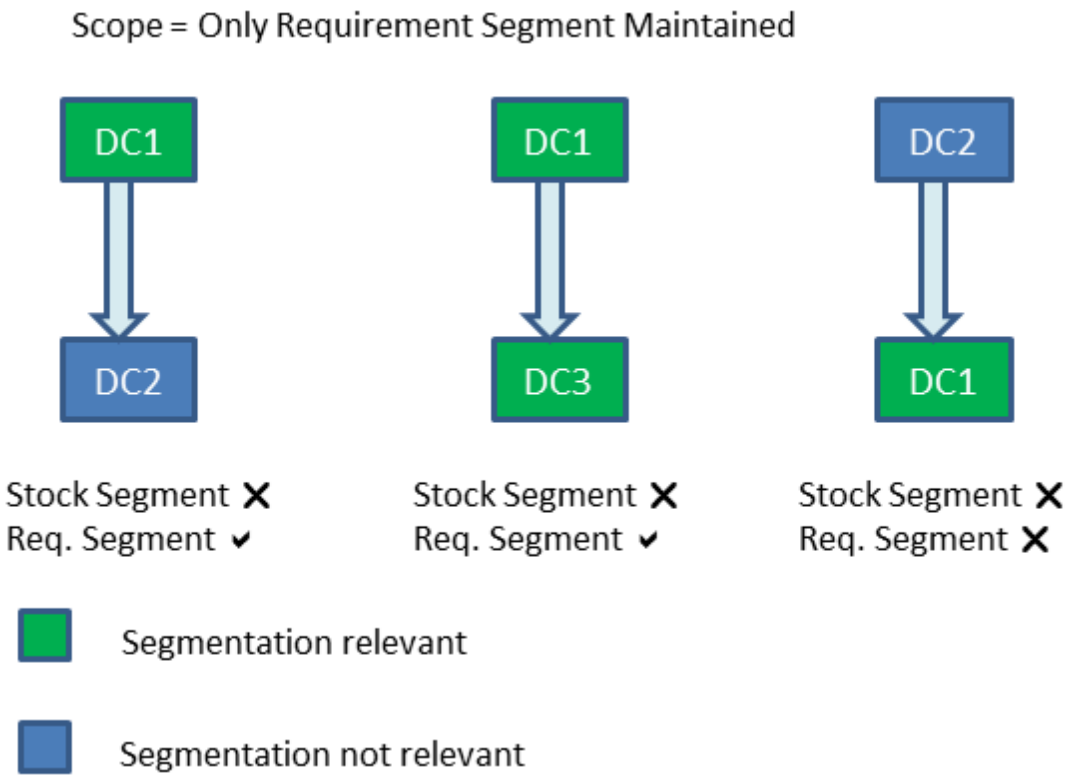


Scope is maintained as Only Requirement Segment is Maintained in the segmentation strategy

Scenario 4: DC3 supplies to DC1

Both the supplying plant and the receiving plant are relevant for segmentation. However, as the scope is Only Requirement Segment is Maintained in the segmentation strategy, you do not define the stock segment. The system determines the requirement segment based on the segment value assignment you define in the Customizing activity Define Segment Value for Automatic Determination under Logistics – General > Segmentation.

The following image illustrates the transfer of stock between plants with different segmentation strategies:



MRP Area

Definition

The MRP area represents an organizational unit for which material requirements planning is carried out independently. The following are the three different types of MRP area:

- Plant MRP Area
- MRP Areas for Storage Locations
- MRP Areas for Subcontractors

Structure

You can define the MRP areas in Customizing for MRP. You can thereby assign the following to an MRP area:

- One or more storage locations (for example, you want to carry out planning for a particular assembly line and therefore assign a production storage location to the MRP area).
- Subcontractor
- You can assign the various MRP areas to the materials that are relevant for segmentation in the material master. For this, you create an MRP area segment for a material for every MRP area, in which it is used. In this MRP area segment, you can define MRP parameters such as the lot-size or MRP type. This allows you to plan the materials with segmentation differently in the MRP area than in the plant MRP area.

i Note

If you have not assigned a material that is relevant for segmentation to an MRP area (you have not created an MRP area segment in the material master) the material will continue to be planned in the plant MRP area only. If you have assigned an MRP area to it, the system can plan it in the plant MRP area and in the assigned MRP area.

Integration

You have activated the business function `LOG_SEGMENTATION_03`.

- The system supports MRP areas for materials that are relevant for segmentation in MRP Live (`MD01N`).
- You can view the stock/requirements situation for materials relevant for segmentation in transaction `MD04` for the selected MRP area.

More Information

For more information about the MRP areas in the standard ERP system, see **MRP Areas**.

Segment Level MRP Data

Use

You can maintain the following MRP relevant parameters for the individual segments of the material in the material master.

- Minimum Lot Sizes
- Safety Stock
- Safety Time Indicator
- Valid From (applicable for ATP/MRP Status)
- Rounding Value
- Purchasing Group
- Safety Time
- Planned Delivery Time
- In-house Production Time
- Total Replenishment Lead Time
- ATP/MRP Status for Material and Segment

Prerequisites

You have activated the business function `LOG_SEGMENTATION_03`.

More Information

Lot-Sizes

Safety Stock

Rounding Value

Lot-Sizes in MRP

Use

During MRP, the system creates procurement proposals according to the lot-size you have defined if there is a material shortage. By determining lot-size, you can plan optimal quantities for your procurement proposals. The planning of lot-size influences the planning of the available capacity, and the planning of production or procurement costs. For materials that are relevant for segmentation, you can plan the quantities to be produced for the individual segments of the material.

- If you have maintained the minimum lot size for the individual segments of the material, the system considers this value in the lot-sizing procedures.
- If you have not maintained the minimum lot-size for the individual segments of the material, the system considers the minimum lot-size maintained for the material and after determining the distribution ratio, adjusts the procurement quantity.

Example

The following materials are configured in material master:

Material	Segment	Minimum lot size
SHIRT		22
SHIRT	WHS	40
SHIRT	RET	30

After executing the MRP run, the system adjusts the procurement quantity as follows according to the minimum lot-size you specified in the material master:

Item No.	Material	Segment	Original Qty	Adjusted Qty
10	SHIRT	RET	12	30
20	SHIRT	WHS	23	40
30	SHIRT	ECO	3	3

- For items no. 10 and 20, as you have maintained the minimum lot size for the individual segment the system adjusts the quantity to 30 and 40, respectively, after executing the lot-size procedure.
- For item no. 30, as you have not maintained the minimum lot-size for the segment, the system retains the original quantity.

If you have not maintained the minimum lot-size for the segments, the system performs the lot-sizing procedure at document level after determining the requirements distribution ratio. As you have not maintained minimum lot-size for the segments `COM` and `SEG`, the system considers the minimum lot-size maintained for the material and after determining the distribution ratio, adjusts the procurement quantity to 16 and 6, summing up to 22.

Item No.	Material	Segment	Original Qty	Adjusted Qty
10	SHIRT	COM	8	16
20	SHIRT	SEG	3	6

More Information

Segment Level MRP Data

Safety Stock

Use

During the MRP run, the system must consider the safety stock requirements. You can maintain safety stock for the following in material master:

- Individual segment
- Material

Features

- If you have maintained a safety stock for the individual segment of the material, the system creates the safety stock requirements at segment level.
- If you have not maintained a safety stock for the individual segment of the material, the system creates the safety stock requirements at the segment level using equal distribution ratio.

Example

You have maintained a safety stock as follows in the material master:

Material	Segment	Safety Stock
SHIRT		200
SHIRT	WHS	22
SHIRT	RET	30
SHIRT	ECO	40

The following table shows that the system created the safety stock requirements at segment level:

Material	Segment	Quantity
SHIRT	WHS	22
SHIRT	RET	30
SHIRT	ECO	40

The following table shows that the system created the safety stock requirements at the material level:

Material	Segment	Quantity
SHIRT	WHS	66
SHIRT	RET	67
SHIRT	ECO	66

More Information

Segment Level MRP Data

Rounding Value

Use

During MRP, when new procurement proposals are created, the quantity of the items must be rounded according to a rounding value.

- If you have maintained the rounding value for the individual segment of the material, the system rounds the quantity of the items to the rounding value maintained at the segment level.

- If you have not maintained the rounding value for the individual segment of the material, the system considers the rounding value maintained at the material level and after determining the distribution ratio, adjusts the quantity at the segment level.

Example

You have maintained a rounding value as follows in the material master:

Material	Segment	Rounding Value
SHIRT		50
SHIRT	WHS	22
SHIRT	RET	30

The following table shows that the system rounds the quantity of the items to the rounding value maintained at segment level:

Material	Segment	Original Quantity	Rounded Quantity
SHIRT	WHS	23	44
SHIRT	RET	30	30
SHIRT	ECO	40	40

The following table shows that the system considers the rounding value maintained at material level and after determining the distribution ratio, adjusts the quantity at segment level:

Material	Segment	Original Quantity	Rounded Quantity
SHIRT	COM	20	33
SHIRT	ECO	10	17

More Information

Segment Level MRP Data

Segmentation: Availability Check

Use

If the scope is **Requirement and Stock Segment Maintained** in the segmentation strategy, the system checks the material availability at the requirement segment level for materials with segmentation.

If one of the following is true, the system does not check the material availability at segment level :

- The segmentation strategy is assigned to a material but the scope is **Only Requirement Segments Maintained**
- The segmentation strategy is not assigned to the material

Integration

The availability check with segmentation is enabled in the following processes:

- Creating or changing sales orders
- Creating or changing stock transport orders
- Creating or changing outbound deliveries
- Rescheduling of sales and stock transfer documents
- Availability overview (`C009`)
- Backorder processing (`C006`)

Prerequisites

- You have created a segmentation structure and a segmentation strategy with scope **Requirement and Stock Segment Maintained** in Customizing for **Logistics – General** > **Segmentation** > **Define Segmentation Structures** and **Define Segmentation Strategies** respectively
- You have created and assigned the segmentation strategy to a material in the material master
- You have defined the mapping rule for ATP in the **ATP: Requirement/Stock Allocation** table in Customizing for **Define Segmentation Strategies** under **Logistics – General** > **Segmentation**

Activities

The availability check uses the mapping rules defined in Customizing for **ATP: Requirement/Stock Allocation** in the segmentation strategy. Here, you must have defined the following:

- **Pool Segment**: Specifies an individual or a combination of valid requirement or stock segment values
- **Seg. Check Level**: Specifies the level at which the system performs the availability check

i Note

- The value **Blank** as a valid segment is considered for an availability check with **Seg. Check Level** as blank.
- The value **Blank** as a pool segment is considered for an availability with **Seg. Check Level** as blank.
- You cannot maintain different **Seg. Check Level** for a pool segment.
- After changing the customizing, you have to perform rescheduling using transaction `V_V2`.

Safety Stock

If the safety stock is to be excluded from the availability check, the safety stock at material level is distributed equally across the valid segment combinations. Depending on the **Seg. Check Level**, safety stock is reduced from plant stock or the segment-level plant stock.

→ Recommendation

To ensure consistent results, SAP recommends that you maintain similar mappings for MRP and ATP.

Example

The following mapping rule is defined for ATP in the segmentation strategy:

Segment value	Pool Segment	Seg. Check Level
A1	A1A2	1
A2	A1A2	1
A3		

- Issues/receipts from segments A1 and A2 are mapped to pool segment A1A2 and the system performs the availability check at multi-level.
- Issues/receipts from segment A3 are mapped to a blank pool segment and the system performs the availability check at organizational level (segmentation is not considered).

	Quantity Received	Issue Quantity	ATP Quantity
Plant Stock	1000 pc.		300 pc.
Receipt 1 (Segment A1)	500 pc.		
Issue 1 (Segment A3)		1200 pc.	
Segment A1	800 pc.		
Issue 2 (Segment A1)		500 pc.	
Segment A2	200 pc.		

- A plant stock of 1000 pieces exists (800 pieces from segment A1 and 200 pieces from segment A2)
- A future receipt of 500 pieces for Segment A1 exists
- For a requirement of segment A3 (Issue 1) amounting to 1200 pieces, the system performs the availability check at organizational level and segmentation is not considered. Issue 1 can be completely covered by 500 pieces from receipt 1 (segment A1) and 700 pieces from plant stock. The remaining plant stock can provide an ATP quantity of 300 pieces for future requirements.
- Later, for a requirement of segment A1 (Issue 2) amounting to 500 pieces, the system performs the availability check at multi-level (plant and plant + segment). However, issue 2 cannot be completely covered, as the stock available at segment A1 is already consumed by Issue 1. Therefore, only 300 pieces available at plant stock level is confirmed for Issue 2.

Vertical Stock Protection in ATP

Use

In Vertical stock protection, your market areas have a strict priority. Higher priority areas can sell stock that was planned for equal or less important area. Vertical stock protection ensures that important areas get stock, even if it results in a shortage for less important areas.

Vertical stock protection is particularly useful if some of your market areas are much more important than others, and you want to ensure that they can get as much supply as they can sell.

Prerequisites

In addition to the steps described under the Prerequisites of Stock Protection, you must define the priority for requirements segments. Also, set the stock protection flag in the segmentation structure for all requirement segments that you want to protect.

Features

When checking availability for an order, the system considers unconsumed PIRs with a higher priority similar to the sales orders confirmed earlier.

Available quantity = Supply – Confirmed quantity of existing orders – Unconsumed quantity of PIRs with higher priority

Contracts and call-off orders in vertical stock protection

If your sales contracts are relevant to requirements and you have activated the availability check for them, they will participate in stock protection like normal sales orders.

Call-off sales orders can have a different requirement date, therefore a new stock protection check is carried out. If a call-of sales order is created from a partially or fully allocated contract, it simply takes over the allocation and does not participate in the stock protection check.

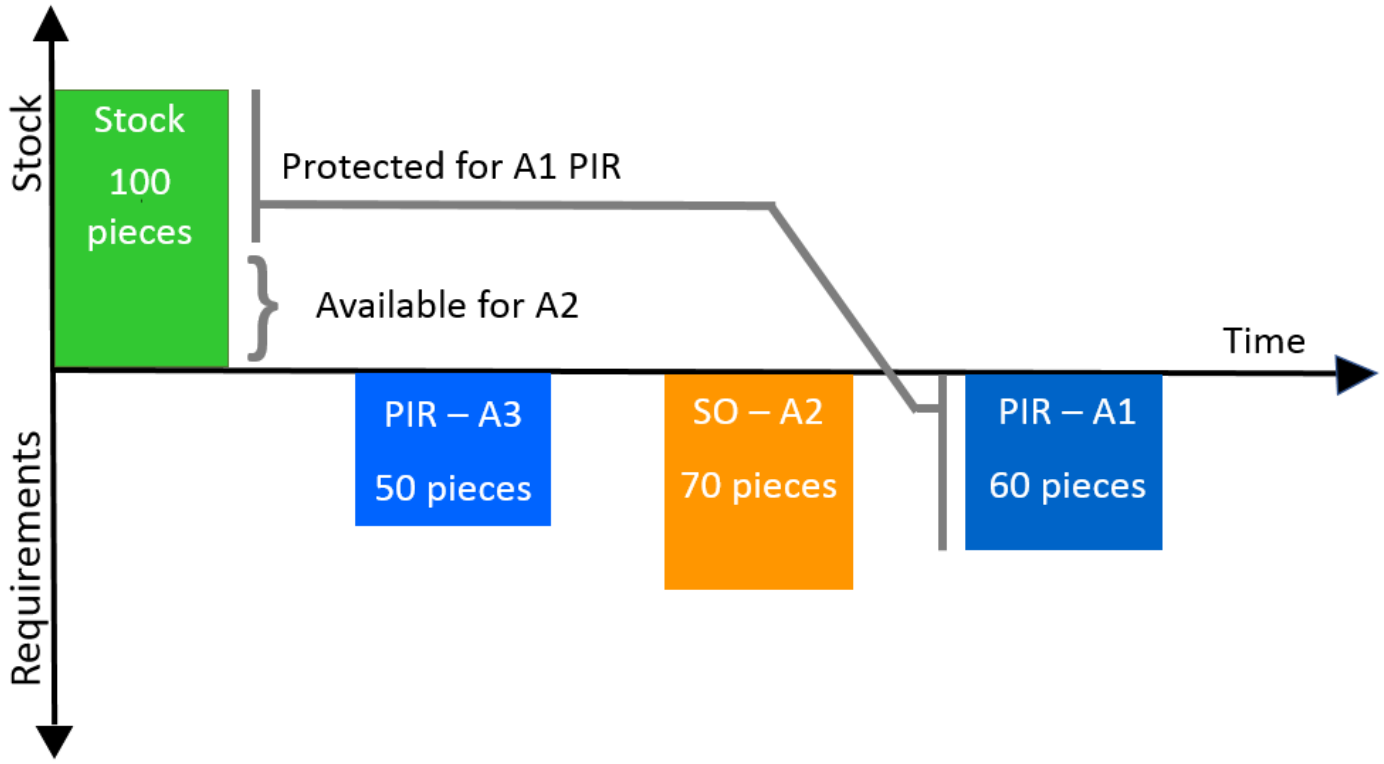
Example

Let's consider the following priorities in the segmentation strategy:

Segmentation Strategy	Segmentation Value	Short Text	Priority Sequence	Stock Protection
Strategy 1	A1	Segment A1	1	X
Strategy 1	A2	Segment A2	2	X
Strategy 1	A3	Segment A3	3	X

A plant stock of 100 pieces is available.

Segment A1 has the highest priority. You can protect the stock for this segment by entering a PIR of 60 pieces in the system.



If a requirement is received for segment A2, an availability check only confirms 40 pieces, because it protects 60 units for the higher priority A1 PIR.

i Note

The system does not protect any quantity for the A3 PIR, because A3 has the lowest priority.

Related Information

Sales Contracts

Stock Protection Up to Plan in ATP

Use

Similar to Vertical stock protection, your areas have a priority. Higher priority areas can always sell the quantity what they planned, even if it results in a shortage for less important areas. However, selling more than planned is only possible, if there is excess supply.

Protection up to plan is particularly useful if some of your market areas are more important than others, but you want to ensure that less important ones get their fair share of supply, too.

Prerequisites

In addition to the steps under the Prerequisites section of Stock Protection, you must have:

- Maintained PIR consumption rules in the segmentation strategy. While maintaining consumption rules, ensure that the priority deciding characteristic is not 1:N.

- Defined the priority for requirements segments and selected the stock protection checkbox in the segmentation structure for all requirement segments that you want to protect.
- Maintained PIR consumption parameters in the article master.

Features

When checking availability for an order, the system first calculates whether the order quantity was planned. This is done by simulating PIR consumption:

- The quantity that the order would consume is the **planned sales quantity**.
- The remaining quantity of the order that could not consume any PIRs is the **unplanned sales quantity**.

Planned and unplanned sales quantities are checked with different priorities. The planned sales quantity has the same priority as the requirement segment of the order item, while the unplanned sales quantity has the lowest priority. This means, that the unplanned portion can be confirmed only if all PIRs relevant to stock protection can be fully confirmed.

The availability check calculation is performed as follows:

1. The supply is reduced by the confirmed quantity of existing orders.
2. The remaining quantity is further reduced by unconsumed quantity of higher priority PIRs.
3. Availability is checked for the **planned** quantity of the current order.
4. The remaining quantity is further reduced by unconsumed quantity of all remaining stock protection relevant PIRs of same and lower priority
5. Availability is checked for the **unplanned** quantity of the current order.

Total availability of the order item is the sum of (3) and (5).

Contracts and call-off orders in stock protection up to plan

If your sales contracts are relevant to requirements and you have activated the availability check for them, they can participate in stock protection like normal sales orders. You can activate stock protection for the contracts in Customizing where you maintain stock protection type.

Call-off sales orders can have a different requirement date, therefore a new stock protection check is carried out. If a call-of sales order is created from partially or fully allocated contract, it simply takes over the allocation and does not participate in stock protection check.

Example

Let's consider the following priorities in segmentation strategy:

Segmentation Strategy	Segmentation Value	Short Text	Priority
Strategy 1	A1	Segment A1	1
Strategy 1	A2	Segment A2	2
Strategy 1	A3	Segment A3	3

You plan 30 units in each requirement segment, that is, 90 units in total.

Stock protection up to plan in shortage situation

Assume that the supply is 80 units, which is not enough to cover the demand.

If a sales order with segment A2 is received for 50 units, the following calculation is made:

Calculation step	Supply/Requirement quantity	Rolling sum
Initial supply	80	80
Higher priority open demand	-30	50
Planned portion of order	-30	20
All other open demand	-30	-10
Unplanned portion of order	-20	-30

This means, that the planned portion of the sales order – 30 units – is confirmed. Because of the shortage, the unplanned portion is rejected, leaving some stock for the A3 planned demand.

Stock protection up to plan in excess situation

Let's assume that the supply is 100 units, which is 10 units more than total demand.

If a sales order with segment A2 is received for 50 units, the following calculation is made:

Calculation step	Supply/Requirement quantity	Rolling sum
Initial supply	100	100
Higher priority open demand	-30	70
Planned portion of order	-30	40
All other open demand	-30	10
Unplanned portion of order	-20	-10

This means that 40 units will be confirmed, 30 units as the planned quantity and 10 unplanned units that make use of excess stock. Please note, that lower priority segments are fully protected in this case.

Related Information

Sales Contracts

Horizontal Stock Protection in ATP

Use

In Horizontal stock protection, the system protects supply for demand plans of all unrelated areas. To prevent deadlocks in stock shortage, orders that were fully planned don't participate in stock protection.

Horizontal stock protection can be useful, if your market areas are equally important and you want to prevent cannibalization of supply.

Prerequisites

In addition to the steps under the Prerequisites section of Stock Protection, you must have:

- Maintained PIR consumption rules in the segmentation strategy.
- Maintained PIR consumption parameters in the article master.

Although it is technically not a prerequisite, you should ensure that demand and supply are synchronized. If demand plans exceed supply, stock protection may not be effective, and some of your market areas can run into shortages.

ℹ Note

Segment priorities don't play any role in this strategy. Protection is controlled by consumption rules.

Contracts and call-off orders in horizontal stock protection

Sales contracts never participate in horizontal stock protection.

Call-off sales orders created from unallocated contracts are treated exactly as normal sales orders. If a call-of sales order is created from a partially or fully allocated contract, it simply takes over the allocation and does not participate in the stock protection check.

Features

When checking the availability for an order, the system first calculates whether the order quantity was planned. This is done by simulating PIR consumption.

If the order is fully consuming PIRs, availability is checked only by the normal ATP logic.

If the order can't fully consume PIRs, that is, it exceed the demand plan, stock protection is active for the unplanned portion of the order. In this case, the system protects stock for all unrelated demand plans. A PIR is considered to be an unrelated demand plan, if either of the following is true:

- Based on the consumption rules defined in segmentation strategy, the requirement segment of the order can't consume the segment of the PIR.
- Based on the consumption rules defined in the article, the PIR is outside the consumption window of the order.

The availability check calculation is performed as follows:

1. The supply is reduced by the confirmed quantity of existing orders.
2. Availability is checked for the planned portion of the current order.
3. The remaining quantity is further reduced by the unconsumed quantity of all unrelated PIRs.
4. Availability is checked for the unplanned portion of the current order.

Example

Let's consider the following consumption rules in the segmentation strategy:

Sales order requirement segment	Consumption sequence	PIR requirement segment
A	1	A
A	2	B
B	1	B
C	1	C

You plan 30 units in each requirement segment, that is, 90 units in total. Your supply is 100 units.

Protection for equally important market areas

Assume that you receive a sales order for 50 units in segment C. Because it exceeds the planned 30 units, horizontal stock protection is active.

A sales order with requirement segment C can consume only PIRs in segment C. This means that the system will first give a higher priority to the planned quantity for segment C, then to A and B, and then to the unplanned quantity for segment C.

Calculation step	Supply/Requirement quantity	Rolling sum
Initial supply	100	100
Planned portion of order	-30	70
PIRs that the current SO can't consume (Unrelated PIRs)	-60	10
Unplanned portion of sales order	-20	-10

Taking stock planned for less important market areas

Let's assume that you receive a sales order for 50 units in segment A. Although the PIR for segment A is only 30 units, the order can also consume PIRs for segment B. This means, that the orders for segment A can take stock that was originally planned for segment B.

A sales order with requirement segment A can't consume PIRs in segment C. This means that the system will protect stock for segment C.

Related Information

Sales Contracts

Basic Data

Use

The basic data in the area of production planning and control include the bill of material and the routing. When you define the basic data for a material associated with segmentation characteristics, you must also specify additional data for the material.

In a bill of materials definition, you specify which component(s) are included in the bill of material when the material gets associated with a specific segment value. In a routing or production task list definition, you must specify which set of task list

(Group & Group Counter) is relevant for production of the segmented material when the material is associated with a specific segment value.

More Information

- Bill of Materials
- Routing

Bill of Materials

Use

In the SAP System, you can use bill of material (BOM) to represent different objects (such as materials, equipment, functional location BOMs, and documents) and to maintain object-specific data. Before you create a BOM with the component parts of an material, the material must have a valid master record in your system.

Features

A BOM consists of a header material and a structure that holds the components which comprise a material. Using segmentation, materials are associated with certain characteristics such as **Quality**, and **Country of Origin**. Material of different segmentation values, such as high quality material or a low quality material can have different components or different component quantity.

The enhancement to the BOM definition allows you to specify which components are applicable or included/excluded in the BOM structure. During the BOM usage and explosion (planned or production order), based on the stock segment specified in the order, the right set of components and their quantity are determined.

The following kinds of bill of material have been enhanced to support the segmentation:

- Material Bill of Materials
- Sales Order BOM

Constraints

The following limitations apply for bill of materials for materials with segmentation:

- Creation of Variant BOM for materials with segmentation is not supported.
- The Engineering Workbench (`CEWB`) for BOM and routing for materials with segmentation is not supported.
- For multiple plants, the segmentation strategy can either be blank for one plant and cannot be blank for another plant. The allocation of BOM to multiple plants should be checked before the assignment. This assignment is not checked.

Material Bill of Material

Use

BOM is a formally structured list of components that make up a product or assembly. The list contains the object number of each component, together with the quantity and unit of measure. Each finished product has its own BOM. Segmentation relevant materials contain information about country of origin and quality and are specified as segment values. The BOM structure can vary

between materials with different segment values. The subsequent sections explain how to specify segment value based BOM structure variations.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.
- You have created the characteristics for segmentation using transaction `CT04`. See Maintain Segmentation Relevance for Functions for more details.

  You have selected the application relevance for **BOM** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance** >).
- You have defined and maintained segmentation structure and strategy for a material and have assigned it to the header material. See (Segmentation Structure and Segmentation Strategy).

Features

The BOM maintenance programs have been enhanced to define and maintain the structure specification. You can create, modify, and display a BOM using the transactions `CS01`, `CS02`, and `CS03` respectively

The maintenance program provides a new ( ) button on the header screen to specify whether a component needs to be included or excluded based on segmentation. You can select a component and click the button to view the possible segment values. Select the **Include** checkbox for a segment value if you want to include the component in the BOM structure (only when that segment value is specified in planning/production). If the **Include** checkbox is not selected for any of the segment values, by default for all segment values, the component is included. For any specific segment value, if the component needs to be excluded, then select the **Exclude** checkbox for the segment value.

Example

You want to create a BOM for a jacket in quality 1 and for customer segment A. For this jacket you need the following components: Fabric, zipper and buttons. Components in the BOM are dependent on the segment value of the header material (jacket). You have assigned the following stock segments to your jacket: **001A**, **002B** and **003C**. You either exclude both segments **002B** and **003C** from the bill of material or you include segment **001A** in the BOM.

More Information

- Creating a Material BOM
- Changing a Material BOM
- Displaying a Material BOM

Creating a Material BOM

After you a create a BOM for different segmentation values, the same BOM is listed in the production order based on different stock segments. You can view only those components which were included when BOM was created for a particular stock segment.

Follow these steps to create a material BOM:

1. On the SAP Easy Access Menu, choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Bill of Material** > **Create** > and enter the required data. On this screen, you maintain data that identifies the bill of material, as well effectivity data.

Alternatively, you can use the transaction `CS01` to create a material BOM.

2. In the **Material** field, enter the material for which you want to create the BOM. If you do not know the material number, you can use the help (F4).

3. Enter a **Plant**.

   If you want the BOM to be effective in a specific plant, enter the plant. You can allocate the BOM to additional plants if required.

4. Enter a **BOM Usage**.

   If a parameter is defined in your user master record as a default value for the **Usage** field, the system enters this default. When you select a BOM usage, you define the maximum range of item statuses that can be assigned to all items in the BOM. For each usage, you can only maintain certain item status indicators.

5. When you create a simple BOM, you do not need to enter an alternative.

   - If you do not make an entry when you first create a BOM, the system automatically creates alternative 01
   - If you already know when you first create a BOM that you want to extend the BOM to make a multiple BOM, and that this alternative is not alternative 01, enter an alphanumeric value.

6. Enter a change number if you want to associate the modified structure to an identifier. You can later view that change by specifying the change number.

7. Press **Enter** to go the **Item Overview** page.

8. Enter the **Item Category** of the BOM. This field is used to control field selection, default values for BOM maintenance, triggering of specific system activities, and so on.

9. In the **Component** field, enter the material number of the components that you specify as part of the BOM.

10. Enter the **Quantity** of the component.

11. Enter the **Unit** for the component quantity.

12. Select one or more components and click the button to enter the segmentation values for a material.

13. In the **Component Selection by Segmentation** screen, enter the following fields:

    a. **Segmentation Value**: Click the **Fast Entry** button to enter a segmentation value.

    b. **Include Segment Value**: Select this checkbox if you want to include the segment value in the header of a material. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.

    ### ℹ Note

    You can also double-click on a component to add a segment value.

    This field is used to set the requirement segment for a component in a planned and production order (during BOM explosion) and is enabled based on the following conditions:

    - Header material is not segmentation relevant.
    - Scope of segmentation strategy for header material is 2 (only requirement segmentation).
    - Scope of segmentation strategy for a component material is 1 (stock and requirement segmentation) or 2 (only requirement segmentation).
    - Scope of segmentation strategy for a header component is 1 (stock and requirement segmentation) and the quantity is negative (if the output material is a by-product).

c. **Exclude Segment Value**: Select this checkbox if you want to exclude the segment value in the header of a material.

14. Click **Accept** to confirm the changes.

Changing a Material BOM

Follow these steps to change a material BOM:

1. On the SAP Easy Access Menu, choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Bill of Material** > **Change** >. The **Change Material BOM: Initial Screen** screen appears. On this screen, you maintain data that identifies the bill of material, as well effectivity data.

   Alternatively, you can use the transaction `CS02` to change a material BOM.

2. In the **Material** field, enter the material for which you want to create the BOM. If you do not know the material number, you can look for the material using the entry help (F4).

3. Enter a **Plant**.

   If you want the BOM to be effective in a specific plant, enter the plant. You can allocate the BOM to additional plants later on under certain circumstances

4. Enter a **BOM Usage**.

   If a parameter is defined in your user master record as a default value for the **Usage** field, the system enters this default. When you select a BOM usage, you define the maximum range of item statuses that can be assigned to all items in the BOM. For each usage, you can only maintain certain item status indicators.

5. If you do not make an entry when you first create a BOM, the system automatically creates alternative 01

6. Enter a change number if you want to associate the modified structure to an identifier. You can later view that change by specifying the change number.

7. Press Enter to go the **Item Overview** page.

8. Enter the **Item Category** of the BOM. This field is used to control field selection, default values for BOM maintenance, triggering of specific system activities, and so on.

9. In the **Component** field, enter the material number of the components that you specify as part of the BOM.

10. Enter the **Quantity** of the component.

11. Enter the **Unit** for the component quantity.

12. Select one or more components and click the button in the tool bar to enter the segmentation values for a material.

13. In the **Component Selection by Segmentation** screen, enter the following fields:

    a. Segmentation Value: Click the **Fast Entry** button to enter a segmentation value.

    b. **Include Segment Value**: Select this checkbox if you want to include the segment value in the header of a material. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.

    > **Note**
    >
    > You can also double-click on a component to add a segment value.
    >
    > This field is used to set the requirement segment for a component in a planned and production order (during BOM explosion) and is enabled based on the following conditions:

- Header material is not segmentation relevant.
- Scope of segmentation strategy for header material is 2 (only requirement segmentation).
- Scope of segmentation strategy for a component material is 1 (stock and requirement segmentation) or 2 (only requirement segmentation).
- Scope of segmentation strategy for a header component is 1 (stock and requirement segmentation) and the quantity is negative (if the output material is a by-product).

c. **Exclude Segment Value**: Select this checkbox if you want to exclude the segment value in the header of a material.

14. Click the **Accept** button to confirm the changes.

Displaying a Material BOM

Follow these steps to display a material BOM:

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Bill of Material** > **Display**. The **Create Material BOM: Initial Screen** appears. On this screen, you can view the data that identifies the bill of material, as well effectivity data.

   Alternatively, you can use the transaction `CS03` to display a material BOM.

2. A parameter is defined in your user master record as a default value for the **Usage** field, the system enters this parameter or the last usage entered as a default value for display.

   In the standard R/3 System, the default valid-from date and valid-to date are the current date.

3. Enter the following data that identifies the BOM on the initial screen:
   - In the **Material** field, enter the material whose BOM you want to display.
   - Enter the **Plant** for which you want to display the BOM. If you want to display a group BOM, do not make an entry in the Plant field.
   - If there is no value or an incorrect default value in the **BOM Usage** field, enter the usage for which you want to display the BOM.
   - You do not need to make an entry in the **Alternative BOM** field for a simple material BOM.
4. Define the validity period.
5. Confirm your entries to go to the **Item Overview** page.
6. Click the button to view the segment values of the material.

Displaying Change Documents From the BOM

Use

When you are processing a BOM with segment values, you can display change documents at any time. You follow the same steps as for calling the function from the main menu. However, you do not need to enter the data for identifying the BOM.

Change documents are created for the following BOM categories:

- Material BOM
- Sales Order BOM

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.
- You have created the characteristics for segmentation using transaction `CT04`. See Maintain Segmentation Relevance for Functions for more details.

  You have selected the application relevance for **BOM** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance**).
- You have defined and maintained segmentation structure and strategy for a material and have assigned it to the header material. See (Segmentation Structure and Segmentation Strategy).
- If you want to extend the existing characteristics for application Relevance of BOM and ROUTING, then execute the report **SGT_SETRELEVANCE** (using transaction `SE38`). All associated characteristics in the structure without relevancy will be checked and extended or application relevance. This is a one-time report executed for a particular characteristic and therefore, all the corresponding relevancies needs to be selected while executing this report.

Activities

Follow these steps to display change documents from a BOM:

1. On the SAP Easy Access menu, choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Reporting** > **Change Documents** > **<BOM Category>**.

   Alternatively, you can use the transaction `CS80` for material BOM and `CS82` for sales order BOM to display change documents.
2. On the initial screen, enter the BOM for which you want to display change document, enter the following fields:
   - Material
   - Plant
   - Usage (optional)
3. Confirm your entries.
   - If the BOM data you enter identifies one individual BOM, you see the **Display change documents: <BOM Category>** screen.
   - If BOMs of different usages exist for the Material and Plant you entered, and you did not enter a usage on the initial screen, you see a list of the BOMs found.
   - If BOMs exist in different plants for the Material and Usage you entered, and you did not enter a plant on the initial screen, you see a list of the BOMs found.
4. Select the BOM(s) whose change documents you want to display and choose **BOM**. The **Display change documents: <BOM Category>** appears.
5. Enter the selection date. The system lists the change documents generated after this selection date.
6. The change documents are displayed with the following fields:
   - Document Number

- Date and Time
- Changed by
- Object ID: From this list, you can display individual change documents. To do this, place the cursor on a line containing the document you require and choose **Display Documents**. You see the document you selected.

7. Double click the document number to view the BOM details. The following values appear:
   - Table: Displays which particular table/tables are updated
   - Table Key: System generated value which is combination of Object ID and Item ID.
   - Field Changed: Displays which particular field has been changed in BOM.
   - Action: Displays the action performed in the BOM like “Entry” , “Change” and “Delete”.
   - Field Description: Description of the changed field.
   - Old Value: Value which exists earlier before the change.
   - New Value: Value which exists after the change.
8. For material with segmentation values, you can view the changes when a BOM is changed during the following scenarios:
   - Creation of BOM: You add component details and a segment value for each component.
   - Changing a BOM at the component level: Displays the changes in the values at the component level when quantity was changed.
   - Changing a BOM at the segment level: Displays the change in the segment values at the component level when new segment values were included. The BOM changes are also displayed if you have modified the BOM to include or exclude segments values for a component.

Material BOM Browser

Use

You use the material BOM browser to navigate in a multi-level material BOM. The material BOM browser gives you a quick overview of BOM headers with segment values and items that is displayed structurally. You can show or hide those branches in this structure.

Activities

Follow these steps to open a material BOM browser:

1. On the SAP Easy Access menu, choose **Logistics** > **Production** > **Bills of Materials** > **Bill of Material** > **Material BOM** > **Multilevel** > **Material BOM Browser**. The initial screen of material BOM browser appears.

   Alternatively, you can use the transaction `CSMB` to open a material BOM browser.

2. Enter the data for BOM, the BOM application and stock segment.

   Select the **Ignore Seg.** checkbox if you do not want to include stock segment in the material BOM header.

3. Enter a value for **Max Expl Level** checkbox if you are working with large BOMs.
4. Click to specify the settings for explosion.

5. Define a view. For more information on defining views for explosion, see the application help for **Defining a View for BOM Explosion** in SAP help portal under **SAP ERP** > **SAP ERP Central Component** > **Logistics** > **Production Planning and Control (PP)** > **Basic Data (PP-BD)** > **BOMs (PP-BD-BOM)** > **BOM Reporting Functions** > **BOM Explosion** > **Selection Criteria for BOM Explosion**.

6. Start the material BOM browser.

7. Based on the stock segment value provided on the selection screen and on the components included or excluded in the BOM definition, the BOM browser displays the included components.

   If the **Ignore Seg.** checkbook is selected, then the BOM browser will behave as it appears for standard and does not include segmentation values.

Sales Order Bill of Material

Use

An order BOM is generated from a material BOM and is modified for a specific sales order with segmentation values. The material BOM, however, does not change.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.
- You have created the characteristics for segmentation using transaction `CT04`.

  You have selected the application relevance for **BOM** and **ROU** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance**).

-
  - You have activated the business function `LOG_SEGMENTATION_02`.
  - You have created the characteristics for segmentation using transaction `CT04`. See Maintain Segmentation Relevance for Functions for more details.

    You have selected the application relevance for **BOM** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance**).
  - You have defined and maintained segmentation structure and strategy for a material and have assigned it to the header material. See (Segmentation Structure and Segmentation Strategy).

Features

You use an order BOM if you want to change the bill of material for a material, to meet the requirements of a single sales order. Since the changes only affect this one sales order, you do not want to change the material BOM. However, you do not want to maintain any new material master record.

Structure An order BOM has the same structure as the material BOM from which it is generated.

A BOM that is created with reference to a specific sales order is described as an order BOM. A sales order number, a sales order item and a material number uniquely identify an order BOM.

Both material BOMs and order BOMs can be contained in the multi-level BOM for the material that you entered in a sales order item. Material BOMs are used for order-independent assemblies, in other words, assemblies that are not changed.

More Information

- Creating a Sales Order BOM
- Changing a Sales Order BOM
- Displaying a Sales Order BOM

Creating a Sales Order BOM

Use

You can create a sales order BOM if you want to make customer-specific changes to the BOM of a segmentation relevant material that you configure in the sales order.

Procedure

Follow these steps to create a sales order BOM:

1. Choose **Logistics** > **Production** > **Master Data** > **Bill of Material** > **Order BOM** > **Create**. The **Create Order BOM** initial screen appears.

   Alternatively, you can use the transaction `CS61` to create a order BOM.

2. Enter the sales order number, the order item, the material number and the BOM usage.

3. If you want to use a material BOM, or an existing order BOM to copy from, go to step 4. If you want to create an order BOM without a reference, click ✔ and go to step 7.

4. Click **Copy Order BOM** or **Copy Material BOM**. The Copy material BOM or Copy order BOM dialog box appears.

5. Enter the data required and click ✔.

   The item overview appears, for the bill of material you are using as a reference.

6. Select the items you want to copy to the order BOM and click ✔.

   The item overview for your new order BOM appears. The items you transferred are displayed in the item overview.

7. Enter the **Item Category** of the BOM. This field is used to control field selection, default values for BOM maintenance, triggering of specific system activities, and so on.

8. In the **Component** field, enter the material number of the components that you specify as part of the BOM.

9. Enter the **Quantity** of the component.

10. Enter the **Unit** for the component quantity.

11. Select one or more components and click the button in the tool bar to enter the segmentation values for a material.

12. In the **Component Selection by Segmentation** screen, enter the following fields:

    a. Segmentation Value: Click the **Fast Entry** button to enter a segmentation value.

    ℹ Note

    This field is used to set the requirement segment for a component in a planned and production order (during BOM explosion) and is enabled based on the following conditions:

    - Header material is not segmentation relevant.

- Scope of segmentation strategy for header material is 2 (only requirement segmentation).
- Scope of segmentation strategy for a component material is 1 (stock and requirement segmentation) or 2 (only requirement segmentation).
- Scope of segmentation strategy for a header component is 1 (stock and requirement segmentation) and the quantity is negative (if the output material is a by-product).

b. **Include Segment Value**: Select this checkbox if you want to include the segment value in the header of a material. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.

c. **Exclude Segment Value**: Select this checkbox if you want to exclude the segment value in the header of a material.

13. Click the **Accept** button to confirm the changes.

Changing a Sales Order BOM

Follow these steps to change a sales order BOM:

1. Choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Order BOM** > **Change** . The **Change Order BOM** initial screen appears.

   Alternatively, you can use the transaction `CS62` to change a order BOM.

2. Enter the following data that identifies the BOM on the initial screen:
   - Sales Order
   - Item
   - Material
   - BOM Usage
3. Confirm your entries to go to the **Item Overview** page.
4. Enter the **Item Category** of the BOM. This field is used to control field selection, default values for BOM maintenance, triggering of specific system activities, and so on.
5. In the **Component** field, enter the material number of the components that you specify as part of the BOM.
6. Enter the **Quantity** of the component.
7. Enter the **Unit** for the component quantity.
8. Select one or more components and click the button in the tool bar to enter the segmentation values for a material.
9. In the **Component Selection by Segmentation** screen, enter the following fields:

   a. Segmentation Value: Click the **Fast Entry** button to enter a segmentation value.

   Note

   This field is used to set the requirement segment for a component in a planned and production order (during BOM explosion) and is enabled based on the following conditions:

   - Header material is not segmentation relevant.
   - Scope of segmentation strategy for header material is 2 (only requirement segmentation).

- Scope of segmentation strategy for a component material is 1 (stock and requirement segmentation) or 2 (only requirement segmentation).
- Scope of segmentation strategy for a header component is 1 (stock and requirement segmentation) and the quantity is negative (if the output material is a by-product).

b. **Include Segment Value**: Select this checkbox if you want to include the segment value in the header of a material. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.

c. **Exclude Segment Value**: Select this checkbox if you want to exclude the segment value in the header of a material.

10. Click the **Accept** button to confirm the changes.

Displaying a Sales Order BOM

Follow these steps to display a sales order BOM.

1. Choose **Logistics** > **Production** > **Master Data** > **Bills of Material** > **Order BOM** > **Display**. The **Display Order BOM** initial screen appears.

   Alternatively, you can use the transaction `CS63` to create a order BOM.

2. Enter the following data that identifies the BOM on the initial screen:
   - Sales Order
   - Item
   - Material
   - BOM Usage
3. Confirm your entries to go to the **Item Overview** page.

Sales Order BOM Browser

Use

You use the order browser when you, as a make-to-order manufacturer, work with order BOMs. After you have created a sales order and, if required, executed variant configuration, it is necessary to technically post-process the BOMs. The changes that you carry out during this process are only valid for this sales order.

Activities

Follow these steps to open a order BOM browser:

1. On the SAP Easy Access menu, choose **Logistics** > **Production** > **Bills of Materials** > **Bill of Material** > **Order BOM** > **Order BOM Browser**. The initial screen of order BOM browser appears.

   Alternatively, you can use the transaction `CSKB` to open an order BOM browser.

2. Enter the data for sales order, item and the BOM application.
3. Enter a value for **Max Expl Level** checkbox if you are working with large BOMs.

4. Click to specify the settings for explosion.

5. Define a view. For more information on defining views for explosion, see the application help for **Defining a View for BOM Explosion** in SAP help portal under **SAP ERP** > **SAP ERP Central Component** > **Logistics** > **Production Planning and Control (PP)** > **Basic Data (PP-BD)** > **BOMs (PP-BD-BOM)** > **BOM Reporting Functions** > **BOM Explosion** > **Selection Criteria for BOM Explosion**.

6. Start the order browser.

7. Based on the stock segment value provided on the selection screen and based on the components included or excluded in the BOM definition, the BOM browser displays the included components.

Routing

Use

In SAP system, you use routing to create one or more task list for production. The steps necessary to produce a material are generally saved as operations and suboperations in a routing. The routing is selected on order creation. The individual operations and suboperations as well as other routing data is copied to the production order.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.
- You have created the characteristics for segmentation using transaction `CT04`. See Maintain Segmentation Relevance for Functions for more details.

  You have selected the application relevance for **BOM** and **ROU** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance**).
- You have defined and maintained segmentation structure and strategy for a material and have assigned it to the header material. See (Segmentation Structure and Segmentation Strategy)

Features

Segmented materials require different task lists for different segment values. For example, if the segmentation characteristic **Quality** associated with the material and high quality materials requires additional operations such as lining or stitching for fashion materials, and low quality materials may not require them. Therefore, the task list needs to be associated to segment value of the finished good material. The production order for that material also determines the right task list based on the stock segment assigned while creating the order.

Selection Criteria for Routing

Apart from the standard selection fields, you can also use stock segment as a criteria to determine the right task list for production. Based on the stock segment assignment for the finished good material, only those task list(s) which are marked as **Included** in the routing maintenance program can be used in the production order.

Constraints

The following limitations apply for routing for materials with segmentation:

- You cannot create rate routings for segmentation relevant materials.
- Segmentation does not support standard rate routings.

- A reference to standard routings is not possible for segmentation-specific data. The engineering change management is not supported for segments.
- MRP does not consider alternative and parallel procedures.
- The Engineering Workbench (`CEWB`) for BOM and routing for materials with segmentation is not supported.
- When you change routing from dialog (using the transaction `CA02`) using change number, the following limitations are apply:
    - Cannot maintain or change segmentation values for routing using change number.

      Reason: Any change in routing with change number creates a new entry at header table `PLKO` with new Internal counter `ZAEHL`, and this field is also a key in segmentation table `SGT_PLKOS`. Therefore, to maintain segmentation value for routing with new internal counter, you have change header information after changing the segmentation details.

      Workaround: You have to modify header (Task list counter) level information to add correct segmentation inclusion/exclusion entries in database.
    - Segmentation records are not be displayed for the routing when you change the routing at the header level, and the material associated with routing has segmentation records in database.

      Reason: Any change in routing by change number creates a new record in `PLKO` table with new internal counter `ZAEHL`.

      Workaround: When you change the routing, click **Segmentation** button and select the segment inclusion/exclusion indicator that should be maintained for new internal counter. You can also follow the same approach when maintaining the segmentation values for a new counter.

Creating a Routing

Use

You use this procedure to create a completely new routing for segmentation relevant materials. If a routing already exists in your system with similar data, you can copy this data into the new routing.

Prerequisites

The following data already exists in the system:

- If you want to maintain the routing with history: The corresponding change master record.
- If you want to link the routing with a revision level: The relevant revision level of the material to be produced.
- Select the relevance option for BOM and Routing when you create a characteristic. To do this, go to **CT04** > **Additional Data** > **Maintain Relevance** > **Relevance**. Select the **Relevance** checkbox for BOM and Routing.

Activities

Follow these steps to create a routing for a segmentation relevant material:

1. On the SAP Easy Access menu, choose **Logistics** > **Production** > **Master Data** > **Routings** > **Routings** > **Standard Routings** > **Create**. Alternatively, you can also use the transaction `CA01` to create a routing.
2. Enter the following mandatory fields:

- Material
- Plant

3. Click to edit the header details. You can now create the first task list.
4. Enter the following fields on the header screen:
   - Usage: This field specifies the areas in which the task list can be used (for example, in production or plant maintenance).
   - Status: This field indicates the processing status of a task list. For example, you can indicate whether the task list is still in the creation phase or has already been released.
5. Click the icon to enter the following values for an operation:
   - Work Center
   - Control Key
   - Description
   - Base Quantity
   - Unit
   - Production Time
   - Set up
6. Click the button to assign the operation to a component.
7. You can create additional task list (group counters) using the steps 3 to 5. After you have created multiple task list for a routing, go to the routing view, select a task list and use the segmentation button to assign segment values to the task list. Click the icon to enter following segment values for the material:
   - **Include Segment Value**: Select this checkbox if you want to make the task list applicable to segment values. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.
   - **Exclude Segment Value**: Select this checkbox if you want to do not want the task list to be applicable for segment values.

   Note

   If you select the include/exclude checkbox for one segment value of the header material, the system automatically excludes/include all other segment values.

8. Choose **Accept** to save the changes.
9. Select the component that should be allocated to the operation and choose **New Assignment**.
10. On the **New Assignment** screen, enter the activity that you want to use to allocate the component. Click to save the entry.

    Note

    If you do not perform the step 10, the BOM components will be assigned to the first operation in the routing by default.

    You can also reassign a component to another operation using the **Reassign** option.

More Information

For more information about creating a routing by copying in standard ERP, see the SAP Library on SAP Help Portal under SAP ERP > SAP ERP Central Component > Production Planning and Control (PP) > Basic Data (PP-BD) > Routings > Maintaining Routings and Rate Routings > Steps in Routing Maintenance > First Creating a Routing by Copying.

Changing a Routing

Use

Follow these steps to change routing for a material with segment values:

1. On the SAP Easy Access menu, choose Logistics > Production > Master Data > Routings > Routings > Standard Routings > Change. Alternatively, you can also use the transaction `CA02` to change an existing routing.
2. Enter the following mandatory fields:
   - Material
   - Plant
3. Click to edit the header details.
4. Enter the following fields on the header screen:
   - Usage: This field specifies the areas in which the task list can be used (for example, in production or plant maintenance).
   - Status: This field indicates the processing status of a task list. For example, you can indicate whether the task list is still in the creation phase or has already been released.
5. Click the icon to enter following segment values for the material:
   - Include Segment Value: Select this checkbox if you want to include the segment value in the header of a material. If you select this checkbox for one segment value of the header material, the system automatically excludes all other segment values.
   - Exclude Segment Value: Select this checkbox if you want to exclude the segment value in the header of a material.

   Note
   If you select the include/exclude checkbox for one segment value of the header material, the system automatically excludes/include all other segment values.

6. Choose Accept to save the changes.
7. Click the icon to enter the following values for an operation:
   - Work Center
   - Control Key
   - Description
   - Base Quantity
   - Unit
   - Production Time
   - Set up

8. Click the button to assign the operation to a component.
9. Select the component that should be allocated to the operation and choose **New Assignment**.
10. On the **New Assignment** screen, enter the activity that you want to use to allocate the component. Click to save the entry.

    Note

    If you do not perform the step 10, the BOM components will be assigned to the first operation in the routing by default.

    You can also reassign a component to another operation using the **Reassign** option.

More Information

Routing

Displaying a Routing

Follow these steps to display a routing for a material with segment values:

1. On the SAP Easy Access menu, choose **Logistics** > **Production** > **Master Data** > **Routings** > **Routings** > **Standard Routings** > **Display**. Alternatively, you can also use the transaction `CA03` to create a routing.
2. Enter the following mandatory fields:
    - Material
    - Plant
3. Click to view the header details.
4. Click the icon to view following segment values for the material:
5. Click the icon to view the values for an operation:
6. Click the button to assign the operation to a component.
7. Select the component that should be allocated to the operation and choose **New Assignment**.

Task List Changes and Delete Task List for Routing

Use

This sections explains the change documents and the task list delete for routing

- `CA60`: Routing Changes
- `CA61`: Display Change Documents for Routing
- `CS98`: Deletion of Task List without Archiving

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.

- You have maintained a routing entry and a task list with relevant segmentation values and operations using the transaction `CA 01` or `CA02`.

Features

Changing Task Lists

Follow these steps to change a task list for routing:

1. On the SAP Easy Access menu, choose Logistics > Production > Master Data > Routings > Routings > Reporting > Task List Changes. Alternatively, you can also use the transaction `CA60` to change a task list for routing.
2. On the Task List Changes screen, enter the following fields:
   - Task List Type
   - Material
   - Plant
   - Task List/Recipe Group
   - Group Counter/Recipe Period of Examination

     Note

     The Group Counter field is updated only when more than one group counter is created.
3. Under the Detailed Change Documentation For section, select all the checkboxes to view detailed description.
4. Choose to view the task list details.

Display Change Documents for Routing

You can use the `CA61` transaction to view the overall details of change documents (such as changed dates, user, and so on).

Follow these steps to display change documents for routing:

1. On the SAP Easy Access menu, choose Logistics > Production > Master Data > Routings > Routings > Reporting > Change Documents > Routing. Alternatively, you can also use the transaction `CA61` to display change documents for routing.
2. On the initial screen, choose a Routing Group.
3. Under the Overview Selection, choose a view.
4. Choose . An overview of the routing changes done when you had created (using the transaction `CA01`) and changed (using the transaction `CA02`) a routing is displayed.

   Double-click an entry in the table to view the complete details.

Deletion of Task List without Archiving

You can also delete a task list without archiving first.

Note

When you delete without archiving, you may lose data unintentionally. You should only use this form of deletion for the data you have created for testing

Follow these steps to delete a task list from the database:

1. On the SAP Easy Access menu, choose Logistics > Production > Master Data > Routings > Routings > Extras > Delete Task Lists > Without Archiving. Alternatively, you can also use the transaction CA98 to delete a task list..
2. On the Deletion of Task List without Archiving screen, enter the following fields:
   - Material Number
   - Plant for Materials
   - Group
   - Group Counter Plant Status Task List Usage Planner Group

     Note

     Enter the Group Counter only if multiple group counters exist for a task list. Otherwise, after execution, the system displays the all the group counters associated with the task list.
3. Choose the option Complete Task Lists to delete a task list.
4. Choose to delete a task list.

BOM Explosion

Use

A BOM explosion answers the question: What is the product made of? This question arises in various situations, such as when you want to:

- Determine material requirements for a product
- Look at the overall structure of a product in a design department
- See an overview of the parts and materials required
- Calculate the effects of changes to costs

During production planning, the system explodes the bills of materials down to the lowest level. Then the system spreads the costs, separated by cost components, from the lowest level to the highest level, to determine the planned costs of the finished product. The BOM explosion can be carried out from any production level.

When you explode a BOM "top down", the system finds all the components that are in the BOM. Some individual components may also be assemblies. These subordinate BOMs are included in reporting functions. There is no limit to the number of levels of BOMs you can have. You can use selection criteria to either include additional information or restrict the list displayed, so you can define precisely how you want to explode a BOM.

Using this report, you can obtain:

- details about a component and assembly data to be collected for the component
- Alternate components at the material and BOM levels for each component
- BOM status

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_02`.
- You have created the characteristics for segmentation using transaction `CT04`.
- You have selected the application relevance for **BOM** and **ROU** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance**).

Features

When Bill of Material is defined for materials with segmentation, you would have specified which components would be inclusive or exclusive in the bill of material based on the segment value that would apply in a BOM usage. BOM explosion is a typical usage simulation. When one of the BOM explosion program is run, the selection screen allows you to specify the segment value for which you can simulate BOM explosion. Alternatively you could also choose not to provide any particular segment value and therefore check on "Ignore Segmentation" checkbox.

How do you explode a BOM?

For each individual BOM type various reporting functions are available:

1. For the structural explosion:
    - BOM level by level
    - Multi-level BOM
2. Summarized BOM: The summarized BOM function is a quantity-based reporting function.

Example

The BOM explosion can be explained using the following example:

Displaying Level by Level BOM Explosion

Context

This BOM reporting function shows which components make up the individual assemblies across the product structure. Components that are also assemblies are exploded lower down.

This reporting function is available to you for the following BOM types:

- Material BOMs
- Order BOMs

Procedure

1. On the SAP Easy Access Menu, choose **Logistics** > **Production** > **Master Data** > **Reporting** > **Material BOM/Order BOM** > **Level by Level**.

    Alternatively, you can use the transactions `CS11` (for a material BOM) and `CSK1` (for a sales order BOM) to explode the BOM level by level.

2. Enter the data that identifies your BOM (such as **Material**, **Plant**, and **Alternative BOM**, if appropriate). In case of a sales order BOM, enter the **Sales Order** number and the corresponding details.
3. In the **Application** field, enter the key of the procedure for automatic alternative determination

4. Enter the data for selecting a BOM.
5. If you are exploding a material BOM, go to next step. If you are exploding a sales order BOM, go to step 7.
6. For a material BOM:
    - Enter the relevant **Stock Segment**. This value depends on depends on whether you included or excluded the segment values during the creation of the BOM.
    - If you select the **Ignore Seg.** checkbox, the BOM is displayed for the specified segment.
7. Click to determine the selection criteria for the BOM explosion and display the output accordingly.
    - If you specify a segment value in the **Stock Segment** field, and do not select the **Ignore Seg.** checkbox, the BOM displays the details for the selected segment value.
    - If you have not specified any value in the **Stock Segment** field, and do not select the **Ignore Seg.** checkbox, the BOM displays the details for all segment values. You can click and icons to view different segments of the header material.
    - If you have not specified any value in the **Stock Segment** field, and select the **Ignore Seg.** checkbox, the BOM is displayed without any segmentation values.
8. Choose .

Results

At each level, you will see only those list of components that fulfill your selection criteria. The quantities refer to the required quantity displayed in the list header.

In the reporting list for BOM level by level, you first see all components on the first level. Components that are assemblies have an indicator in the **Asm** (assembly) field. However, they are not exploded down immediately.

A dividing line shows where the next explosion level begins.

If a component is also an assembly, the system explodes the next level (for example, level 2) in a separate block, provided that you select the multi-level indicator in the view data. You see all the assemblies that exist for components of the previous level.

An assembly is not exploded under certain circumstances. The next level down is not displayed until all the components on the level above have been listed.

Displaying Multi-Level BOM Explosion

Context

This reporting function determines all components (assemblies and individual parts) in a product and displays them in their structural context for production.

This reporting function is available to you for the following BOM types:

- Material BOMs
- Order BOMs

Procedure

1. On the SAP Easy Access Menu, choose Logistics > Production > Master Data > Reporting > Material BOM/Order BOM > Multilevel BOM .

   Alternatively, you can use the transactions CS12 (for a material BOM) and CSK2 (for a sales order BOM) to display the BOM level by level.

2. Enter the data that identifies your BOM (such as Material, Plant, and Alternative BOM, if appropriate). In case of a sales order BOM, enter the Sales Order number and the corresponding details.
3. In the Application field, enter the key of the procedure for automatic alternative determination
4. Enter the data for selecting a BOM.
5. If you are displaying a material BOM, go to step 6. If you are displaying a sales order BOM, go to step 7 directly.
6. For a material BOM:
   - Enter the relevant Stock Segment. This value depends on depends on whether you included or excluded the segment values during the creation of the BOM.
   - If you select the Ignore Seg. checkbox, the BOM is displayed for the specified segment.

     Note

     - If you specify a segment value in the Stock Segment field, and do not select the Ignore Seg. checkbox, the BOM displays the details for the selected segment value.
     - If you have not specified any value in the Stock Segment field, and do not select the Ignore Seg. checkbox, the BOM displays the details for all segment values. You can click and icons to view different segments of the header material.
     - If you have not specified any value in the Stock Segment field, and select the Ignore Seg. checkbox, the BOM is displayed without any segmentation values.

7. Click to determine the selection criteria for the BOM explosion and display the output accordingly.
8. Choose .

Results

You see only those list of components that fulfill your selection criteria. In the Multi-level BOM list, you immediately see whether a component is an assembly – the main difference between this and a level-by-level BOM explosion.

The quantities refer to the required quantity displayed in the list header.

The components of the subordinate assembly are inset directly below, with the next number up to show that this is the next level. If a component is an assembly, the explosion continues immediately.

The list of components on any level is interrupted each time a component that is an assembly occurs. After the list of items in the subordinate BOM, the list is continued on the superior level.

Note

Assemblies and components that occur more than once in the structure are shown more than once.

Displaying a Summarized BOM

Context

In certain application areas in a company, such as material requirements planning (MRP) and costing, you need a quick overview of all the parts required to make a product. This reporting function produces a complete count of the assemblies and individual components, which you can restrict using various criteria.

This reporting function is available to you for the following BOM types:

- Material BOMs
- Order BOMs

Procedure

1. On the SAP Easy Access Menu, choose **Logistics** > **Production** > **Master Data** > **Reporting** > **Material BOM/Order BOM** > **Level by Level** >.

   Alternatively, you can use the transactions CS13 (for a material BOM) and CSK3 (for a sales order BOM) to display a summarized BOM.

2. Enter the data that identifies your BOM (such as **Material**, **Plant**, and **Alternative BOM**, if appropriate). In case of a sales order BOM, enter the **Sales Order** number and the corresponding details.
3. In the **Application** field, enter the key of the procedure for automatic alternative determination
4. Enter the data for selecting a BOM.
5. If you are displaying a material BOM, go to next step. If you are displaying a sales order BOM, go to step 7.
6. For a material BOM:
   - Enter the relevant **Stock Segment**. This value depends on depends on whether you included or excluded the segment values during the creation of the BOM.
   - If you select the **Ignore Seg.** checkbox, the BOM is displayed for the specified segment.

     Note

     - If you specify a segment value in the **Stock Segment** field, and do not select the **Ignore Seg.** checkbox, the BOM displays the details for the selected segment value.
     - If you have not specified any value in the **Stock Segment** field, and do not select the **Ignore Seg.** checkbox, the BOM displays the details for all segment values. You can click and icons to view different segments of the header material.
     - If you have not specified any value in the **Stock Segment** field, and select the **Ignore Seg.** checkbox, the BOM is displayed without any segmentation values.

7. Click to determine the selection criteria for the BOM explosion and display the output accordingly.
8. Choose .

Results

You will see only those list of components that fulfill your selection criteria. The quantities refer to the required quantity displayed in the list header.

- This reporting function produces a complete count of the assemblies and other components across all levels in a product structure.
- This list does not show you the structure of the product. Assemblies and individual components which occur more than once in the product are displayed as one item, with a sum total item quantity.

- Displaying an assembly as a component is optional.

Planned Orders

Use

Planned orders are created in material requirements planning to cover the requirements. Planned orders represent a demand to procure or produce a material. Planned orders for materials that are to be produced in-house are converted to production orders. The material components required for production are contained as items in the planned order and are copied directly when the planned order is converted to a production order.

The standard planned order has been enhanced to include segment values for segmentation-relevant material to meet the needs of the fashion industry. The stock segment is determined based on the requirement segment specified for a material in a sales order or production order, and based on the segmentation strategy rule **ATP: Requirement/Stock allocation**. See Segmentation Strategy for more information about segmentation strategies

Features

- If the segmentation strategy for a material is of scope 1 (stock and requirement segmentation), the stock and requirement segment that is defined during the creation of a planned order should be copied to all the subsequent documents.

  Example: During the creating of a planned order, you define the **Quality1 (Q1)** and **Country1 (DE)** as the **Stock Segment (Q1DE)** . When you convert a planned order into production order, the same values (Q1DE) is copied to the stock segment of the production order.

- If the segmentation strategy for a material is of scope 2 (only requirement segmentation), the stock segment at the header item and requirement segment for component is available in the planned order. However, you cannot enter any valid or invalid value for stock segment.

- For a material which is not relevant to segmentation, the stock segment for the header and requirement segment for component is available in the planned order. However, you can save the orders without entering any values to these fields and you cannot enter any value to stock and requirement segment.

See the following for more information about the creation and conversion of planned orders:

- Creating a Planned Order
- Changing a Planned Order
- Displaying a Planned Order
- Partial Conversion of Planned Orders to Production Orders
- Collective Conversion of Planned Orders to Production Orders
- Converting Planned Orders into Purchase Requisitions

Creating a Planned Order

Use

As a rule, planned orders are generated automatically during a planning run. However, it may be necessary to create planned orders manually, for example, if you want to procure materials independently from requirements.

Automatic Creation of Planned Orders

During the planning run, the system automatically calculates the materials to be procured as well as the requirements quantity and date. The system then creates the corresponding planned order. The system also explodes the BOM for materials that are produced in-house and uses the BOM components as material components for the planned order. The system creates a corresponding dependent requirement for these components. If the quantity or the date of the planned order changes or if the bill of material changes, the bill of material is re-exploded in the next planning run and the dependent requirements of the material components are adjusted accordingly.

The stock segment is determined based on the requirement segment specified for a material in a sales order or production order, and based on the segmentation strategy rule **ATP: Requirement/Stock Allocation**. See Segmentation Strategy for more information about segmentation strategies

Creating a planned order manually

You can create planned orders manually, if you want to procure materials independently from requirements. For this, you determine which material should be procured, the quantity of material to be procured, the date it should be available and whether it is to be procured externally or internally. If you create or change a planned order manually, you can also explode the BOM manually and adjust the material components.

While creating a planned order, you should specify the stock segment. The BOM explosion and determination of the component is based on the specified stock segment and inclusion or exclusion of components for the specified stock segment. The requirement segment for a component is determined based on the segmentation strategy **Dependent Requirements/Stock Transfer**. See Segmentation Strategy for more details about segmentation strategies.

Creating a planned order from sales order (Planning strategy 81)

When a sales order is created and saved, a planned order will be generated for that material. The requirement segment from the sales order will be converted to stock segment using the segmentation strategy **Requirement to Stock Conversion**.

The demand elements specify the requirement segment of the material and MRP determines the stock segment (based on the segmentation strategy **Requirement to Stock Conversion**) and creates the planned order.

If the material has sub-components, BOM explosion occurs and the components are determined based on the stock segment of the main material. BOM explosion and determination of stock segment also depends on the stock segments and the inclusion or exclusion of components for that stock segment.

The requirement segment for the components determined depends on the segmentation strategy **Dependent Requirements/Stock transfer**.

Procedure

Follow these steps to manually create a planned order for a segmentation relevant material:

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **MRP** > **Planned Order** > **Create**.

   Alternatively, you can use transaction `MD11`.

   The initial screen for creating a planned order appears.

2. Enter a **Planned Order Profile**.

   The planned order profile determines the order type, the procurement type, and the special procurement type of the planned order.

3. You can also enter the number of a planned order, from which you want to copy data. The data from this planned order is copied into the new planned order. You can overwrite this data.

4. Choose .

   The detail screen for the planned order appears.

   The system sets the **order type** and **special procurement type** automatically according to the planned order profile. You cannot change these fields.

5. Enter the **Order Quantity** to be produced or ordered according to the planned order.

6. Enter the **End Date**, according to the planned order, that you should expect either the completion or the delivery of the planned order quantity.

7. Enter the earliest **Start Date**, according to the planned order, that the production of the planned order quantity starts.

8. Enter the **Opening Date** when the planned order will be opened.

9. Enter the **Stock Segment** that is assigned to a stockable item.

   ### i Note

   - The planned order is automatically flagged as convertible in the **Conversion Indicator** field. This enables the later conversion to a production order for in-house production, or to a purchase requisition for external procurement.
   - As you manually created the planned order, it is automatically flagged as firmed in the Firming screen area. Firmed planned orders are not changed by the system in a subsequent planning run.

10. Choose to enter component details.

    The component overview appears. The system displays the component processing status that was saved last. If you have recreated the planned order, the system explodes the BOM and copies the BOM items generated into the planned order as components. .

    After the BOM explosion, you can view the header item and component with relevant stock and requirement segment in the planned order.

    ### i Note

    - You can add only those stock and requirement segments which are maintained during creation of Segmentation Strategy.
    - BOM can be fixed for particular stock segment by selecting the **Component** checkbox under **Firming** options. After it is fixed, you cannot change the stock segment.

11. Double-click component and enter the following fields:

    - **Semi-Finished or Raw Material**
    - **Requirement quantity and unit of measure**
    - **Plant**
    - **Requirement Segment**

12. Save the planned order.

Changing a Planned Order

Use

Changes to existing planned orders can be necessary if the planned order receives a rescheduling proposal during the planning run.

Procedure

Follow these steps to change a planned order for a segmentation relevant material:

1. In the SAP Easy Access Menu, choose Logistics > Production > MRP > Planned Order > Change. The initial screen for creating a planned order appears.

   Alternatively, you can use the transaction `MD12`.

   The initial screen for changing a planned order appears.

2. Enter the number of the planned order to be changed and press Enter.

   The detail screen for changing a planned order appears.

3. Make the required changes.

4. Save the changes.

   i Note

   If you change a planned order, it is considered firmed for the planning run and it will not be changed automatically during subsequent planning runs..

Displaying a Planned Order

Context

You can display planned orders maintained with segment values using the individual display.

Procedure

1. In the SAP Easy Access Menu, choose Logistics > Production > MRP > Planned Order > Display > Individual Display.

   Alternatively, you can use transaction `MD13`.

   The initial screen for the individual display appears.

2. Enter the number of the corresponding planned order and press Enter.

Conversion of Planned Orders to Production Orders

Use

You use this function to partially convert a planned order to one or more production order(s), with an option to retain the planned order with accordingly reduced quantities.

The planner can use this function to:

- Convert the planned order from the stock/requirements list
- Create production order with reference to planned order

Prerequisites

You have activated the business function `LOG_SEGMENTATION_02`.

Features

This function has the following features:

- Context menu entry in the stock/requirements list

  When you select the context menu entry **Partially Convert Planned Order** >> **Production Order** for a planned order in the stock requirement list, the **Query Subset/Order Correction** screen is displayed for the selected planned order.

- Query Subset/Order Correction

  When you select the **Partial Conversion** checkbox on the initial screen for a segmentation planned order, the quantity of the planned order is split into several partial quantities which are then converted to production orders or process orders.

  If a partial lot quantity is maintained in the planned order, the query subsets screen displays equally split partial quantities in the **Subsets** table.

- Generation of partial quantity production orders

  When you select the **Generate** button, temporary production order numbers are created for each row in the **Subsets** table. You can also view or change each of the generated temporary orders before saving. While viewing a temporary order, you can carry out the functions like **Material Availability Check**, **Capacity Availability Check**, **Change SKU Quantities**, **View Component Overview** and **Release the Order**.

- Deletion of planned order

  If the quantities of all the temporary orders are equal to or greater than the quantities of the planned order, then the planned order is automatically marked for deletion. However, if you reduce the quantities of the partial orders, such that some open SKU quantities remain with the planned order, then you can deselect the checkbox **Delete Planned Order** to prevent the planned order from deletion. You can also mark the planned order for deletion irrespective of the quantities of the partial quantity orders.

- Saving the converted production orders

  On choosing **Save** on the **Query Subset/Order Correction** screen, all the temporary production orders are saved with new order numbers. If you had selected the **Delete Planned Order** checkbox, the original planned order is deleted after the production orders are saved. However, if you had not selected the **Delete Planned Order** checkbox, the original planned order is updated with the open quantities after the production orders are saved. When updating the planned order quantities, the **Firming** indicator is set for the planned order automatically.

Activities

- To convert the planned order from the stock/requirements list:

  In the SAP Easy Access Menu, choose **Logistics** > **Production** > **MRP** > **Evaluations** > **Stock/Requirements List**.

  Alternatively, you can also use the transaction `MD04`.

- To create production order with reference to planned order:

In the SAP Easy Access menu choose, Logistics > Production > MRP > Planned Order > Convert to Production Order > Individual Conversion.

Alternatively, you can also use the transaction `CO40`.

- To partially convert a segmentation planned order to one or more production order(s)

  in the SAP Easy Access menu, choose Logistics > Production > MRP > Planned Order > Convert to Production Order > Partial Conversion.

  Alternatively, you can also use the transaction `CO48`.

  The converted production order is created. The stock and requirement segment of the finished product is transferred or copied from the planned order to the production order.

Collective Conversion of Planned Orders to Production Orders

Use

You use this function to convert several planned orders into production orders. You can also select the planned orders that you want to convert together according to the date of their creation. The stock segment from planned order should be copied to production order.

Prerequisites

You must have activated the business function `LOG_SEGMENTATION_02`.

Activities

Follow these steps to use this function:

1. In the SAP Easy Access Menu, choose Logistics > Production > MRP > Planned Order > Convert to Production Order > Collective Conversion.

   Alternatively, you can use the transaction `CO41`.

2. Enter the **Planning Plant** and **Material**.

3. Choose the type of planned orders that you want to convert based on **Status** and **Availability**.

4. Choose . The system lists all planned orders that fit the given selection criteria.

More Information

Partial Conversion of Planned Orders to Production Orders

Planned Orders

Converting Planned Orders into Purchase Requisitions

Use

For materials procured externally, you should convert the planned order into a purchase requisition. During this conversion, the stock segment from planned order is copied to purchase requisition.

The following options are available:

- Individual conversion (individual planned order)

  You can convert the total planned order quantity or just part of it. Partial conversion is useful if the requirements situation has changed or you want to produce part of the planned order in-house.

- Collective conversion (several planned orders)

  Collective conversion is also possible as a background job.

  All purchase requisitions that were created as described above are firmed automatically. They are no longer changed by the system in subsequent planning runs.

After the conversion of planned order to purchase requisition, the stock segment of the main material is transferred or copied from the planned order to the purchase requisition line item. The components of the requirement segments are also displayed.

Procedure

Total Conversion or Partial Conversion of Individual Planned Order

1. In the SAP Easy Access Menu, choose Logistics > Production > MRP > Planned Order > Convert to Purchase Requisition > Individual Conversion.

   Alternatively, you can use transaction `MD14`.

2. Enter the number of the planned order you would like to convert into a purchase requisition and press Enter.

   The detail screen for the individual conversion of planned orders now appears.

   The first part of the screen displays the data of the planned order and the second part of the screen shows the data of the purchase requisition. This data is partly drawn from the planned order and partly from the material master. If necessary, you can overwrite this data.

3. Enter data as required.

4. In the case of partial conversion, the Converted quantity field the system automatically proposes the planned order quantity. Overwrite this value with the quantity that is actually to be converted and transferred to the purchase requisition and choose Enter.

   The system displays the message Planned order only partially converted.

   The Planned order quantity is automatically reduced to the remaining quantity. You can correct or delete the planned order quantity. If you delete the planned order quantity, the system automatically deletes the planned order during conversion.

5. Enter data as required.

   The planned order is converted totally or partially into a production order.

Collective Conversion of Individual Planned Order

1. In the SAP Easy Access Menu, choose Logistics > Production > MRP > Planned Order > Convert to Purchase Requisition > Collective Conversion.

   Alternatively, you can use transaction `MD15`.

   The initial screen for the collective conversion of planned orders now appears.

2. Enter the **MRP controller number** and the **plant**.

3. Limit the selection further, if necessary, using the opening date and the procurement type. Press **Enter**.

4. The system displays a list of all the planned orders that fulfill the selection criteria.

5. Select the required planned orders.

6. Choose **Convert online** or **Do not convert online**.

   With **Convert online**, the system displays the screen for individual conversion once again before the conversion for each planned order and gives you the opportunity to edit it.

   With **Do not convert online**, the system converts the planned orders selected directly into purchase requisitions.

   If necessary information is missing for creating a purchase requisition, the system interrupts the conversion procedure and displays the screen for individual conversion. You can then add the missing data here. If you choose **Enter**, the system continues with the conversion.

   i Note

   To avoid the interruption of the conversion procedure, you can predefine default data for the purchase requisition. To do this, choose **Goto** > **Purchase requisition default data** and enter the required data. This data applies to all purchase requisitions created in the conversion.

Production Orders

Use

Production orders contain all the information you need for the production process, for example quantity, basic date, information about BOM and routing, and segment values (stock and requirements segments).

Features

You create a production order for a material with segment values in the SAP standard application. In a production order for a segmentation relevant material, you can maintain the quantity to be produced at the segment level. You can find information from both the segmentation BOM (see Bill of Material) and routing (see Routing).

Tolerances

Excess quantities or shortages can result during production. For these situations you can maintain tolerance limits for each segment value. You can use this so that upon goods receipt in your warehouse you can consider quantity deviations per segment value that arise during the production process.

Deviation of the segment quantity in the production order from the bundle quantity

There are the following cases:

- There is a sales order or there are planned independent requirements. A quantity shortage of one or more dimensions occurs.

  A new planned order is generated for the missing quantity after a new material requirement planning (MRP) run. The MRP temporarily assigns it to the planned independent requirement.

- There is a planned independent requirement. There is a quantity excess of one or more dimensions.

The excess quantities become freely available stock. They can be consumed by a subsequent planned independent requirement if the dates allow. MRP temporarily assigns the freely available stock to the planned independent requirement.

- There is no planned independent requirement. There is a quantity excess of one or more dimensions.

  The production order is changed accordingly.

- There is no planned independent requirement. There is a quantity shortage of one or more dimensions.

  The production order is changed accordingly.

More Information

- For more information about production orders in standard ERP, see **SAP Library on SAP Help Portal** under **SAP ERP** > **SAP ERP Central Component** > **Logistics** > **Production Planning and Control (PP)** > **Production Orders (PP-SFC)**.

Creating a Production Order

Context

A production order specifies which material is to be produced, where it is to be produced, which operations are required to do this and on which date production is to take place. It also defines how the order costs are to be settled.

Production orders can be created manually without being previously requested. When you create a production order manually, you should then manually maintain the stock segment.

Alternatively, they can be automatically created by converting a planned order (see Conversion of Planned Orders to Production Orders). During material requirements planning (MRP run), planned orders are created at every BOM level to cover requirements and segment values. For materials produced in-house, a secondary requirement is also generated when the BOM is exploded, which is necessary for producing the end product or assembly. For externally produced materials, an ordering transaction is initiated when a purchase requisition is generated.

Planned orders generated in the MRP run can be converted individually into production orders from the current stock/requirements list. They can also be grouped together by the MRP run and converted into production orders together. These production orders can be released together.

While creating a production order, you should specify the stock segment. The BOM explosion and determination of the component is based on the specified stock segment and inclusion or exclusion of components for the specified stock segment. The requirement segment for a component is determined based on the segmentation strategy Dependent Requirements/Stock Transfer. See Segmentation Strategy for more details about segmentation strategies.

Procedure

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **Shop Floor Control** > **Order** > **Create** > **With Material** . The initial screen for creating a planned order appears.

   Alternatively, you can use the transaction `CO01` to create production order.

2. On the **Create Production Order** screen, enter the following (mandatory) fields:

   - Material you want to produce
   - **Production plant** where the material is to be produced
   - **Order type** (controls whether an internal or external number range is used)

- If you have selected an order type for which an external number assignment is required then you must also specify an order number in the field **Order**.

3. Press **Enter** to continue.
4. Enter the following fields:
   - **Stock Segment** of the header material for which production needs to be done
   - Required quantity and dates for the production.
5. Go to the **Component Overview** tab and choose the production version as 01 (for normal sequence).
6. On the component overview page, the component details are displayed based on the data maintained in the segment BOM.
7. Save the order.

   To reduce the number of steps you have to work through, you can already release an order when you create it. To release a production order, choose . In Customizing for Shop Floor Control (**Master Data** > **Define Production Scheduling Profile**, you can set up automatic release to take place when an order is created.

Next Steps

- Production Orders
- Conversion of Planned Orders to Production Orders
- Segmentation Strategy

Changing a Production Order

Context

Maintenance of the status is the most common reason for changing an order. Unless defined otherwise in a profile, the order receives the status CRTD (created) after it has been created. The order cannot be executed until it has been released (status REL). Other reasons for changing an order are changes to the order header, components or operations. This mostly occurs when an order is to be created for rework, with the current material as the component.

Procedure

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **Shop Floor Control** > **Order** > **Change**. The initial screen for changing a production order appears.
2. Enter the **Order** number and press **Enter** to change the header details.
3. Make the required changes in the available tabs on the order header.
4. Click to go to the operation overview page. Make the required changes to the operation data.
5. Click to go to the component overview page. Make the changes to the component data (such as, requirement and stock segment).
6. Save the order.

Displaying a Production Order

Procedure

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **Shop Floor Control** > **Order** > **Change**.

Alternatively, you can use the transaction `CO02` to display a production order.

The initial screen for display a production order appears.

2. Enter the **Order** number and press **Enter** to view the header details.
3. Click to go to the operation overview page. The operation data details are displayed on the screen.
4. Click to go to the component overview page. The component data (such as, requirement and stock segment) on the screen.

Creating a Production Order from a Sales Order

Context

When a sales order is created and saved, a production order will be generated for that material. The requirement segment from the sales order will be converted to stock segment using the segmentation strategy **Requirement to Stock Conversion**.

Procedure

1. In the SAP Easy Access Menu, choose **Logistics** > **Production** > **Shop Floor Control** > **Order** > **Create** > **For Sales Order** > .

   Alternatively, you can use the transaction `CO08` to create a production order from a sales order.

   The initial screen for creating a planned order appears.

2. Enter the following data:
   - **Sales order number**
   - **Order number**
   - **Order type** (it controls whether an internal or external number range is used.)
   - If you have selected an order type for which an external number assignment is required then you must also specify an order number in the field **Order**.

   Note

   - Material number, quantities and dates are copied from the sales order. You can change the plant, configuration, quantities and dates in the production order.
   - Changes in the sales order or the production order are not automatically copied to the other object.
   - If a sales order BOM and a sales order routing exist, then they are copied to the production order.

Next Steps

- Production Orders
- Creating a Production Order

Posting a Goods Issue for Component of a Production Order

Use

The final step in the material staging process is to post a goods issue for consumption. When a production order is completed, the materials have been removed from the production storage bins in the warehouse and the final product has been fully assembled, it is still necessary to balance the stock quantities in IM and WM.

You post the use of materials to the database either by “backflushing” with the completion confirmation or directly in Inventory Management. This posting is necessary to reduce the stock quantities in the production storage bins of the WM and IM components.

Activities

Follow these steps to post a goods issue for a component of a production order:

1. On the SAP Easy Access Menu, choose **Logistics** > **Materials Management** > **Inventory Management** > **Goods Issue (MIGO)** >.
2. Choose **To order** from the application toolbar.

   The system displays a dialog box in which you must enter the production order number as a minimum.

   When you post the goods issue, it is important that you enter the production order number. The system will only reduce the quantities in the production storage bins if you enter the production order number on this screen. Otherwise, it will post the material to the general interface and create transfer requirements.
3. Enter the movement types as 261 for a production order or operation under the **Where** tab,
4. Enter the stock segment under the **Batch** tab. The stock and requirement segment mapping is done based on the segmentation strategy **Requirement/Stock Allocation**.
5. Post the goods issue.

Since the transfer requirement for the production order was created far in advance of this goods issue posting, the materials are delivered in a timely manner to the production storage bins. You post the actual goods issue in IM (or in PP) after the product has been assembled in production. Since the production storage bin was noted in the production order when it was released, the material usage posting reduces the quantity in the production bins and no further action is required in WM.

Confirmations

Use

The confirmation is used for the order monitoring. It documents the status of production orders and operations in the production cycle.

You can carry out the order monitoring for segmentation relevant materials.

With a confirmation you specify

- The quantity in an operation that was produced as yield, scrap and the quantity to be reworked
- How much work was actually done
- Which work center was used for the operation
- Who carried out the operation

Exact confirmation shortly after completion of an operation is essential for realistic production planning and control.

Features

For a component of a material with segment values (stock and requirement segments), the **Backflush** indicator is set. The stock segment and batch is identified and saved in the production order.

You can carry out the confirmation for a production order (`CO15`) or for an operation (`CO11` or `CO11N`). Specify the order and operation. and choose **Goods Movement**. The component quantity, stock segment and batch information is transferred from the production order and you can now fix batch to an operation.

The stock segment of the finished product is determined using the segmentation strategy **Requirement to Stock Conversion**.

If the confirmation is not saved, you can view the error logs using the transaction `COGI`.

You can enter a confirmation for the following:

- For an operation (using transactions `CO11` or `CO11N`)

  You have the following options:

  - Confirm quantities, durations, activities, or personnel data, enter a **time ticket confirmation** or a **progress confirmation**.
  - To make a confirmation at a particular point in time (for example, at the start of setup or the finish of processing), enter a confirmation for a **time event**.
  - When you confirm time events, the system calculates the required duration. For example, the setup duration is the time between setup start and setup finish.

- For an order (using `CO15`)

  If you enter a confirmation at order header level, the system confirms all the operations that have a control key in which confirmation is optional or necessary. The quantities confirmed in the operations are proportional to the quantities confirmed in the order header.

- By using another confirmation as a reference

  You can copy the data from a confirmation that has already been made to the current confirmation. Enter the number of the reference confirmation to identify it.

ℹ Note

When you process confirmation for a segmentation relevant production order, the stock and requirement segment values are determined automatically for components that have the **Backflush** indicator set.

Collective Orders with Segmentation

Use

In a collective order, planned orders or production orders are linked to one another over several production levels. Each order in the collective order has its own order number. If subassemblies are produced directly for superior orders within a production process, without physically entering the warehouse, it is useful to have a representation via collective orders.

The components for which separate production orders are created in the collective order are called directly produced components.

Features

Segmentation support

Based on the Customizing activity for segmentation in the Segmentation Structure (`SGTC`) transaction, the `Stock Segment` value for the header article, and the `Requirement Segment` value for the components are determined automatically for the header article and for its child orders respectively (directly produced components of collective orders).

Example

Assume that you want to produce a suit. The BOM for a suit contains a jacket and trousers. You want to enter these two components in separate production orders, but you do not want them to be posted to stock.

You set the special procurement type to direct production in the material master record for the jacket and the trousers, so that production occurs using a collective order. When you create a production order for the suit, a collective order is automatically created. It contains subordinate production orders for the jacket and the trousers.

More Information

Collective Orders

Order Progress Report

Use

The order progress report shows you which documents, MRP elements, stocks, and deliveries exist for products and their components that have been ordered by a customer. You can enter the relevant production or process order to view the segment details.

The report provides you with the following standard information:

- An overview of the status of production and procurement
- Statements about the adherence to the delivery dates or delay
- Reporting functions for production and planned orders
- View stock segments at the header level and requirement segment at the component level

An order progress report can be displayed for more than one sales orders or a WBS element.

i Note

You can also use the order progress report just like the standard order progress report for the make-to-order production only.

Prerequisites

You have activated the business function `LOG_SEGMENTATION_02`.

Features

The order progress report considers segmentation strategies and sort methods for segmentation-relevant materials and their components during stock, receipt, or issue mapping. You can enter Production/Process order number and view segment details at header, item, and component level.

The following information is displayed in addition to the SAP standard data:

- The segment values of the material
- Operations for production orders
- Quantities in the operations with information about the confirmations

See Examples for Order Progress Report to understand how order progress report works for segmentation relevant materials.

Examples for Order Progress Report

Use

The following examples explain the order progress report calculates availability based on receipt dates and ATP logic:

Prerequisites

- You must maintain a profile using the Customizing activity **Define Profiles for Order Report** under **Production** > **Material Requirements Planning** > **Evaluation**.

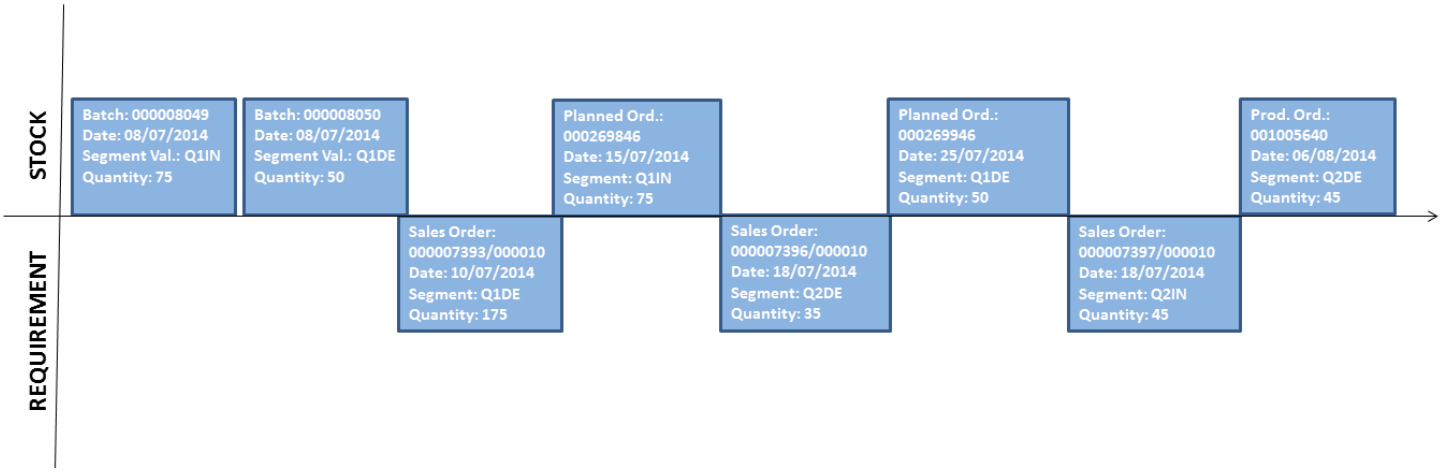
  For the following example, you can use the profile `SAP000000001` (Standard Profile MD4C) for availability calculation based on the Receipt Dates and profile `SAP000000002` (SAP Standard Profile CO46) for availability calculation based on the ATP Logic.

- You must the following values for **Sort Stock based on Segment** checkbox in **Material Master** > **Logistics: DC Fashion**:
  - If you select this checkbox, segments have higher priority than the requested delivery date. The stock availability based on the segmentation strategy **Requirement to Stock Allocation** is of higher priority and the stock with stock segment matching the requirement segment (by the segmentation strategy rule **Requirement to Stock Allocation**) is automatically assigned to the order.
  - If you do not select this checkbox, the requested delivery date has a higher priority than the segments. The stock availability by date takes the priority and stock available closer to the delivery date is automatically assigned to the order.

Features

Availability Calculation based on Receipt Dates

Consider the following stock and requirement scenario:



Requirement/Stock Allocation Rules:

Requirement Segment	Priority	Stock Segment
Q1DE	1	Q1DE
Q1DE	2	Q1IN
Q2DE	1	Q2DE
Q2DE	2	Q1IN
Q2IN	1	Q2IN
Q2IN	2	Q1IN

CASE 1: Sort Method is "X" (Segment Higher than Requested Delivery Date)

1. **For sales order with quantity 175 for segment value Q1DE**

   The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Stock (0000008050)	Q1DE	1	50	0
Pld Order (0000269946)	Q1DE	1	50	2014/07/25
Stock (0000008049)	Q1IN	2	75	0
Pld Order (0000269846)	Q1IN	2	75	2014/07/15

   The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007393/000010)	175	Q1DE	Stock (00000008050)	50	Q1DE
			Pld Order (0000269946)	50	Q1DE
			Stock (00000008049)	75	Q1IN

   Since the segment value is higher than the delivery date, the first batch stock with quantity 50 is allocated, then the planned order with quantity is allocated (with segment Value Q1DE). The batch stock quantity 75 with segment value Q1IN is then allocated. Thus, quantity 175 is consumed.

2. **Sales Order with Quantity 35 for Segment Value Q2DE**

   The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Prd Order (01005640)	Q2DE	1	45	2014/08/06
Stock (0000008049)	Q1IN	2	0	0
Pld Order (0000269846)	Q1IN	2	75	2014/07/15

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007396/000010)	35	Q2DE	Stock (00000008050)	35	Q2DE

Here, the production order (01005640) is allocated as segment value for requirement and stock are the same, even though the Stock (0000008049) and PldOrd (0000269846) are created on previous dates.

3. Sales Order with Quantity 45 for Segment Value Q2DE

   The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Stock (0000008049)	Q1IN	2	0	0
Pld Order (0000269846)	Q1IN	2	75	2014/07/15

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007397/000010)	45	Q1IN	Pld Order (0000269846)	45	Q1IN

Quantity for batch stock (0000008049) is already consumed, therefore, quantity 45 is allocated to PldOrd (0000269846) and remaining 30 can be filled with other requirement.

Sort Method is “ ” (SPACE) (Requested Delivery Date higher than Segment)

1. Sales Order with Quantity 175 for Segment Value Q1DE

   The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Stock (0000008050)	Q1DE	1	50	0
Stock (0000008049)	Q1IN	2	75	0

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000269846)	Q1IN	2	75	2014/07/15
Pld Order (0000269946)	Q1DE	1	50	2014/07/25

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007393/000010)	175	Q1DE	Stock (0000008050)	50	Q1DE
			Stock (0000008049)	75	Q1IN
			Pld Order (0000269846)	50	Q1DE

In this case, the batch stock will be consumed first, since Stock (0000008050) and Stock (0000008049) are available in the system and then will be allocated to open order as per the delivery date. Therefore, the planned order (0000269846) will be allocated for quantity 50 and rest quantity 25 still open to fill from any other requirement.

2. Sales Order with Quantity 35 for Segment Value Q2DE

   The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Stock (0000008049)	Q1IN	2	0	0
Pld Order (0000269846)	Q1IN	2	25	0
Prd Order (01005640)	Q2DE	1	45	2014/07/15

   The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007396/000010)	35	Q2DE	Pld Order (0000269846)	25	Q1IN
			Prd Order (01005640)	10	Q2DE

   In continuation of case 1, Stock (0000008049) is completely consumed and Pld Ord (0000269846) still has open quantity 25. Therefore, the first quantity 25 will be allocated to Pld Ord (0000269846) and according to the checking rule, the remaining quantity 10 is allocated to the Pld Order (01005640) and quantity 35 will be open.

3. **Sales Order with Quantity 45 for Segment Value Q2IN**

The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Stock (0000008049)	Q1IN	2	0	0
Pld Order (0000269846)	Q1IN	2	0	2014/07/15

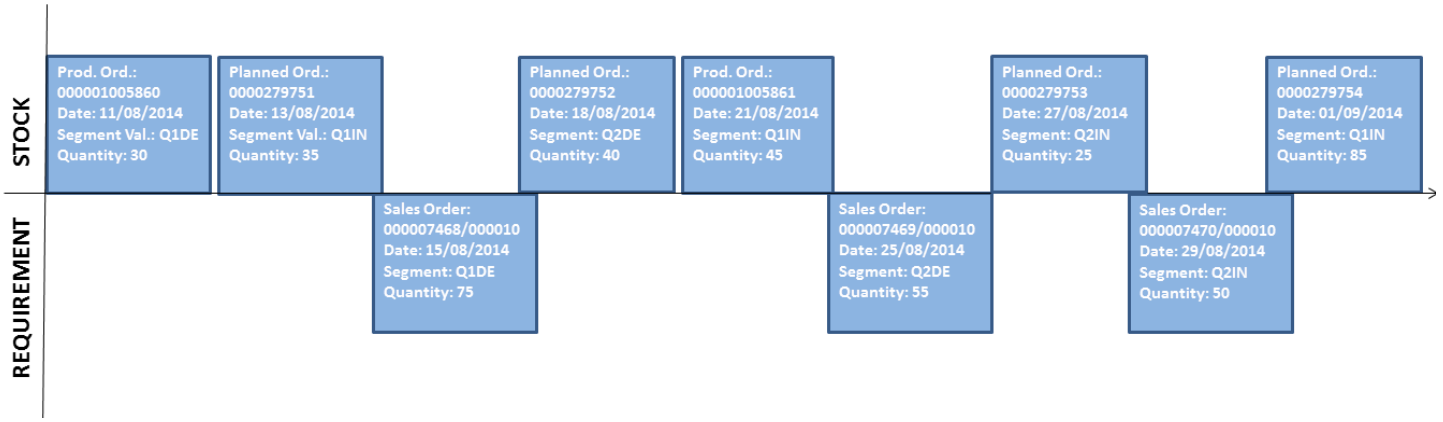
The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007397/000010)	45	Q2IN			

This requirement will not be consumed by any open stock since the allocation rule does not support the allocation from segment Q2IN to Q1IN.

Availability Calculation based on ATP Logic

Consider the following stock and requirement scenario:



Requirement/Stock Allocation Rules:

Requirement Segment	Priority	Stock Segment
Q1DE	1	Q1DE
Q1DE	2	Q1IN
Q2DE	1	Q2DE
Q2DE	2	Q1IN
Q2IN	1	Q2IN
Q2IN	2	Q1IN

Sort Method is "X" (Segment Higher than Requested Delivery Date)

1. **Sales Order with Quantity 75 for Segment Value Q1DE**

The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Prd Order (0001005860)	Q1DE	1	30	11/08/2014
Pld Order (0000279754)	Q1IN	2	85	2014/01//09
Prd Order (0001005861)	Q1IN	2	45	2014/08/21
Pld Order (0000279751)	Q1IN	2	35	2014/08/13

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007468/000010)	75	Q1DE	Prd Order (0001005860)	30	Q1DE
			Pld Order (0000279751)	35	Q1IN

In this case, the segment value has a higher priority, therefore, the first Prd Ord (0001005860) with segment Q1DE will be consumed and then the Pld Order (0000279751) will consume quantity 35 and remaining 50 will be open stock.

2. Sales Order with Quantity 55 for Segment Value Q2DE

The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000279752)	Q2DE	1	40	2014/08/18
Pld Order (0000279754)	Q1IN	2	85	2014/01/09
Prd Order (0001005861)	Q1IN	2	45	2014/08/21

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007469/000010)	55	Q2DE	Prd Order (0001005861)	15	Q1IN
			Pld Order (0000279752)	40	Q2DE

Based on the checking rule, Pld Order (0000279752) is allocated first with quantity 40, but Pld Order (0000279754) and Prd Order (0001005861) have same segment value and same priority, therefore the order which was created first will be allocated, that is, Prd Order (0001005861) will consume quantity 15.

3. Sales Order with Quantity 50 for Segment Value Q2IN

The following table displays the valid stock for particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000279753)	Q2IN	1	25	2014/08/27
Pld Order (0000279754)	Q1IN	2	85	2014/01/09
Prd Order (0001005861)	Q1IN	2	30	2014/08/21

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007470/000010)	50	Q2IN	Pld Order (0000279753)	25	Q2IN
			Prd Order (0001005861)	25	Q1IN

Based on the checking rule, the Pld Order (0000279753) is consumed first for segment value Q2IN and then the Pld Ord (0000279754) is allocated. The Prd Order (0001005861) has same segment value and priority, but PrdOrd (0001005861) will get allocated for quantity 25 since it was created before PldOrd (0000279754). Also After scenario 2 and 3, open quantity will of PrdOrd (0001005861) will be 5.

Sort Method is “ ” (SPACE) (Requested Delivery Date higher than Segment)

1. Sales Order with Quantity 75 for Segment Value Q1DE

The following table displays the valid stock for a particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000279754)	Q1IN	2	85	2014/09/01
Prd Order (0001005861)	Q1IN	2	40	2014/08/21
Pld Order (0000279751)	Q1IN	2	35	2014/08/13
Prd Order (0001005860)	Q1DE	1	30	2014/08/11

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007468/000010)	75	Q2IN	Pld Order (0000279751)	35	Q2IN
			Prd Order (0001005860)	30	Q1IN

In this sort method with ATP, stock will be checked backwards and will be used. Therefore, for SO (0000007468/000010), nearest stock is Pld Order (0000279751) and will be consumed first for quantity 35 and then the Prd Order (0001005860) will be consumed with quantity 30.

a. Sales Order with Quantity 75 for Segment Value Q1DE

The following table displays the valid stock for a particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000279754)	Q1IN	2	85	2014/09/01
Prd Order (0001005861)	Q1IN	2	40	2014/08/21
Pld Order (0000279752)	Q2DE	2	40	2014/08/18

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007469/000010)	55	Q2IN	Prd Order (0001005861)	45	Q1IN
			Pld Order (0000279752)	10	Q2DE

In this sort method with ATP, the Pld Order (0000279754) will be consumed for segment value Q1DE while checking backwards and PldOrd (0000279752) will then be consumed for quantity 10..

i. Sales Order with Quantity 50 for Segment Value Q2IN

The following table displays the valid stock for a particular segment value:

Valid Stock	Segment Value	Priority	Quantity	Date
Pld Order (0000279754)	Q1IN	2	85	2014/09/01
Pld Order (0000279753)	Q2IN	1	25	2014/08/27

The allocation will be done as follows:

Requirement	Quantity	Segment Value	Stock Allocation	Allocated Quantity	Segment Value
SO (0000007470/000010)	50	Q2IN	Pld Order (0000279753)	25	Q2IN

In this sort method with ATP, the most recent stock backward was Pld Order (0000279753), therefore quantity 25 is consumed.

Production Order Information System

Use

You use this tool for shop floor control with reporting functions for production orders and planned orders. You can view all the orders in the system, including the orders with deletion indicators. Archived orders are however not taken into account.

The segmentation production orders selected include additional segmentation information like requirement and stock segments.

You can view segmentation information based on header level, item level and component level are displayed for relevant production orders. At the header or item level, you can view the stock segment of the finished goods. At the component level, you can view the requirement segment of the part or component.

At the component level, the stock segment is displayed only if you have entered the stock segment in the order and assigned the batch directly in the production order.

Prerequisites

You have activated the switch `LOG_SEGMENTATION_02`.

Features

This function provides the following features:

- Additional Segmentation Selection Criteria

  The selection screen includes additional selection criteria (stock and requirement segments). The selection options are provided only for the following list types:

  - Object Overview
  - Order Headers
  - Items
  - Components
  - Confirmations
  - Purchase Orders

- Additional segmentation relevant information displayed

  Information about segmentation based on header level, item level and component level are displayed for relevant production orders. At the header or item level, you can view the stock segment of the finished goods. At the component

level, you can view the requirement segment of the part or component.

Activities

1. To filter records based on segment fields, in the SAP Easy Access menu, choose Logistics > Production > Shop Floor Control > Information System > Order Information System.

   Alternatively, you can use the transaction `COOIS` to execute a production order progress report.

2. Specify the filtering values and choose ✓.

Order Information System

Use

The order information system is a tool for shop floor control and process industries with reporting functions for production orders and planned orders. You can view all the orders in the system, including the orders with deletion indicators. Archived orders are however not taken into account.

The production orders selected are for segmentation relevant materials and consist of additional segment relevant information such as requirement segment and stock segment.

Features

This function provides the following features:

- Additional segmentation relevant selection criteria.

  The selection options are provided only for the following list types:

  - Order Headers
  - Items
  - Components
  - Confirmations

  When you execute the order information system report, the production order is expanded from the header level to the component level. The segment relevant data is also displayed.

- The order information system has the following display options:

  - In an individual object list, a list is generated for the selected object (for instance, order header, operation, and component). You can define the structure and appearance of the individual object list with a layout.
  - On the object overview the individual orders are displayed with their subordinate objects in a hierarchical structure. With an overall profile, you can define which fields are displayed and which objects (order headers, items, sequences, and so on) are read or displayed. In addition, you can define which fields are displayed for the individual objects.

When you execute the order information system report, the production order is expanded from the header level to the component level. Click the icon to include the requirement segment. The requirement segment is then displayed for the component or part.

Selection of Individual Object List

Individual object lists contain detailed information about objects in planned and production orders. For example, you want to display all production resources and tools (PRTs) used within a particular time period.

The segmentation values are displayed at the item and component level for a material with segment values.

Follow these steps to display **stock segment** at the item and component level for a segmentation relevant material:

1. For the production order, choose Logistics > Production > Shop Floor Control > Information System > Order Information System.

   Alternatively, you can use the transaction C028.

2. Specify the filtering criteria as Item.

3. Select a profile and provide the order number range.

4. On the Overview Items screen, select the header material and click the icon to include the stock segment.

5. Choose stock segment and include in the list of selected fields.

   The stock segment for the header material appears on the overview screen.

Follow these steps to include **requirement segment** at the item and component level for a segmentation relevant material:

1. For the production order, choose Logistics > Production > Shop Floor Control > Information System > Order Information System.

   Alternatively, you can use the transaction C028.

2. Specify the filtering criteria as Component.

3. Select a profile and provide the order number range.

4. On the Overview Items screen, select the header material and click the icon to include the requirement segment.

5. Choose the requirement segment and include in the list of selected fields.

   The requirement segment for the header material appears on the overview screen.

For more information about Individual Object Lists in standard ERP, see SAP Library on SAP Help Portal under SAP ERP > SAP ERP Central Component > Logistics > Production Planning and Control (PP) > Production Orders (PP-SFC) > Order Information System > Individual Object List.

Missing Parts

Use

If, during an availability check, the system determines that a component is not available on a date on which it is required, the production order is automatically given the status PRT shortage (MSPR). This status remains active in the order header until a subsequent availability check (overall check) determines that all components have become available (for example, due to a goods receipt).

An individual check has no effect on the status of components.

You can display missing parts in the SAP system for a specific order, collective order, or several orders using the missing parts information system.

Features

Missing Parts in a Production Order

If you check the availability of all the material components in the network header or the activity overview and material is not available, a dialog box appears from which you can go directly to the log or the missing parts list and missing parts overview.

In the **missing parts overview**, you can:

- Edit missing parts data (for example, by changing the requirements quantity, batch or issue storage location)
- Check the availability of individual components
- Filter or sort components
- View requirement segment

The missing parts overview always displays the results of the last availability check carried out in the order, regardless of whether you have since exited the order.

Follow these steps to display requirement segment for a missing part during availability check:

1. On SAP Easy Access menu, choose Logistics > Production > Shop Floor Control > > Production > Order > Create or Change. The production order create or change screen appears.

   Alternatively, you can also use the transaction `CO01` or `CO02`.

2. From the toolbar, choose Goto > Missing Parts > Missing Parts Overview. The requirement segment is displayed on screen under Overview Missing Parts.
3. From the toolbar, choose Goto > Missing Parts > Missing Parts List. The availability check screen appears.
4. Click to display the requirement segment for the component in a production order.
5. Choose the requirement segment and include in the list of selected fields.

   The requirement segment for missing part is displayed on the screen.

In the **missing parts list** you can:

- Display the missing parts of production orders which have been checked
- Check the availability of individual components
- Filter or sort components
- Print the missing parts
- Vary the field selection

The missing parts list is not saved if you exit order processing.

Missing Parts Information System

Use

You can use the missing parts information system to display the missing parts list for a selection of a segmentation relevant materials, or for all materials. You can also restrict your selection of missing parts according to a specific plant, MRP controller or

requirements date.

A profile determines what data is displayed on the missing parts list and how it is displayed. The SAP System contains two standard profiles. They sort the missing parts list according to order number or material number. You can, however, define your own profiles in Customizing for **Shop Floor Control**.

If you often enter the same selection criteria when you create missing parts list, you can save the criteria in the form of a variant. You can then call up the variant any time you want via the menu.

Features

In addition to the data from the SAP standard, you can see segmentation relevant data for reservations with missing quantities.

Activities

Calling Up a Missing Parts List

Follow these steps to call a missing parts list:

1. On SAP Easy Access menu, choose **Logistics** > **Production** > **Shop floor control Information system** > **Missing parts info system** >. The system branches to the initial screen for generating a missing parts list.

   Alternatively, you can also use the transaction `C024` to call a missing part list.

2. Select a Profile.

   In Customizing for Shop Floor Control, you can create your own profiles (**Information system** > **Define Profile for Missing Parts Information System** >.

3. If required, maintain additional criteria for creating the missing parts list. You can enter the following criteria: Material, plant, MRP controller, requirements date (or interval), WBS element, sales order.

   You include or exclude data that meets these criteria in the list. You can also specify that a particular field is to be taken as the upper or lower (alphabetic/numeric) limit for the selection process. To do this, position the cursor on the relevant field and choose **Edit** > **Selection options** >. Enter the required restrictions and choose ✔.

   If you want to save your entries as a variant, select **Goto** > **Save as variant** >.

4. Choose .

   The system lists all reservations with missing parts that meet the specified selection criteria.

   The requirement segment column holds the requirement segment of the part that is in the production order.

Product Cost Planning and Cost Object Controlling

Use

There are special structures available in the system for mapping logistical business processes. These structures enable you to manage materials and their characteristic or segmentation values.

Segmentation values (for example, quality) used to map the characteristic values of materials also influences the product controlling of the R/3 system. However, for costing, you must have stored a segment value for the costing at material level, in the material master of the material to be cost.

During costing, the system uses this preference segment value to determine and during the explosion of the quantity structure (BOM and routing). The results of the costing can be selected and released at material level.

Prerequisites

- You have activated the business function `LOG_SEGMENTATION_03`.
- You have created the characteristics for segmentation using transaction `CT04`. See Maintain Segmentation Relevance for Functions for more details.

  You have selected the application relevance for **BOM** when creating a characteristic for segmentation (under **Additional Data** > **Maintain Relevance** >).
- You have defined and maintained segmentation structure and strategy and have assigned it to the header. See (Segmentation Structure and Segmentation Strategy).

Product Cost Planning

In a fashion industry, the material usage and work steps in production generally depend on the segmentation values (for example, quality) that should be produced.

High Quality finished products require components of different standard than what the low quality products need. Similarly, if we consider Country of Origin as the segmentation characteristic, the cost of production in developing countries is less compared to cost of production in developed countries.

The costs of the produced stock unit (segmentation) of a product are therefore sometimes very different. In order to get a near accurate cost, you must cost the production costs of a product at the level of produced unit by considering a segment value.

There are generally two different cases regarding the costing of a product:

- The product is not split valuated. All segment values should have the same standard price (like in core standard). All segment values should have the same standard price (like in core standard).
- The product is split valuated. The valuation takes place at segment level.

The products that are not split valuated should be costed based on a segment value. For this, the costing functions used are same as that of the R/3 Standard system. However, the prerequisite for costing is to store a segment value for the costing at material level, in the material master of the material to be costed.

Product Costing at Valuation Type Level

Use

When costing a valuation type, the system determines the quantity structure that should be used (bill of material and routing) based on the respective preference segment value. You should therefore combine the segment values of a product that have the segment-dependent bills of material/routing definitions into one valuation type.

A prerequisite for the costing at valuation type level is that the accounting views that belong to the valuation types are created in the material master. There is a mass processing available for this that automatically creates the accounting views at valuation type level at the material level.

You can generally define your own valuation type for each segment value of a product. Note that the number of valuation types is decisive for the performance of the costing run (see Costing Run). Therefore, the number of valuation types per product should be as few as possible. In uncertain situations, you should therefore not consider minor differences in the production costs at segment level when defining the valuation types. Instead you should treat them as variances in the cost object controlling.

If the number of valuation types (sum of all products) that should be processed in a costing run exceeds 10,000, you should contact your SAP consultant.

Activities

To maintain costing for a material with split valuation, perform the following steps:

1. In Material Master > Accounting view, maintain the Valuation Category as X (automatic split valuation).
2. In the material master, under the Segment Valuation Data view, maintain the preference segment values and the associated valuation type.
3. Using the report Material Valuation: Create/Change (Transaction: `SGT_MAINTAIN_VALN`), you extend the material for the valuation type.
4. Enter a Default Stock Segment value in the Costing at Material Level in the MRP view. The costing at material level is also required for split valuated materials due to technical reasons.

Quantity Structure of Product Costing

Use

The basis of the costing of a product at the valuation type level is its quantity structure. In the product costing the quantity structure is based on the specifications of the BOM and the routing.

Features

The quantity structure of a product can be defined as segment-dependent. Product costing considers these segment-dependent specifications. When costing a certain segment, the system explodes the bill of material for the corresponding segment value and determines the relevant routing.

The material and production costs of a material can be applied with overhead costs within the framework of product costing. For this, you can define a costing sheet with overhead rates in the used costing variant/valuation variant (Customizing).

However, the overhead rates cannot be defined based on segment values. That means that all valuation types of a material are debited with the same overhead rates.

Definition of Bills of Material and Routing

Use

Bills of material and routings make up the quantity structure of a product. The product costing determines material and production costs of the product based on this quantity structure.

Features

Bills of Material for Segmentation Relevant Material

The definition of the quantity structure of material (product) enables you to define segment-dependent specifications.

These specifications can also be cost-relevant. For the definition of the bills of material, depending on the produced segment of the finished product, you can usually specify (for all segment values) whether or not certain components should take part in the BOM explosion by excluding or including the component for the produced segment of the finished product.

Routing for Segmentation Relevant Material

The production operations are often different depending on the segment produced. You can therefore create several alternative routing for one material within the system.

In the routing header of a material, you can specify whether or not the routing alternative should be used in the production of a certain segment by excluding or including the routing for the produced segment of the finished product.

Explosion of Multi-Level Bills of Materials

Use

When costing a finished product, the system explodes the bills of materials down to the lowest level. Then the system spreads the costs, separated by cost components, from the lowest level to the highest level, to determine the planned costs of the finished product.

The BOM explosion can be carried out from any production level.

Features

When costing a finished product, the system explodes the bills of materials down to the lowest level. Then the system spreads the costs, separated by cost components, from the lowest level to the highest level, to determine the planned costs of the finished product. The BOM explosion can be carried out from any production level.

There is one special feature that is relevant for semi-finished products whose segment values or valuation types deviate from the finished product: The product costing always explodes multi-level BOMs according to the definitions of valuation types. That means that only the preference segment value of a valuation type is used for the further explosion of a multi-level BOM.

i Note

Using the function valuated BOM in the costing transaction, you can follow the BOM explosion. For this choose a display variant (function variants in the BOM display) that displays the field valuation type.

Component Prices in Costing

Use

When costing a finished product, the system determines the material costs based on the prices of the used components (material master record). When defining a costing variant/valuation variant in Customizing you can use a valuation strategy to specify which valuation approach the system should use for costing (for example current price, future price).

Features

Basically, the system handles component prices of a segmentation relevant material in the same way as the prices of a standard material. If a component is split valuated, the system can use component prices at valuation type level in costing.

When costing a certain segment value of a finished product, the system considers segment-dependent component prices during the entire cost rollup (using all production levels). According to the specifications of the respective BOM, the system determines the component that is necessary for the segment of the finished product. Then the system determines the price of the corresponding component valuation type. If you multiply the price of the valuation type by the price of the input quantity, you get the material costs for the respective component.

Costing with Reference to the Purchasing Info Record

Use

Procuring raw materials or trading goods results in considerable delivery costs (such as freight and customs costs). These delivery costs can often be activated, but in product controlling they should be displayed separately.

The R/3 system therefore enables you to create a cost component split without quantity structure, by using the raw material costing.

In addition, the price information of the purchasing info record for the costing of subcontracted materials is also relevant (besides the component prices).

Features

- In raw material costing, you can refer to a purchasing info record of the respective material. With the conditions contained in the info record, you can create a cost component split with all cost components (including delivery costs).
- You must set the valuation strategy L in the valuation variant used and enter a corresponding sub-strategy. With the sub-strategies A and B you can refer to the condition tables. You can define the valuation variant in Customizing under **Controlling** > **Product Cost Controlling** > **Product Cost Planning** > **Material Cost Estimate with / without Quantity Structure** > **Costing Variant: Components** > **Define Valuation Variants**.
- Assign the used conditions to an origin group. The system uses this origin group to determine the cost component with which the related value should appear in the product costing.
- When costing subcontracted materials, the prices of the purchasing info record represent the costs of the subcontractor's service. You must add on the costs of the provided raw materials.
- In the raw material costing and the costing of subcontracted materials you can include conditions that are defined at the segment level.

Costing (Individual Processing)

Use

In general, the finished products of a company are costed during periodic mass processing (see Costing Run). There are also transactions in the SAP system for costing individual materials. You can save, select, and release the created costings. You can also analyze individual costings with these transactions.

Features

You can use the standard transaction for the costing of individual materials for segmentation relevant materials as well (Under SAP Easy Access Menu ▶ Logistics ▶ Production ▶ Product Cost Planning ▶). After you have exited the Material Costing: Initial Screen, the system considers the first valuation type and reads its corresponding preference segment value to execute the costing for the valuation type.

> **ℹ Note**
>
> If you choose Create Material Costing with Quantity Structure, enter the valuation type only. That means you can no longer enter the segment values because the system determines them from the valuation type.

With Create Material Costing: Initial Screen, you can only cost one segment value. The default value is the segment value that you have entered in the material master (Valuation view) for the costing at material level.

The costing is automatically carried out based on the segment that is entered in the valuation view for the costing at material level (see Costing without Split Valuation).

In the main screen of the costing display, the system shows the valuation type of the costing.

Costing Run

Use

The costing run in the R/3 System is available for the periodic costing of a big number of products/materials (for example all materials of one plant). With help of the costing run, the high-performing costing of a great number of materials per collective processing is possible.

Features

You can use the Standard costing run of the R/3 System for carrying out costing at the valuation type for segmentation relevant materials as well (Under SAP Easy Access Menu ▶ Logistics ▶ Production ▶ Product Cost Planning ▶). You can process segmentation relevant materials and standard materials in the same costing run.

You can divide the costing run into the following phases:

- Create costing run
- Select materials
- Explode quantity structure
- Execute costing run
- Mark/release costing run

These phases are the same for both segmentation relevant materials and Standard materials.

The Select materials phase offers all selection possibilities of the R/3 Standard System (plant, low-level code). The phases Execute costing run and Mark/release costing run allows the system to execute valuation type level for materials with split valuation. The costing run always processes all valuation types that are defined for a material. A costing is created at material level (valuation type BLANK). The costing results of the valuation types the system releases the accounting views at valuation type level (see Accounting Data of the Valuation Types). The result of the valuation type BLANK, the system releases in the accounting view at material level.

With the function materials overview in the list display of the costing run, you can view the created costings. For a better overview, choose a display variant that displays the field valuation type.

The explosion of the quantity structure is based on the preference segment of the respective valuation type.

The valuation type **BLANK** is costed on the basis of the segment for the costing at material level from the MRP view (material master). If the material is not split valuated, the system costs the material only on basis of the default values from the MRP view. Marking and releasing of the costing result is only executed at material level.

The phase **Explode quantity structure** is only necessary, if the selection was not carried out via all low-level codes in the phase **Select materials**. If you have selected only finished materials, you can include the low-level codes (semifinished products) into the selection with help of the quantity structure explosion. Costings are then also created, marked and released for these materials.

Activities

Follow these steps to create a costing run:

1. In the SAP Easy Access menu, choose **Accounting** > **Controlling** > **Product Cost Controlling** > **Product Cost Planning** > **Material Costing** > **Edit Costing Run**.

   Alternatively, you can also use the transaction `CK40N` to edit a costing run.

2. Enter the following fields:
   - Costing Run
   - Costing Run Date
   - Description
3. Save the changes. The costing data is displayed.
4. Under the **Cost Estimate** tab page, for each flow step, click parameter to edit costing for specific material and click **Execute**.
5. Under the **Structure Explosion** tab page, click parameter to edit costing for specific material and click **Execute**.

More Information

- Product Cost Controlling and Valuation
- Quantity Structure of Product Costing

Price Update in Material Master

Use

One important function of the product costing is the determination of a Standard price for the material valuation. You can use the Standard costing run of the R/3 System for carrying out costing at the valuation type for segmentation relevant materials Therefore you can release the results of the product costing and automatically enter it in the material master as new Standard price.

Features

Costing of a split valuated product can be saved, marked and released at valuation type level. Besides the Standard price of the accounting views, you can update the planned price fields 1, 2 and 3 with the results of the product costing. For this you can use the corresponding R/3 Standard functions.

In the material master, accounting views exist at valuation type level for the segment-dependent price information. Additionally, all planned price fields of the costing view are available at valuation type level as well. You can view the planned price fields of the costing view with the button product costing in the respective accounting view at valuation type level.

Costing Without Split Valuation

Use

If there are only minor cost differences for a product at the segment level, the product must not be split valuated.

Features

The products that are not split valuated should be costed based on a segment value. For this, the costing functions used are same as that of the R/3 Standard system. However, the prerequisite for the costing is to store a segment value for the costing at material level, in the material master of the material to be costed.

At the costing, the system uses this segment value for the determination and explosion of the quantity structure (BOM and routing). The results of the costing can be selected and released at material level.

Costing without a segment value in the MRP view is not possible if the business function `LOG_SEGMENTATION_03` is activated and the material is segmentation relevant.

For a segmentation relevant material, you must specify the default stock segment at **Material Master** > **MRP View**.

ℹ Note

You can use the costing with and without split valuation mixed. Materials with and without split valuation can be processed in one costing run.

Costing for Sales Orders

Use

In the production scenario of the make-to-order, it is possible to use the sales order items as controlling objects of the cost object controlling. The system collects all actual costs that occur during the processing of a sales order on the respective sales order item.

The actual costs and target costs are used for a target/actual comparison. You can also use the target costs with help of a corresponding condition category in the pricing of the sales order item. Concerning the results analysis of the sales order item, the target costs are important.

Features

You can determine the actual costs of a sales order item with help of product costing or an unit costing for the sales order item (Go to **VA01** > **Sales Order/Item Overview** > **Extras** > **Costing** VA01 >> Sales Order/Item Overview >> Extras >> Costing).

The product costing as well as the order-independent standard cost estimate are carried out on basis of the quantity structure (BOM, routing) of the product to be costed. The system uses the quantity of the order item as costing lot size.

When calling the costing function in the sales order, the system considers the default stock segment maintained for the material on which the costing of the respective sales order item should be based. The quantity structure and thus the planned costs of this segment are now valid for the entire order item quantity (all schedule lines).

With the unit costing you can manually create the quantity structure of a product (for products without BOM/routing).

ℹ Note

In the scenario, make-to-order with order items with costs and revenues, the sales order costing is important for the determination of the planned costs for the sales order. Because of the entered restrictions the planned costs are not determined exactly for a segment, but on basis of a defaulted segment. The following functions are affected:

- The monitoring of the actual costs via the target /actual comparison in the corresponding evaluation for the sales order item.
- The valuation of the sales order stock with valuation strategy (without valuation strategy the system valuates the sales order stock with the Standard price).
- The results analysis for the sales order item at planned costs basis (the results analysis is optional and is used for the periodic determination of delivered but not yet billed goods; is must not be carried out).

Material Cost Estimate with Quantity Structure

Costing with a quantity structure is a tool for planning costs and setting prices for materials without reference to orders. It is used to calculate the cost of goods manufactured and cost of goods sold for each product unit. You can use the results of material cost estimates with a quantity structure to valuate materials at standard prices.

For more information on standard cost estimate with quantity structure, see Material Cost Estimate with Quantity Structure.

Origin of Costing Data

Use

The quantity structure is made up of the data for material consumption and for the activities and business processes used to produce a material. The quantity structure specifies the following:

- Which material components are needed, and which materials are procured externally
- Which operations and sub-operations are carried out to produce the material, and which operations are carried out externally
- Which activities and business processes are consumed to produce the material

Integration

The cost estimate with quantity structure accesses data in the **Production Planning** (`PP and PP-PI`), **Materials Management** (`MM`), and **Controlling** (`CO`) components of the SAP System. The following table provides an overview of the origin of the data used by costing with quantity structure:

Type of data determined by cost estimate with quantity structure	From	In
The price for an externally-procured material with segmentation	The material master record or the purchasing info record . If the pricing is maintained on the segment level then this price is considered. For more information, see Determination of Pricing in Purchasing Documents based on Segmentation.	MM

More Information

Origin of Costing Data

Cost Object Controlling

Use

In the R/3 System the functions of the cost object controlling for materials with segmentation and split valuation is similar to that of materials without segmentation.

During the target cost calculation for materials which are segmented and split valuated, the system always uses the valuation type of the stock segmented that was originally intended to be produced. That means if the order was created for stock segment Q1 but the goods receipt was carried out for stock Q2, the target costs are always calculated based on the valuation type of stock segment Q1 and not that of receiving stock segment's valuation type.

Features

With the cost object controlling for production orders you can compare and analyze the target costs of a production order with the actual costs.

Target costs are the planned costs of the produced product on basis of the lot size of the production order. You must differentiate between these and the planned costs of the product costing. These are based on a (planned-) costing lot size.

Actual costs are caused by the actual resource consumption of the production order (for example material withdrawals). A

After the final delivery of an order the cost variances can be determined and calculated.

⚠ Caution

During the target cost calculation for materials which are segmented and split valuated, the system always uses the valuation type of the stock segmented that was originally intended to be produced. That means if the order was created for stock segment Q1 but the goods receipt was carried out for stock Q2, the target costs are always calculated based on the valuation type of stock segment Q1 and not that of receiving stock segment's valuation type.

Adjusting Screen Sequences

Create a new material master screen sequence to use with the segmentation functionality.

Context

Use the following procedure to adjust the screen sequence for segmentation in Material Master.

Procedure

1. Copy SAP delivered Material Master screen sequence to your own screen sequence.

   i Note

   Do not change any of the screen sequences delivered by SAP (for example, 23, 33). If necessary, you can copy a screen sequence delivered by SAP into the customer namespace.

   a. In Customizing, choose **Logistics – General** > **Material Master** > **Configuring the Material Master** > **Define Structure of Data Screens for Each Screen Sequence**.

   b. Choose the appropriate screen sequence. For example, if you want to use the standard screen sequence **21**, then select screen sequence 21 (Std ind.(short) tab pages) and choose **Copy As... (F6)**.

   c. Enter the target screen sequence, for example, Z3. Provide a meaningful description for the *<Screen seq. description>* field. Select the newly created screen sequence and double click **Data screens**.

   d. Select your newly created screen sequence (Z3).

   e. Double click the folder `Data screens` under Dialog Structure.

   f. Select the screen description Basic Data 2.

   g. Double click the folder `Subscreens`.

   h. Modify any value in the *<Program>* field. For example, provide the value `SAPLLO_SGT_MM` under *<Program>* and `2016` under *<Scr>*.

   i. Choose **Back (F3)**.

   j. In the `Data screens` folder, choose MRP 1 as the *<Screen description>* and double click the `Subscreens` folder.

   k. Modify one of the *<Program>* field values. For example, enter the value `SAPLLO_SGT_MM` under *<Program>* field and `2017` under *<Scr>* field.

   l. Choose **Back (F3)**.

   m. In the `Data screens` folder if you want to use Segment level MRP Data (85), choose **New Entries** and enter the following information:

Screen Sequence number	Z3
Logical Screen	85
Description for logical screen	Segment level MRP data
Screen Type	2
Number of the screen container	3000
Maintenance Status	D
GUI Status	ADD2001
Title type for selection group	1
Alternative screen description for "Goto" menus	Seg. level MRP data

   n. Select the newly created data screen with the logical screen 85 and double-click on the *<Subscreens>* folder to maintain the screen for segment based MRP data.

o. Modify one of the *<Program>* field values. For example, enter the value `SAPLLO_SGT_MM` in *<Program>* field and `2070` for the *<Scr>* field.

p. Choose **Back (F3)**.

q. In the `Data screens` folder if you want to use Segment Valuation Data (VD), choose **New Entries** and enter the following information:

Screen Sequence number	Z3
Logical Screen	VD
Description for logical screen	Segment Valuation Data
Screen Type	1
Number of the screen container	4000
Maintenance Status	B
GUI Status	DATE00
Title type for selection group	2
Alternative screen description for "Goto" menus	Seg. Valuation Data

r. Choose Enter.

s. Select the newly created data screen having logical screen as `VD` and double-click on *<Subscreens>* folder to maintain screen for Segmentation Valuation Data.

t. Modify one of the *<Program>* field values. For example, enter the value `SAPLLO_SGT_MM` in *<Program>* field and `2050` for the *<Scr>* field.

u. Choose **Back (F3)**.

v. Choose save and **Exit (Shift + F3)**.

ℹ Note

With respect to material master, only the following screens are applicable:

- Segmentation MRP data general data - SGT_MRP_GN
- Segmentation MRP data quantity based - SGT_MRP
- Segmentation Dependent Valuation data - SGT_MADKA

2. Assign screen sequences to a user, material type, transaction, or, industry sector.

In this Customizing activity, you can assign a screen sequence to a combination of one or more transactions, users, material types, or industry sectors. This mean the screen sequence appears automatically for the respective users when the transaction, material type, or industry sector is chosen.

ℹ Note

We recommend reading the Customizing activity documentation and making the Customizing settings:

Execute the following steps:

a. In Customizing, choose **Logistics – General > Material Master > Configuring the Material Master > Assign Screen Sequences to User/Material Type/Transaction/Industry Sector**.

b. Double click the `User screen reference` folder and choose **New Entries** .

c. Select the User screen reference folder and maintain the SRef: user= “Newly created custom screen sequence (for example "UN")” against your user name.

d. Select the `Screen sequence control` folder and choose New Entries.

e. Enter your user ID in the *<Name>* filed and the newly created screen sequence in the *<SRef: user>* fields.

f. Double click the `Screen sequence control` under Dialog Structure.

g. Choose New Entries and enter the following information:

SRef: trans.	01
SRef: user	Enter the newly created screen reference depending on the user, in this case Z3.
SRef: matl type	*
SRef: industry	*
SSq	Z3

h. Choose Save.

i. Choose on Exit (Shift + F3).

3. To maintain the entries in the V_T133D view (Maintenance of Logical Screens)

a. Execute transaction SM30.

b. Enter **V_T133D** in the *<Table/View>* field.

c. Choose Maintain.

d. Choose New Entries and enter the following information:

Screen Sequence	Logical Screen	Function Code	Successor Screen	Called indicator	Indicator: Return from call
Z3	08	SEGR	08		
Z3	26	FSDB	26		
Z3	26	FSGT	26		
Z3		PB60	85	X	X

e. Choose Save.

f. Choose Exit (Shift + F3).

4. To maintain the entries in the V_T133E view (Maintenance of Logical Screens)

a. Execute transaction SM30.

b. Enter `V_T133E` in the *<Table/View>* field.

c. Choose Maintain.

d. Choose New Entries and enter the following information:

Screen Sequence (SScrn)	Function Code	Ind:Successor Screen (Spec)
Z3	SEGR	X
Z3	FSDB	X

Screen Sequence (SScrn)	Function Code	Ind:Successor Screen (Spec)
Z3	FSGT	X
Z3	TAB1	X

e. Choose **Save**.

f. Choose **Exit (Shift + F3)**.

You can view the segmentation field in Material Master.